

Supporting Information

The Discovery of GSK3227634: A Potent Inhaled Inhibitor of the KEAP1/NRF2 Protein-Protein Interaction for the Treatment of COPD

James F. Callahan^{1†*}, Thomas G. Davies^{2†*}, Marcus Bantscheff³, Sabrina Bedard¹, Jeffrey Boehm¹, Catherine Booth-Genthe¹, Nestor Concha¹, Roy Copley⁴, Alicia Davis¹, Roderick Davis¹, Dirk Eberhard³, Joseph Foley¹, Pearl Flamberg¹, Nicole C. Goodwin¹, Tom D. Heightman², Steven R. Katchur¹, Jeffrey K. Kerns^{1*}, Casey Kmett¹, Charles J. Kotzer¹, Philip Landis¹, Tindy Li¹, Derrick Meinhold¹, Marcel Mülbaier³, Hong Nie¹, Ruth R. Osborn¹, Torren M. Peakman², Brian Peck¹, Patricia L. Podolin¹, Thomas Sweitzer¹, Hongxing Yan¹, Paris Ward¹, Thilo Werner³, Lawrence A. Wolfe III¹, Alison J.-A. Woolford², Zining Wu¹, Yolanda Sanchez¹ and William L. Rumsey¹

[†] Equal contributors

¹GSK, 1250 South Collegeville Road, Collegeville, PA 19426, USA

²Astex Pharmaceuticals, 436 Cambridge Science Park, Cambridge CB4 0QA, U.K.

³Cellzome GmbH, a GSK Company, Heidelberg, 69117, Germany

⁴GSK, Gunnels Wood Road, Stevenage, Hertfordshire, SG1 2NY, UK

*Corresponding Authors

tom.davies@astx.com

jamescallahan982@gmail.com

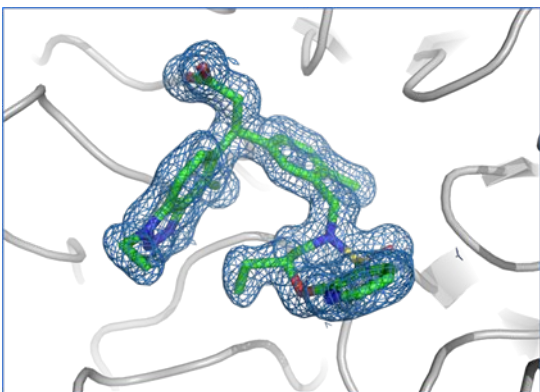
jeffrey.k.kerns@gsk.com

Contents	Page
Protein-ligand crystallography data collection and refinement statistics. 2F _o -F _c map and omit map electron density for the KEAP1/Kelch- 5 crystal structure.	S3-4
SPR data for 5 binding to Kelch domain of KEAP1.	S5
Refinement summary and crystallographic data tables associated with the small-molecule crystal structure of 5 .	S6-15
NMR Data for 5 to support conformational analysis.	S16-27
Proteins affected by 5 in affinity-enrichment based chemo-proteomics (AEC) and thermal proteome profiling (TPP) experiments.	S28-30
¹ H NMR binding data of 5 to KEAP1 and KLHL2 Kelch domains.	S31-32
LC-MS data for compound 5 .	S33
VCD Analysis Compounds 36-S , 27 , and 5 .	S34-40
Preparation of 7-21 , 23-24 .	S41-78

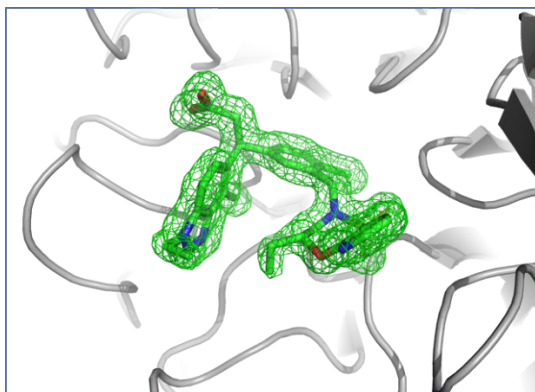
Table S1. X-ray data collection and refinement statistics for the KEAP1/Kelch-5 complex.

Wavelength (Å)	0.9787
Resolution range (Å)	33.69-1.4 (1.42-1.4)
Space group	$P6_1$
Unit cell (Å)	a=b=102.94 Å, c=55.55 Å, $\alpha=\beta=90^\circ$, $\gamma=120^\circ$
Total reflections	471033
Unique reflections	64791 (5653)
Multiplicity	7.3 (6.4)
Completeness (%)	98.0 (86.1)
Mean I/ σ I	27.7 (3.08)
Wilson B-factor (Å ²)	14.49
R-merge	0.069 (0.557)
R-meas	0.074 (0.601)
R-pim	0.027 (0.224)
CC1/2	0.999 (0.91)
Reflections used in refinement	64749 (2364)
Reflections used for R-free	3280 (124)
R-work	0.120 (0.192)
R-free	0.148 (0.246)
Number of non-hydrogen atoms	2510
macromolecules	2200
ligands	50
solvent	260
Protein residues	289
RMS (bonds) (Å)	0.006
RMS (angles) (°)	0.82
Ramachandran favored (%)	96.52
Ramachandran allowed (%)	3.48
Ramachandran outliers (%)	0.00
Rotamer outliers (%)	0.87
Clash score	4.36
Average B-factor (Å ²)	21.20
macromolecules	19.46
ligands	27.82
solvent	34.70

Statistics for the highest resolution shell are shown in parentheses.



(a) Electron density map calculated with weighted $2F_o - F_c$ coefficients and contoured at 1σ level.



(b) Simulated annealed omit electron density map calculated with weighted $F_o - F_c$ coefficients and contoured at 3.5σ level.

Figure S1: KEAP1/Kelch + 5 complex (PDB 30HE). **(a)** $2F_o - F_c$ map; **(b)** omit map electron density

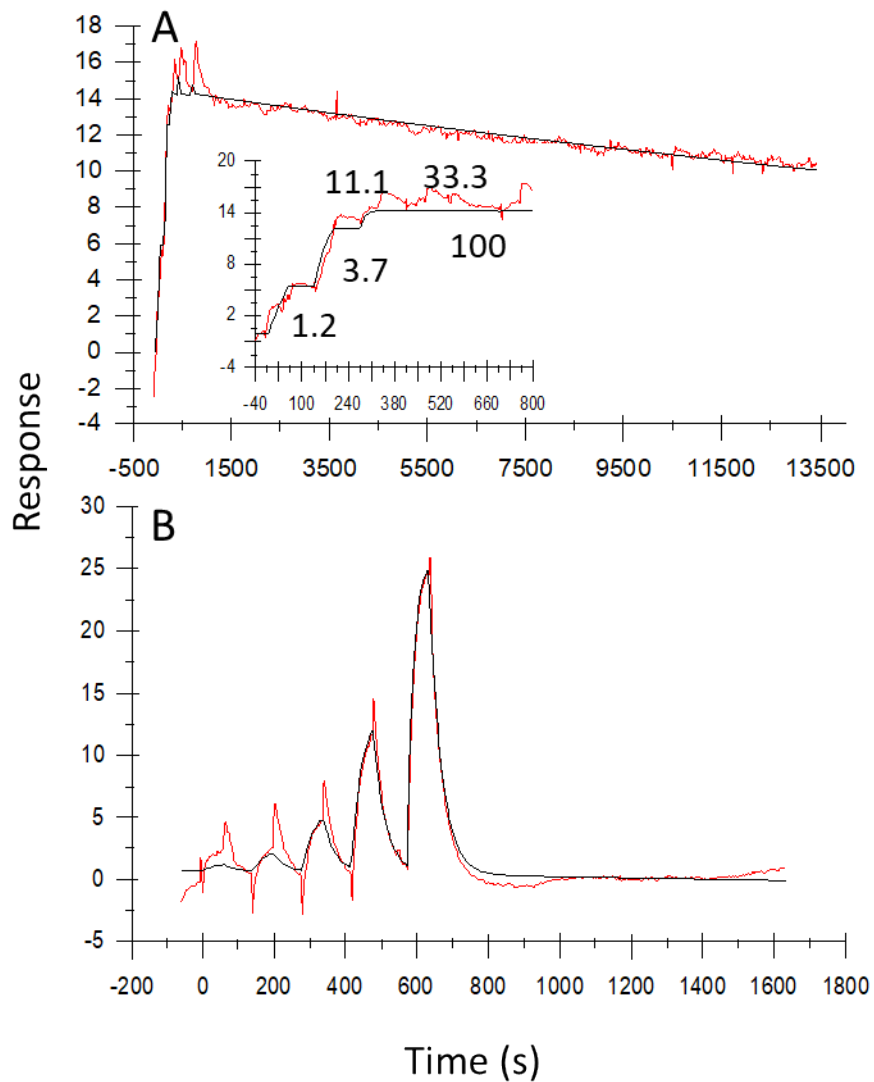


Figure S2. SPR single-cycle kinetic measurement of on-rate (k_{on}), off-rate (k_{off}), and binding affinity (K_d) for compound **5** (A) and the unlabeled NRF2 peptide AFFAQLQLDEETGEFL (B). The inset in panel A shows an expanded view of the association phase to highlight the 5-point 3-fold titration with the concentration in nM shown alongside each injection.

Small Molecule X-ray Diffraction Study of compound 5

Refinement Summary

The structure was solved and refined by full-matrix least squares procedures using the SHELXTL V2008/4 package (Bruker, 2008). Co-ordinates and anisotropic atomic displacement parameters were refined for all non-hydrogen atoms. For the hydrogen of the carboxylic acid group, atomic co-ordinates and an isotropic atomic displacement parameter were freely refined. All other hydrogen atoms were included in calculated positions and were refined using the riding mode. Isotropic atomic displacement parameters for these hydrogens were used as appropriate multiples of U_{eq} for the attached carbon atom. Further details of this data collection and refinement can be found in the crystallographic information file (CIF) deposited as ESI, together with the check CIF output. The CIF and FCF files were also deposited with the Cambridge Crystallographic Data Centre (CCDC 2541002).

Table S2. Sample and crystal data for 5.

Crystallization solvents	Acetonitrile/water	
Crystallization method	Slow evaporation	
Moiety formula	$C_{29}H_{33}N_5O_5S$	
Empirical formula	$C_{29}H_{33}N_5O_5S$	
Formula weight	563.66	
Temperature	150(2) K	
Wavelength	1.54178 Å	
Crystal size	0.38 x 0.25 x 0.19 mm	
Crystal habit	Colorless block	
Crystal system	Trigonal	
Space group	$P3_2$	
Unit cell dimensions	$a = 14.25220(10)$ Å	$\alpha = 90^\circ$
	$b = 14.25220(10)$ Å	$\beta = 90^\circ$
	$c = 12.04696(10)$ Å	$\gamma = 120^\circ$
Volume	2119.20(3) Å ³	
Z	3	
Density (calculated)	1.325 Mg/m ³	
Absorption coefficient	1.413 mm ⁻¹	
F(000)	894	

Table S3. Data collection and structure refinement for 5.

Diffractometer	Oxford Diffraction Gemini A Ultra
Radiation source	Enhance Ultra (Cu) X-ray Source, CuK α
Data collection method	ω scans
Theta range for data collection	3.58 to 66.37°
Index ranges	$-16 \leq h \leq 16, -16 \leq k \leq 16, -14 \leq l \leq 14$
Reflections collected	58212
Independent reflections	4929 [R(int) = 0.0277]
Coverage of independent reflections	99.9 %
Variation in check reflections	N/A
Absorption correction	Analytical
Max. and min. transmission	0.817 and 0.676
Structure solution technique	Direct methods
Structure solution program	SHELXT (Sheldrick, 2014)
Refinement technique	Full-matrix least-squares on F ²
Refinement program	SHELXTL V2008/4 (Bruker, 2008)
Function minimized	$\Sigma w(F_o^2 - F_c^2)^2$
Data / restraints / parameters	4929 / 1 / 370
Goodness-of-fit on F ²	1.042
$\Delta/\sigma_{\max}$	0.001
Final R indices	
4900 data; I>2 σ (I)	R1 = 0.0227, wR2 = 0.0608
all data	R1 = 0.0229, wR2 = 0.0610
Weighting scheme	calc w = 1/[$\sigma^2(F_o^2) + (0.0392P)^2 + 0.3745P$] where P = (F _o ² + 2F _c ²)/3
Absolute structure parameter	0.001(9)
Extinction coefficient	0.00087(11)
Largest diff. peak and hole	0.224 and -0.182 eÅ ⁻³

Table S4. Atomic coordinates and equivalent isotropic atomic displacement parameters ($\text{\AA}^2$) for 5. U_{eq} is defined as one third of the trace of the orthogonalized U^{ij} tensor.

	x	y	z	U_{eq}
O1	0.60607(8)	0.92443(9)	0.71605(9)	0.0305(2)
O2	0.43958(8)	0.81963(10)	0.64885(10)	0.0369(3)
C3	0.53466(11)	0.84896(11)	0.64897(11)	0.0225(3)
C4	0.58704(11)	0.80132(11)	0.57642(11)	0.0220(3)
C5	0.52889(11)	0.75789(11)	0.46440(11)	0.0210(3)
C6	0.56542(10)	0.68694(10)	0.40495(10)	0.0190(3)
C7	0.50359(11)	0.62119(10)	0.31736(11)	0.0194(3)
C8	0.54083(11)	0.55594(11)	0.26824(11)	0.0204(3)
N9	0.49687(9)	0.48024(9)	0.18402(10)	0.0228(2)
N10	0.55914(10)	0.43849(10)	0.16735(10)	0.0260(3)
N11	0.64267(10)	0.48432(9)	0.23867(10)	0.0239(2)
C12	0.63522(11)	0.55858(11)	0.30312(11)	0.0207(3)
C13	0.69916(11)	0.62585(11)	0.38901(11)	0.0231(3)
C14	0.66171(11)	0.68791(11)	0.43798(11)	0.0224(3)
C15	0.40328(11)	0.61853(12)	0.27454(13)	0.0266(3)
C16	0.72677(14)	0.45403(14)	0.23873(14)	0.0352(4)
C17	0.83482(16)	0.5457(2)	0.1988(2)	0.0577(5)
C18	0.54463(11)	0.85367(11)	0.39390(10)	0.0204(3)
C19	0.63752(11)	0.91134(11)	0.33009(11)	0.0205(3)
C20	0.65766(11)	1.00327(11)	0.27081(11)	0.0209(3)
C21	0.58366(12)	1.04134(11)	0.27608(12)	0.0245(3)
C22	0.49048(12)	0.98258(12)	0.33890(13)	0.0270(3)
C23	0.47008(11)	0.88943(12)	0.39595(12)	0.0250(3)
C24	0.60702(14)	1.14524(13)	0.21970(15)	0.0356(4)
C25	0.75965(11)	1.06126(11)	0.20245(12)	0.0239(3)
N26	0.73340(10)	1.04715(9)	0.08180(10)	0.0244(3)
S27	0.83308(3)	1.12678(3)	0.00052(3)	0.02650(9)
O28	0.88202(10)	1.23375(8)	0.04530(10)	0.0367(3)
O29	0.79218(9)	1.10738(9)	-0.11032(9)	0.0350(2)
C30	0.93120(12)	1.08502(11)	0.00974(12)	0.0250(3)
C31	1.03666(12)	1.15853(12)	0.03998(12)	0.0283(3)
C32	1.11283(13)	1.12583(13)	0.03634(14)	0.0344(3)
C33	1.07963(13)	1.02161(15)	0.00322(15)	0.0390(4)
N34	0.97679(11)	0.94775(11)	-0.02255(12)	0.0341(3)
C35	0.90423(12)	0.97908(11)	-0.01750(12)	0.0254(3)
O36	0.80126(8)	0.90262(8)	-0.04565(8)	0.0265(2)
C37	0.71738(12)	0.86561(12)	0.03971(12)	0.0255(3)
C38	0.66490(12)	0.93551(11)	0.04086(12)	0.0244(3)
C39	0.63414(13)	0.74944(13)	0.01016(15)	0.0357(4)
C40	0.68130(16)	0.67509(14)	0.00331(19)	0.0490(5)

Table S5. Selected bond lengths (Å) for 5.

O1-C3	1.3233(18)	O2-C3	1.2019(18)
C3-C4	1.5129(19)	C4-C5	1.5421(18)
C5-C18	1.5261(19)	C5-C6	1.5274(18)
C6-C7	1.3937(19)	C6-C14	1.4223(18)
C7-C8	1.4106(19)	C7-C15	1.5024(19)
C8-N9	1.3820(18)	C8-C12	1.3919(19)
N9-N10	1.3078(18)	N10-N11	1.3433(17)
N11-C12	1.3589(18)	N11-C16	1.4631(19)
C12-C13	1.395(2)	C13-C14	1.374(2)
C16-C17	1.516(3)	C18-C23	1.3895(19)
C18-C19	1.3896(19)	C19-C20	1.3902(19)
C20-C21	1.4083(19)	C20-C25	1.5075(19)
C21-C22	1.388(2)	C21-C24	1.507(2)
C22-C23	1.390(2)	C25-N26	1.4893(18)
N26-C38	1.4747(18)	N26-S27	1.6287(12)
S27-O29	1.4276(12)	S27-O28	1.4277(12)
S27-C30	1.7761(15)	C30-C31	1.384(2)
C30-C35	1.398(2)	C31-C32	1.380(2)
C32-C33	1.374(3)	C33-N34	1.345(2)
N34-C35	1.317(2)	C35-O36	1.3627(18)
O36-C37	1.4609(17)	C37-C38	1.516(2)
C37-C39	1.520(2)	C39-C40	1.514(2)

Table S6. Selected bond angles (°) for 5.

O2-C3-O1	123.96(13)	O2-C3-C4	124.52(13)
O1-C3-C4	111.48(11)	C3-C4-C5	113.52(11)
C18-C5-C6	112.63(11)	C18-C5-C4	108.59(11)
C6-C5-C4	112.55(11)	C7-C6-C14	120.06(12)
C7-C6-C5	119.34(11)	C14-C6-C5	120.60(11)
C6-C7-C8	116.17(12)	C6-C7-C15	123.41(12)
C8-C7-C15	120.41(12)	N9-C8-C12	107.31(12)
N9-C8-C7	130.50(13)	C12-C8-C7	122.19(13)
N10-N9-C8	108.79(11)	N9-N10-N11	108.57(11)
N10-N11-C12	110.42(11)	N10-N11-C16	120.96(12)
C12-N11-C16	128.62(12)	N11-C12-C8	104.91(12)
N11-C12-C13	132.88(13)	C8-C12-C13	122.20(13)
C14-C13-C12	115.49(12)	C13-C14-C6	123.86(12)
N11-C16-C17	112.02(15)	C23-C18-C19	118.03(13)
C23-C18-C5	121.39(12)	C19-C18-C5	120.50(12)
C18-C19-C20	121.89(12)	C19-C20-C21	119.89(13)
C19-C20-C25	119.04(12)	C21-C20-C25	121.06(12)
C22-C21-C20	117.85(13)	C22-C21-C24	120.89(13)
C20-C21-C24	121.20(13)	C21-C22-C23	121.74(13)
C18-C23-C22	120.54(13)	N26-C25-C20	110.54(11)
C38-N26-C25	117.27(11)	C38-N26-S27	115.28(9)
C25-N26-S27	114.83(9)	O29-S27-O28	119.63(7)
O29-S27-N26	107.33(7)	O28-S27-N26	107.88(7)
O29-S27-C30	107.89(7)	O28-S27-C30	106.74(7)
N26-S27-C30	106.71(6)	C31-C30-C35	119.18(14)
C31-C30-S27	120.13(11)	C35-C30-S27	120.65(11)
C32-C31-C30	118.44(14)	C33-C32-C31	118.08(14)
N34-C33-C32	124.38(16)	C35-N34-C33	117.20(14)
N34-C35-O36	115.58(12)	N34-C35-C30	122.58(14)
O36-C35-C30	121.78(13)	C35-O36-C37	118.24(11)
O36-C37-C38	110.65(11)	O36-C37-C39	106.29(11)
C38-C37-C39	110.19(12)	N26-C38-C37	115.74(12)
C40-C39-C37	113.40(14)		

Table S7. Selected torsion angles (°) for 5.

O2-C3-C4-C5	30.2(2)	O1-C3-C4-C5	-152.25(12)
C3-C4-C5-C18	68.24(14)	C3-C4-C5-C6	-166.36(11)
C18-C5-C6-C7	-72.07(15)	C4-C5-C6-C7	164.76(12)
C18-C5-C6-C14	107.70(14)	C4-C5-C6-C14	-15.47(17)
C14-C6-C7-C8	1.75(19)	C5-C6-C7-C8	-178.48(11)
C14-C6-C7-C15	-177.62(13)	C5-C6-C7-C15	2.1(2)
C6-C7-C8-N9	177.37(13)	C15-C7-C8-N9	-3.2(2)
C6-C7-C8-C12	-1.47(19)	C15-C7-C8-C12	177.93(13)
C12-C8-N9-N10	0.01(15)	C7-C8-N9-N10	-178.97(14)
C8-N9-N10-N11	0.27(15)	N9-N10-N11-C12	-0.46(16)
N9-N10-N11-C16	-179.47(13)	N10-N11-C12-C8	0.45(15)
C16-N11-C12-C8	179.36(14)	N10-N11-C12-C13	179.00(14)
C16-N11-C12-C13	-2.1(3)	N9-C8-C12-N11	-0.27(15)
C7-C8-C12-N11	178.80(12)	N9-C8-C12-C13	-179.02(12)
C7-C8-C12-C13	0.1(2)	N11-C12-C13-C14	-177.31(14)
C8-C12-C13-C14	1.0(2)	C12-C13-C14-C6	-0.7(2)
C7-C6-C14-C13	-0.7(2)	C5-C6-C14-C13	179.52(13)
N10-N11-C16-C17	111.74(17)	C12-N11-C16-C17	-67.1(2)
C6-C5-C18-C23	142.72(12)	C4-C5-C18-C23	-91.93(14)
C6-C5-C18-C19	-40.75(16)	C4-C5-C18-C19	84.60(14)
C23-C18-C19-C20	0.99(19)	C5-C18-C19-C20	-175.65(12)
C18-C19-C20-C21	1.12(19)	C18-C19-C20-C25	-179.45(12)
C19-C20-C21-C22	-1.84(19)	C25-C20-C21-C22	178.74(12)
C19-C20-C21-C24	175.44(13)	C25-C20-C21-C24	-4.0(2)
C20-C21-C22-C23	0.5(2)	C24-C21-C22-C23	-176.81(14)
C19-C18-C23-C22	-2.36(19)	C5-C18-C23-C22	174.25(12)
C21-C22-C23-C18	1.7(2)	C19-C20-C25-N26	110.31(13)
C21-C20-C25-N26	-70.26(16)	C20-C25-N26-C38	-52.53(16)
C20-C25-N26-S27	167.30(9)	C38-N26-S27-O29	42.65(11)
C25-N26-S27-O29	-176.38(10)	C38-N26-S27-O28	172.81(10)
C25-N26-S27-O28	-46.22(12)	C38-N26-S27-C30	-72.80(11)
C25-N26-S27-C30	68.17(11)	O29-S27-C30-C31	120.88(12)
O28-S27-C30-C31	-8.89(13)	N26-S27-C30-C31	-124.05(12)
O29-S27-C30-C35	-56.61(13)	O28-S27-C30-C35	173.62(11)
N26-S27-C30-C35	58.46(13)	C35-C30-C31-C32	3.4(2)
S27-C30-C31-C32	-174.11(12)	C30-C31-C32-C33	-0.3(2)
C31-C32-C33-N34	-2.3(3)	C32-C33-N34-C35	1.4(3)
C33-N34-C35-O36	179.11(14)	C33-N34-C35-C30	2.0(2)
C31-C30-C35-N34	-4.5(2)	S27-C30-C35-N34	173.03(12)
C31-C30-C35-O36	178.63(13)	S27-C30-C35-O36	-3.86(19)
N34-C35-O36-C37	115.76(14)	C30-C35-O36-C37	-67.14(17)
C35-O36-C37-C38	89.54(14)	C35-O36-C37-C39	-150.83(13)
C25-N26-C38-C37	-69.95(16)	S27-N26-C38-C37	70.04(14)
O36-C37-C38-N26	-70.02(15)	C39-C37-C38-N26	172.72(12)
O36-C37-C39-C40	58.03(18)	C38-C37-C39-C40	177.96(14)

Table S8. Anisotropic atomic displacement parameters ($\text{\AA}^2$) for 5.The anisotropic atomic displacement factor exponent takes the form: $-2\pi^2[h^2a^{*2}U^{11} + \dots + 2hka^*b^*U^{12}]$.

	U ¹¹	U ²²	U ³³	U ²³	U ¹³	U ¹²
O1	0.0249(5)	0.0358(6)	0.0307(5)	-0.0140(5)	-0.0004(4)	0.0151(5)
O2	0.0223(6)	0.0462(7)	0.0370(6)	-0.0154(5)	0.0023(4)	0.0130(5)
C3	0.0245(7)	0.0228(7)	0.0191(6)	0.0008(5)	0.0032(5)	0.0109(6)
C4	0.0234(7)	0.0229(7)	0.0192(7)	-0.0009(5)	0.0007(5)	0.0111(6)
C5	0.0189(6)	0.0217(7)	0.0204(7)	-0.0019(5)	0.0001(5)	0.0087(5)
C6	0.0188(6)	0.0192(6)	0.0179(6)	0.0038(5)	0.0030(5)	0.0087(5)
C7	0.0199(6)	0.0187(6)	0.0191(6)	0.0021(5)	0.0002(5)	0.0093(5)
C8	0.0216(6)	0.0195(6)	0.0188(6)	0.0017(5)	-0.0011(5)	0.0094(5)
N9	0.0245(6)	0.0218(6)	0.0233(6)	-0.0041(4)	-0.0049(5)	0.0124(5)
N10	0.0288(6)	0.0257(6)	0.0250(6)	-0.0052(5)	-0.0060(5)	0.0147(5)
N11	0.0264(6)	0.0250(6)	0.0249(6)	-0.0041(5)	-0.0039(5)	0.0163(5)
C12	0.0217(7)	0.0216(7)	0.0200(7)	0.0020(5)	0.0000(5)	0.0117(6)
C13	0.0214(7)	0.0269(7)	0.0230(7)	-0.0020(5)	-0.0039(5)	0.0135(6)
C14	0.0225(7)	0.0248(7)	0.0182(7)	-0.0027(5)	-0.0027(5)	0.0105(6)
C15	0.0239(7)	0.0282(7)	0.0303(7)	-0.0065(6)	-0.0068(6)	0.0150(6)
C16	0.0405(9)	0.0422(9)	0.0371(9)	-0.0112(7)	-0.0094(7)	0.0313(8)
C17	0.0410(10)	0.0774(15)	0.0659(14)	-0.0109(11)	0.0079(9)	0.0380(11)
C18	0.0208(7)	0.0217(6)	0.0185(6)	-0.0043(5)	-0.0025(5)	0.0106(6)
C19	0.0220(6)	0.0238(7)	0.0192(6)	-0.0040(5)	-0.0030(5)	0.0141(6)
C20	0.0222(6)	0.0218(7)	0.0188(6)	-0.0032(5)	-0.0040(5)	0.0110(5)
C21	0.0279(7)	0.0221(7)	0.0253(7)	-0.0057(5)	-0.0082(5)	0.0138(6)
C22	0.0245(7)	0.0299(8)	0.0333(8)	-0.0055(6)	-0.0053(6)	0.0186(6)
C23	0.0201(7)	0.0283(7)	0.0270(7)	-0.0030(6)	0.0001(5)	0.0123(6)
C24	0.0368(9)	0.0307(8)	0.0452(10)	0.0041(7)	-0.0036(7)	0.0212(7)
C25	0.0241(7)	0.0241(7)	0.0231(7)	0.0017(5)	-0.0011(5)	0.0118(6)
N26	0.0275(6)	0.0215(6)	0.0233(6)	0.0025(5)	0.0009(5)	0.0115(5)
S27	0.03187(19)	0.02014(16)	0.02633(17)	0.00812(13)	0.00527(14)	0.01212(14)
O28	0.0435(6)	0.0198(5)	0.0432(6)	0.0085(5)	0.0109(5)	0.0130(5)
O29	0.0431(6)	0.0357(6)	0.0278(6)	0.0125(4)	0.0051(5)	0.0210(5)
C30	0.0279(7)	0.0241(7)	0.0201(7)	0.0060(5)	0.0044(5)	0.0107(6)
C31	0.0293(8)	0.0223(7)	0.0238(7)	0.0014(6)	0.0028(6)	0.0057(6)
C32	0.0258(8)	0.0347(8)	0.0323(8)	0.0001(6)	-0.0013(6)	0.0074(7)
C33	0.0286(8)	0.0419(9)	0.0463(10)	-0.0036(7)	-0.0047(7)	0.0175(7)
N34	0.0294(7)	0.0292(7)	0.0424(8)	-0.0025(6)	-0.0015(6)	0.0139(6)
C35	0.0267(7)	0.0237(7)	0.0208(7)	0.0011(5)	0.0013(5)	0.0089(6)
O36	0.0235(5)	0.0234(5)	0.0256(5)	0.0000(4)	0.0021(4)	0.0065(4)
C37	0.0264(7)	0.0236(7)	0.0236(7)	0.0039(5)	0.0037(5)	0.0103(6)
C38	0.0232(7)	0.0233(7)	0.0226(7)	0.0018(5)	-0.0008(5)	0.0087(6)
C39	0.0309(8)	0.0242(8)	0.0462(10)	0.0017(7)	0.0081(7)	0.0094(7)
C40	0.0471(11)	0.0259(8)	0.0714(13)	0.0075(8)	0.0231(9)	0.0163(8)

Table S9. Hydrogen atom coordinates and isotropic atomic displacement parameters ($\text{\AA}^2$) for 5.

	x	y	z	U
H1	0.5745(17)	0.9508(17)	0.7590(17)	0.043(5)
H4A	0.5880	0.7414	0.6171	0.026
H4B	0.6631	0.8577	0.5620	0.026
H5	0.4496	0.7120	0.4800	0.025
H13	0.7642	0.6286	0.4121	0.028
H14	0.7025	0.7342	0.4974	0.027
H15A	0.3497	0.5961	0.3343	0.040
H15B	0.3731	0.5669	0.2130	0.040
H15C	0.4216	0.6909	0.2484	0.040
H16A	0.7039	0.3902	0.1901	0.042
H16B	0.7352	0.4332	0.3149	0.042
H17A	0.8257	0.5701	0.1256	0.087
H17B	0.8870	0.5201	0.1934	0.087
H17C	0.8617	0.6060	0.2514	0.087
H19	0.6887	0.8873	0.3269	0.025
H22	0.4393	1.0066	0.3430	0.032
H23	0.4046	0.8499	0.4367	0.030
H24A	0.5475	1.1593	0.2340	0.053
H24B	0.6141	1.1387	0.1396	0.053
H24C	0.6747	1.2052	0.2489	0.053
H25A	0.7983	1.1394	0.2211	0.029
H25B	0.8080	1.0321	0.2202	0.029
H31	1.0562	1.2298	0.0627	0.034
H32	1.1861	1.1740	0.0562	0.041
H33	1.1328	1.0002	-0.0018	0.047
H37	0.7496	0.8678	0.1141	0.031
H38A	0.6415	0.9390	-0.0356	0.029
H38B	0.5990	0.8994	0.0876	0.029
H39A	0.5759	0.7209	0.0667	0.043
H39B	0.6010	0.7489	-0.0622	0.043
H40A	0.7351	0.6995	-0.0565	0.073
H40B	0.6232	0.6010	-0.0120	0.073
H40C	0.7161	0.6767	0.0740	0.073

Table S10. Selected hydrogen bond information for 5 (Å and °).

D-H...A	d(D-H)	d(H...A)	d(D...A)	<(DHA)
O1-H1...N9#1	0.88(2)	1.85(2)	2.7362(15)	175(2)
#1 -y+1,x-y+1,z+2/3				

NMR Data for 5 to support conformational analysis

NMR chemical shifts for **5** in DMSO-d₆ and in sodium phosphate buffer (pH 7.1) are summarized in **Table S5**. ¹H and 2D spectra in DMSO-d₆ are shown in **Figures S4-S8** and ¹H, 2D and key traces from the ROESY spectrum in sodium phosphate buffer are shown in **Figures S9-S13**. The numbering scheme used is shown below.

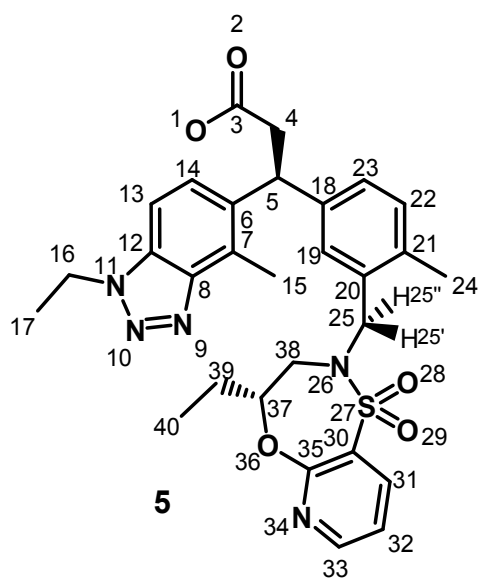


Table S11. NMR chemical shift data for 5 in DMSO-d6 and in sodium phosphate buffer.

Data in DMSO-d6			Data in 50 mM phosphate		
No	δ_H (Multiplicity, <i>J</i>)	δ_C	No	δ_H (Multiplicity, <i>J</i>)	δ_C
1	12.11 (s)	-	1	-	nd
3	-	172.6	3	-	nd
4'	3.04 (m)	39.8	4'	2.94 (d, 8.1 Hz)	45.4
4''	3.04 (m)	39.8	4''	2.94 (d, 8.1 Hz)	45.4
5	4.79 (t, 7.9 Hz)	40.8	5	4.89 (t, 8.1 Hz)	43.9
6	-	136.6	6	-	nd
7	-	126.4	7	-	nd
8	-	145.9	8	-	nd
12	-	130.7	12	-	nd
13	7.59 (m)	107.3	13	7.64 (d, 8.8 Hz)	108.9
14	7.46 (m)	126.4	14	7.61 (d, 8.8 Hz)	128.7
15	2.74 (s)	12.5	15	2.67 (s)	13.8
16	4.66 (q, 7.3 Hz)	42.3	16	4.73 (q, 7.3 Hz)	44.7
17	1.46 (t, 7.3 Hz)	14.7	17	1.57 (t, 7.3 Hz)	15.4
18	-	141.4	18	-	nd
19	7.24 (m)	129.2	19	6.86 (d, 1.9 Hz)	130.1
20	-	132.9	20	-	nd
21	-	135.1	21	-	nd
22	7.12 (d, 7.8 Hz)	130.5	22	7.24 (d, 7.9 Hz)	132.3
23	7.23 (m)	127.4	23	7.38 (dd, 7.9,1.9 Hz)	129.4
24	2.24 (s)	17.9	24	2.27 (s)	18.8
25'	3.93 (d, 14.0 Hz)	48.3	25'	3.87 (d, 14.4 Hz)	50.6
25''	4.43 (d, 13.9 Hz)	48.3	25''	4.52 (d, 14.4 Hz)	50.6
30	-	128.3	30	-	nd
31	8.23 (dd, 7.6,1.9 Hz)	139.0	31	8.27 (dd, 7.7,1.9 Hz)	141.4
32	7.46 (m)	121.1	32	7.45 (dd, 7.7,5.0 Hz)	122.9
33	8.55 (dd, 4.9,1.9 Hz)	152.8	33	8.46 (dd, 5.0,1.9 Hz)	153.6
35	-	160.0	35	-	nd
37	4.34 (tdd, 9.9,4.3,1.3 Hz)	77.2	37	4.05 (m)	80.4
38ax	3.65 (m)	50.8	38ax	3.64 (dd, 15.7,10.3 Hz)	52.7
38eq	2.83 (dd, 15.3,1.3 Hz)	50.8	38eq	2.87 (d, 15.5 Hz)	51.5
39'	1.31 (dq, 14.5,7.3,4.2 Hz)	25.4	39'	1.06 (ddt, 14.6,12.0,7.4 Hz)	26.5
39''	1.51 (ddd, 14.0,8.2,7.2 Hz)	25.4	39''	1.36 (dt, 14.8,7.5 Hz)	26.5
40	0.94 (t, 7.3 Hz)	9.4	40	0.65 (t, 7.4 Hz)	9.8

nd = not determined

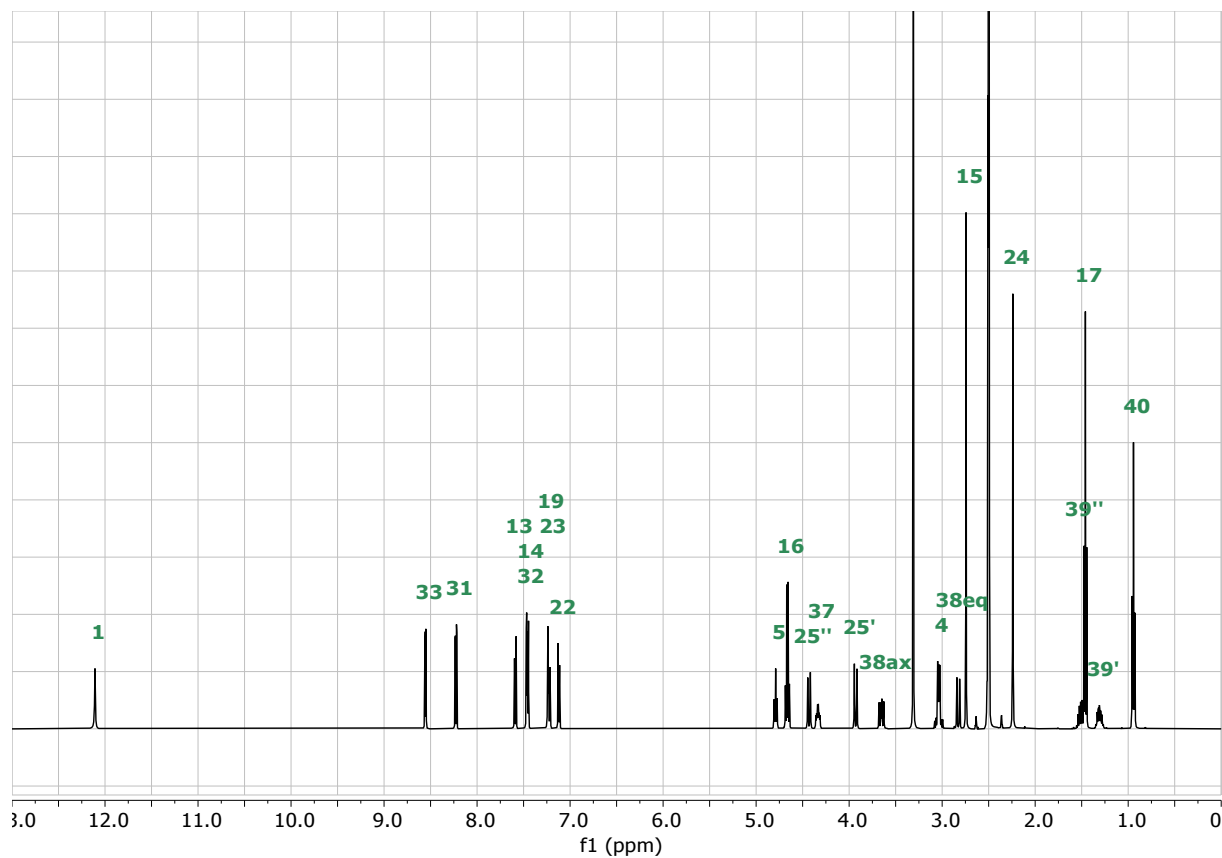


Figure S3. 500 MHz ^1H NMR spectrum for 5 in DMSO- d_6 .

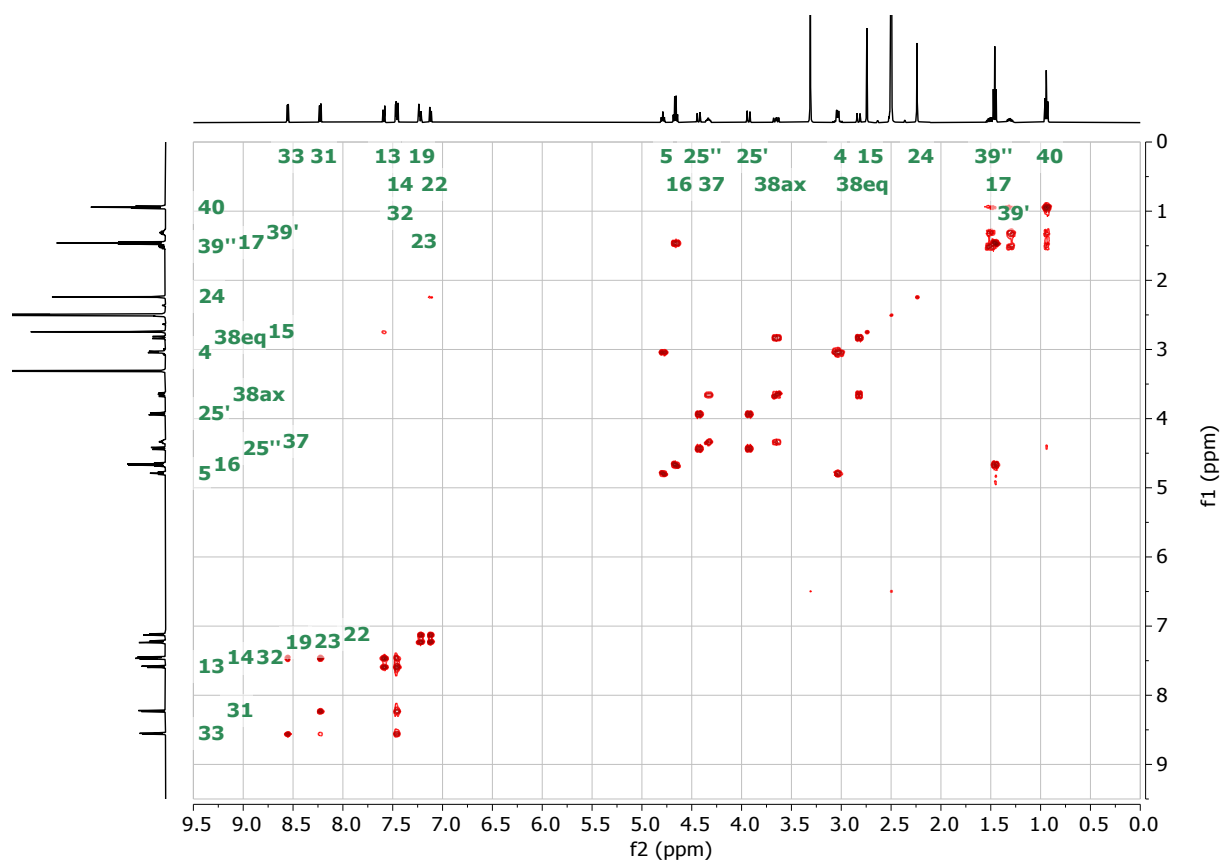


Figure S4. 500 MHz COSY spectrum for 5 in DMSO-d₆.



Figure S5. 500 MHz edited HSQC spectrum for 5 in DMSO-d6.

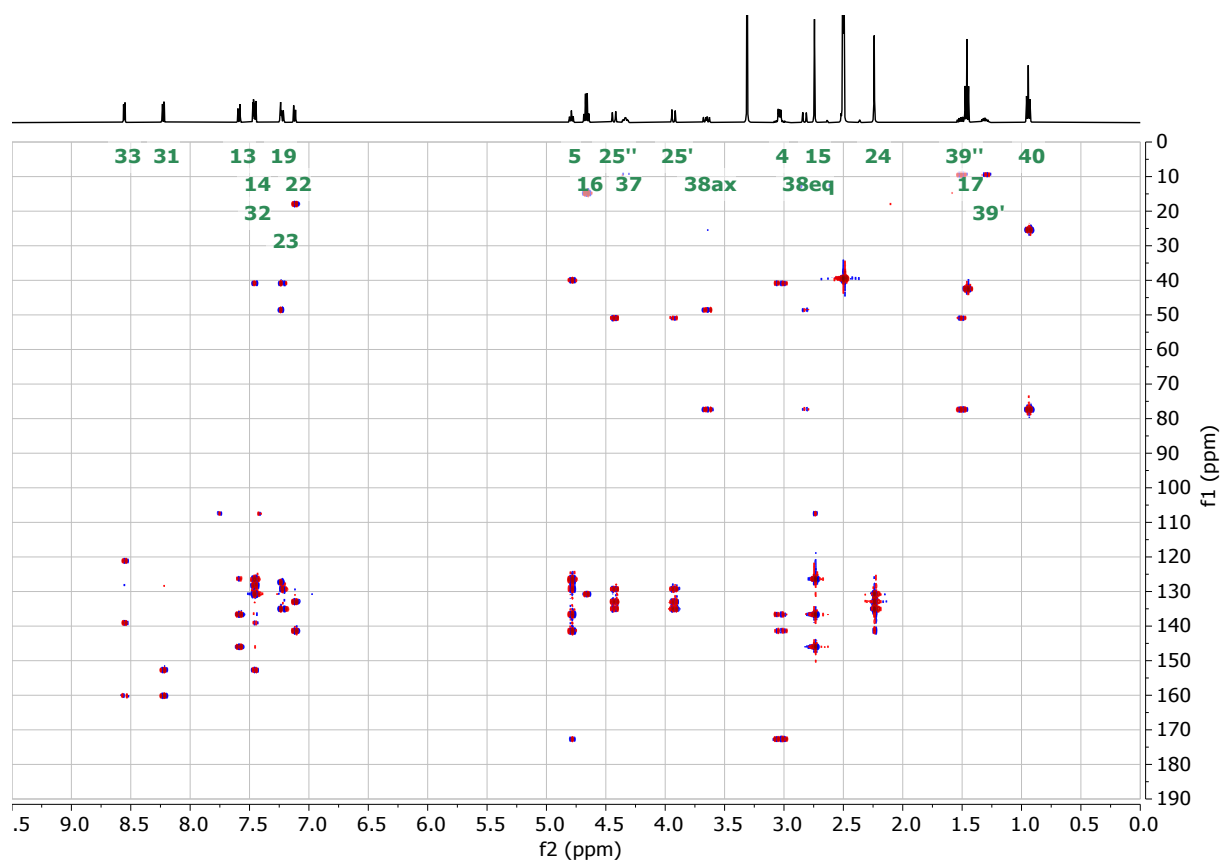


Figure S6. 500 MHz HMBC spectrum for 5 in DMSO-d₆.

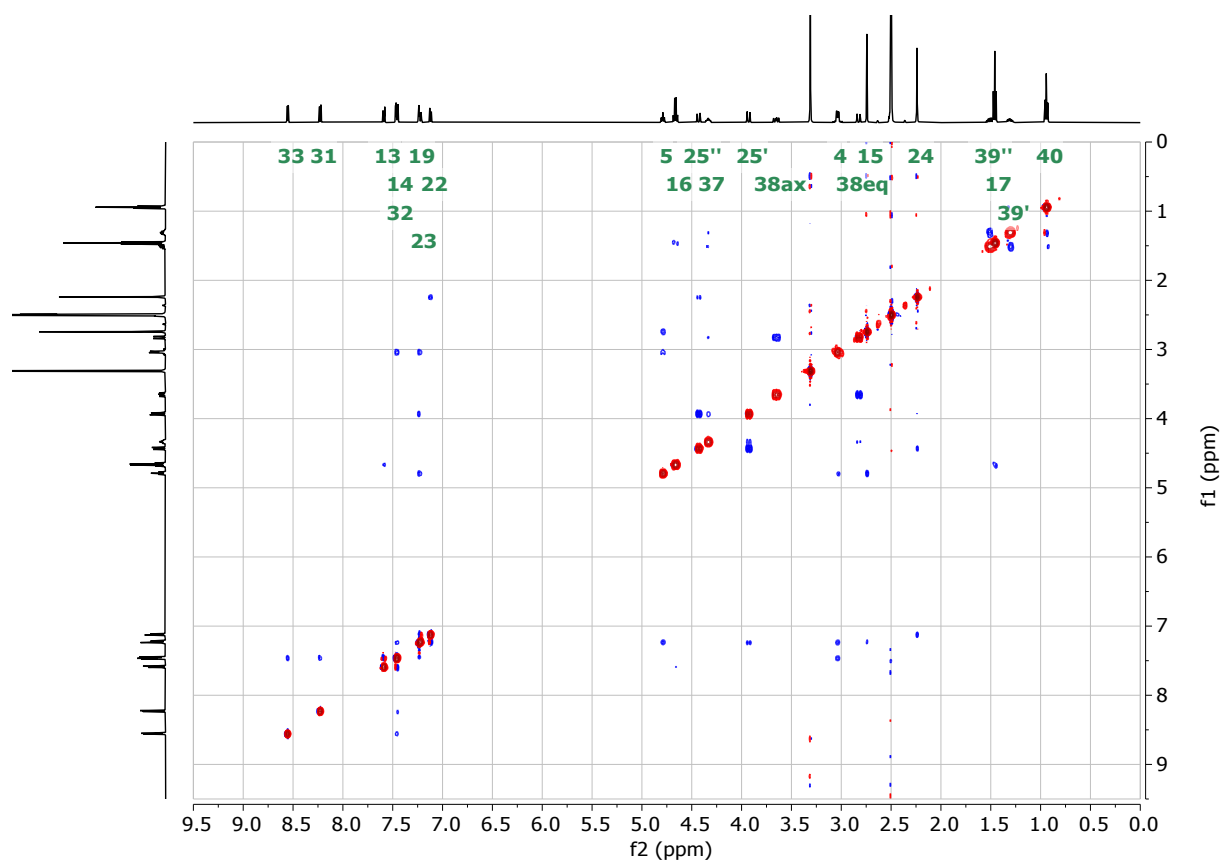


Figure S7. 500 MHz ROESY spectrum for 5 in DMSO-d₆.

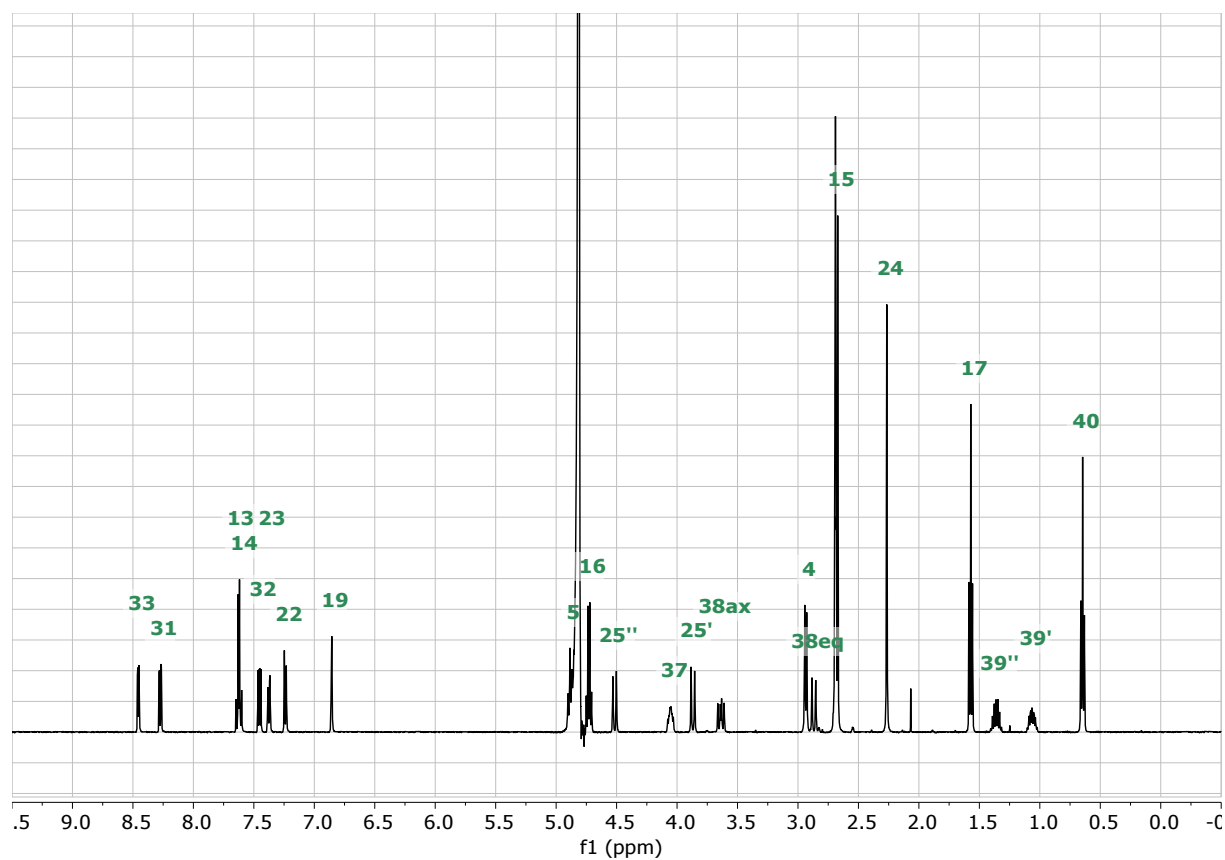


Figure S8. 500 MHz ^1H NMR spectrum for 5 in sodium phosphate buffer with water suppression.



Figure S9. 500 MHz edited HSQC spectrum for 5 in sodium phosphate buffer.

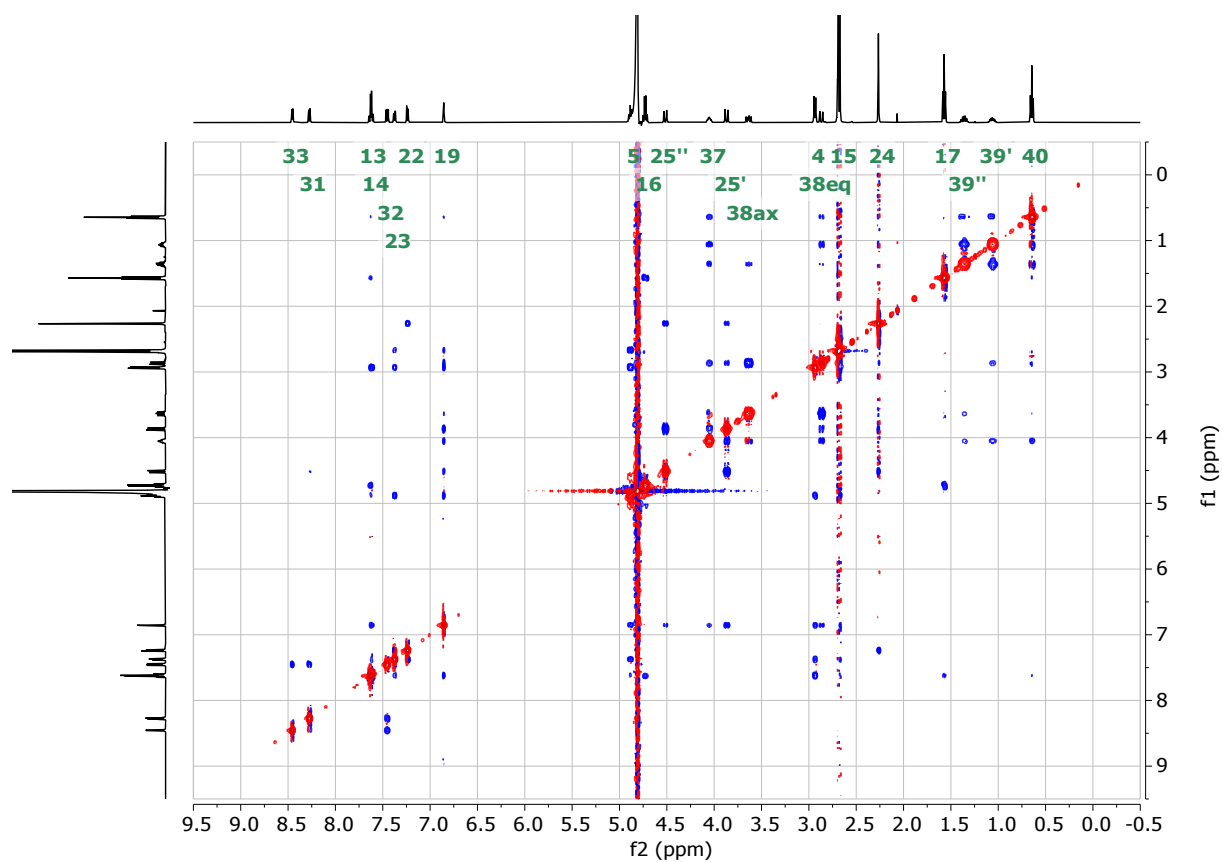


Figure S10. 500 MHz ROESY spectrum for 5 in sodium phosphate buffer with water suppression.

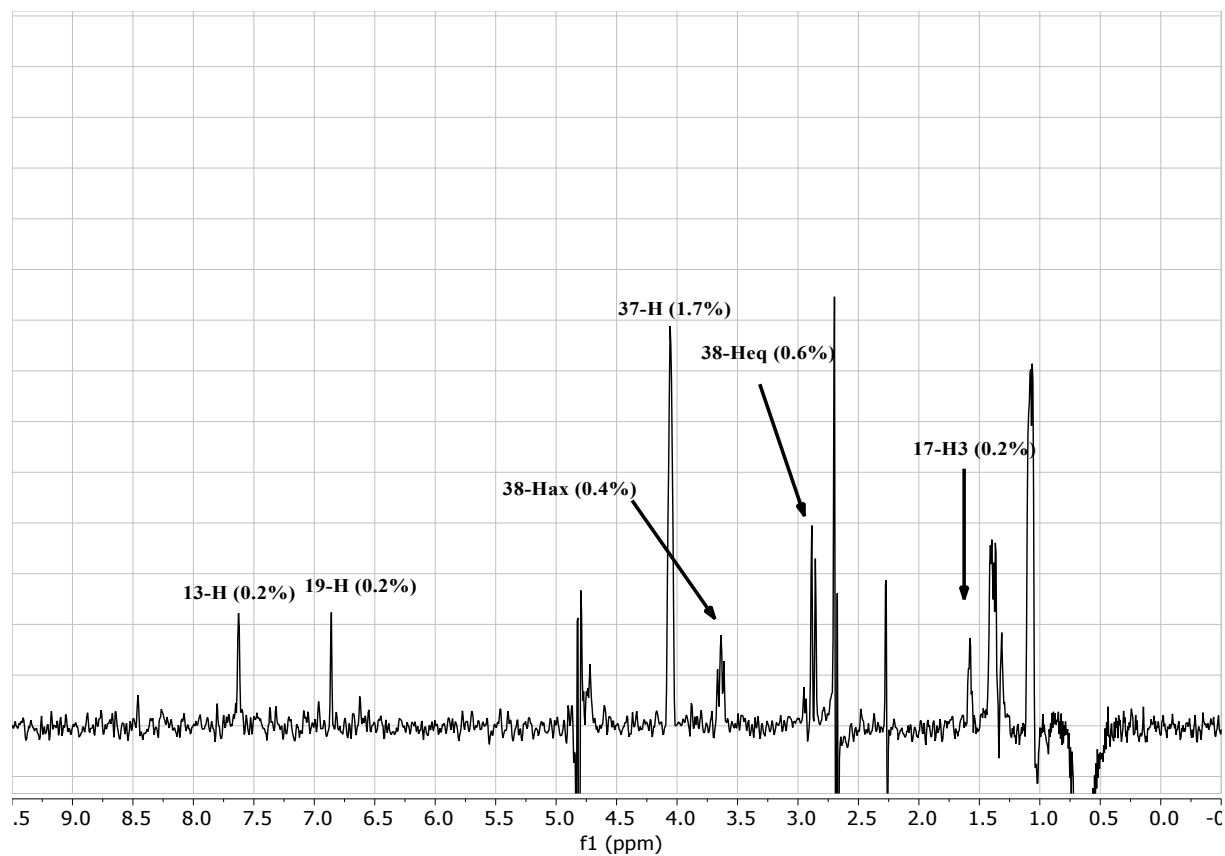


Figure S11. Trace in f2 from the ROESY spectrum for the methyl group (40-H₃) with % enhancements highlighted.

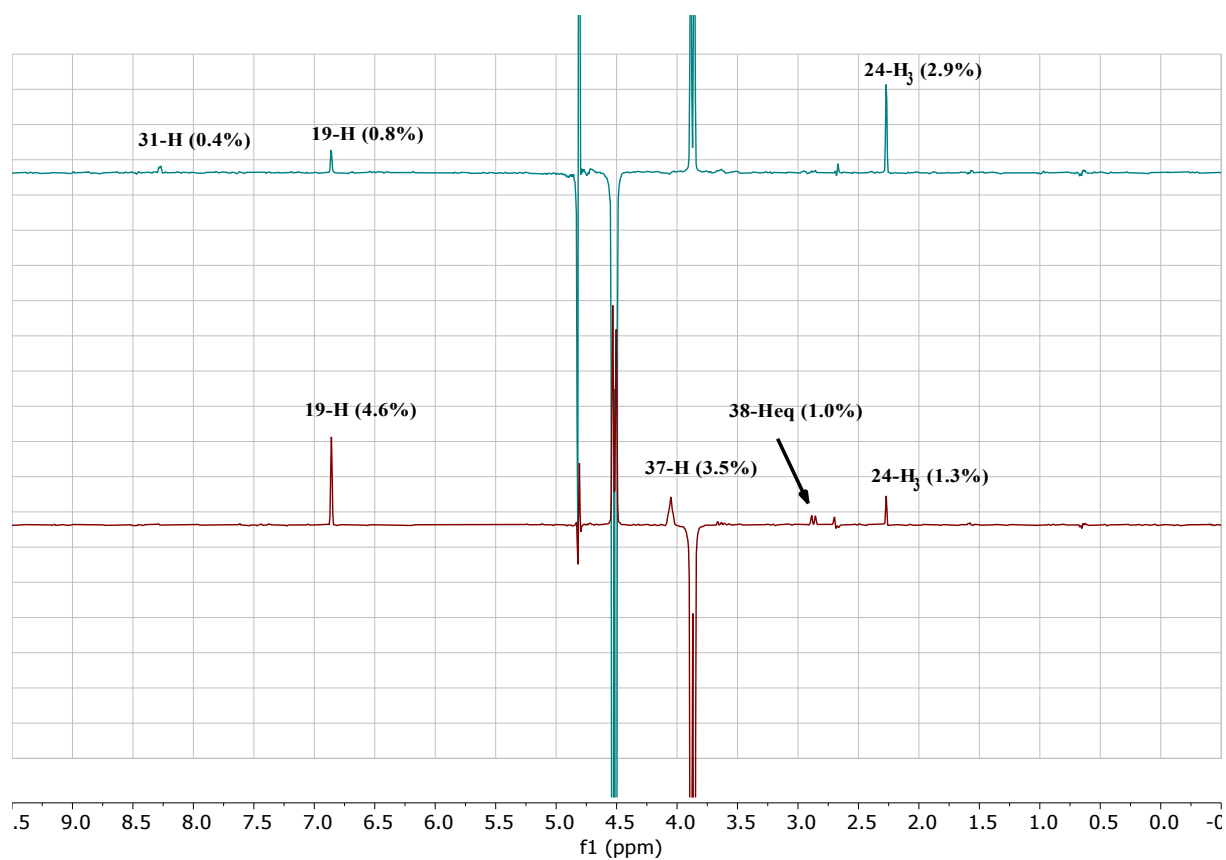


Figure S12. Traces in f2 from the ROESY spectrum for the hydrogens of the methylene linker (upper panel 25-H''; lower panel 25-H') with % enhancements highlighted.

Affinity enrichment chemo-proteomics (AEC) and thermal proteome profiling (TPP) of 5

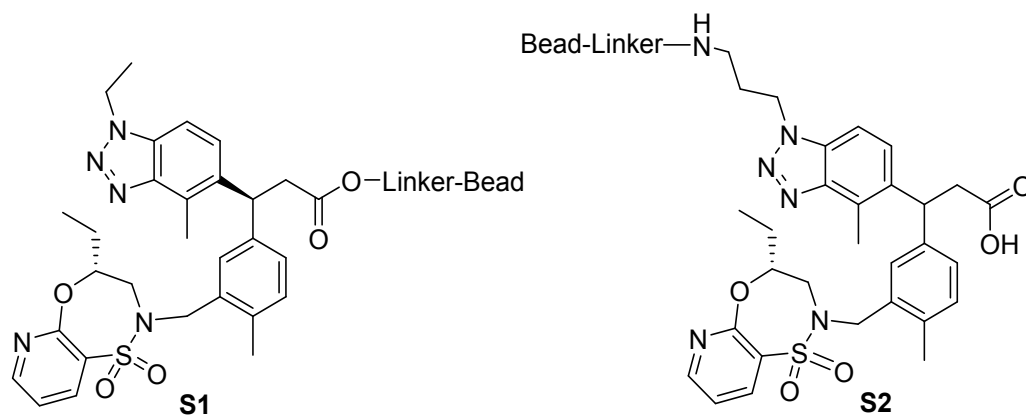
Affinity-enrichment based chemo-proteomics (AEC)

For AEC two compound matrices were generated. Compound **5** was linked to sepharose beads through the carboxylic acid moiety **S1** and **5** was linked to sepharose beads through the amine moiety **S2** to yield a concentration of 1 mM on dry beads. Briefly, 5 mg aliquots of protein extracts from HEK293, K-562 and HepG2 cell and human placenta were pre-incubated with 1 μ M, 0.1 μ M, 0.01 μ M, 1 nM, 0.1 nM and 0.01 nM compound **5** for 45 min at 4°C, respectively. 35 μ L of each of the compound matrices were added to aliquots of the lysate-compound mix (in two separate experiments) and incubated for 1 hour at 4°C. The affinity-matrices were then removed from the lysates by filtration, and the matrix-bound proteins were eluted, digested and quantified by LC-MS/MS. Two consecutive incubations with the affinity-matrices were performed to determine the IC_{50} / pK_d app correction factor F for each captured protein. Firstly, a 5 mg aliquot of the mixed cell/tissue extract was incubated with 35 μ L of the affinity matrix for 1 h at 4°C. Then, the non-bound fraction was subjected to a re-binding step with 35 μ L of fresh affinity-matrix for 1 h at 4°C. Matrix-bound proteins were eluted, digested and quantified by LC-MS/MS. Experiments were performed in duplicate.¹³

Table S12. Affinity enrichment chemo-proteomics (AEC) data

Gene Symbol (HGNC)	Gene Name (HGNC)	pK_d^{app}, S1 matrix	pK_d^{app}, S2 matrix
KLHL2	Kelch-like family member 2	10.3; 10.3	10.4; 10.6
FBXL17	F-box and leucine-rich repeat protein 17	10.4; 10.4	10.2; 10.2
KEAP1	Kelch-like ECH-associated protein 1	9.2; 9.4	9.6; 9.3
PGAM5	Phosphoglycerate mutase family member 5	n.d.	8.1; 8.1
SQSTM1	Sequestosome 1	n.d.	8.1; 8.1
KLHL17	Kelch-like family member 17	10.7; 10.5	n.d.
KLHL7	Kelch-like family member 7	10.3; 9.8	n.d.

Affinities determined in a competition binding assay with endogenous proteins and two different capturing matrices. (n.d.: not determined)



Thermal proteome profiling (TPP)

In short, changes in abundances of extracted proteins upon incubation with compound are computed relative to the vehicle control for the corresponding temperature and data are normalized to the median of the distribution. If a protein's abundance at the highest compound concentration is more than 2-fold increase compared to vehicle treatment, it is considered stabilized by the compound. Conversely, if a protein's abundance at the highest compound concentration is reduced by more than 2-fold, it is considered as destabilized. Subsequently, concentration dependent protein abundances are transformed to range from 0 to 1 for stabilized proteins and from 1 to 0 for destabilized proteins. Sigmoidal curve fitting of compound dependent changes in detected protein quantities enables the determination of EC_{50} values at different temperatures, representing the concentration for which 50% of the respective (off-) target is bound by the compound. Binding data for off-target candidate proteins are required to meet the quality criteria and to be stabilized/destabilized in at least two consecutive temperatures. Affected proteins are classified according to their EC_{50} values. The thermal stability of an off-target protein is considered as strongly affected, if its EC_{50} is within 10-fold of the cellular potency in the respective efficacy-target dependent cellular assay. It is regarded as moderately affected if its EC_{50} is within 10 and 100-fold of the cellular EC_{50} , and as weakly affected if its EC_{50} is more than 100-fold from EC_{50} in the cellular assay.¹⁶

Table S13. Thermal proteome profiling (TPP) data

Gene Symbol (HGNC)	Gene Name (HGNC)	pEC₅₀ (temperature)
KEAP1	Kelch-like ECH-associated protein 1	> 7.7 (50.4 °C); > 7.7 (51.9 °C)
FABP1	Fatty acid binding protein 1, liver	6.8 (56.1 °C); 6.6 (58.2 °C)
YIPF3	Yip1 domain family, member 3	6.1 (62.4 °C)
TUBB	Tubulin, beta class I	6.0 (63.9 °C)
AKR1C3	Aldo-keto reductase family 1, member C3	< 6.0 (50.4 °C); < 6.0 (51.9 °C)
APRT	Adenine phosphoribosyltransferase	< 6.0 (63.9 °C)
CPOX	Coproporphyrinogen oxidase	< 6.0 (62.4 °C); < 6.0 (63.9 °C)
COASY	CoA synthase	< 6.0 (48.1 °C); < 6.0 (50.4 °C)
PPOX	Protoporphyrinogen oxidase	< 6.0 (54 °C)

Proteins with an altered thermal stability in 2-dimensional thermal profiling experiment.

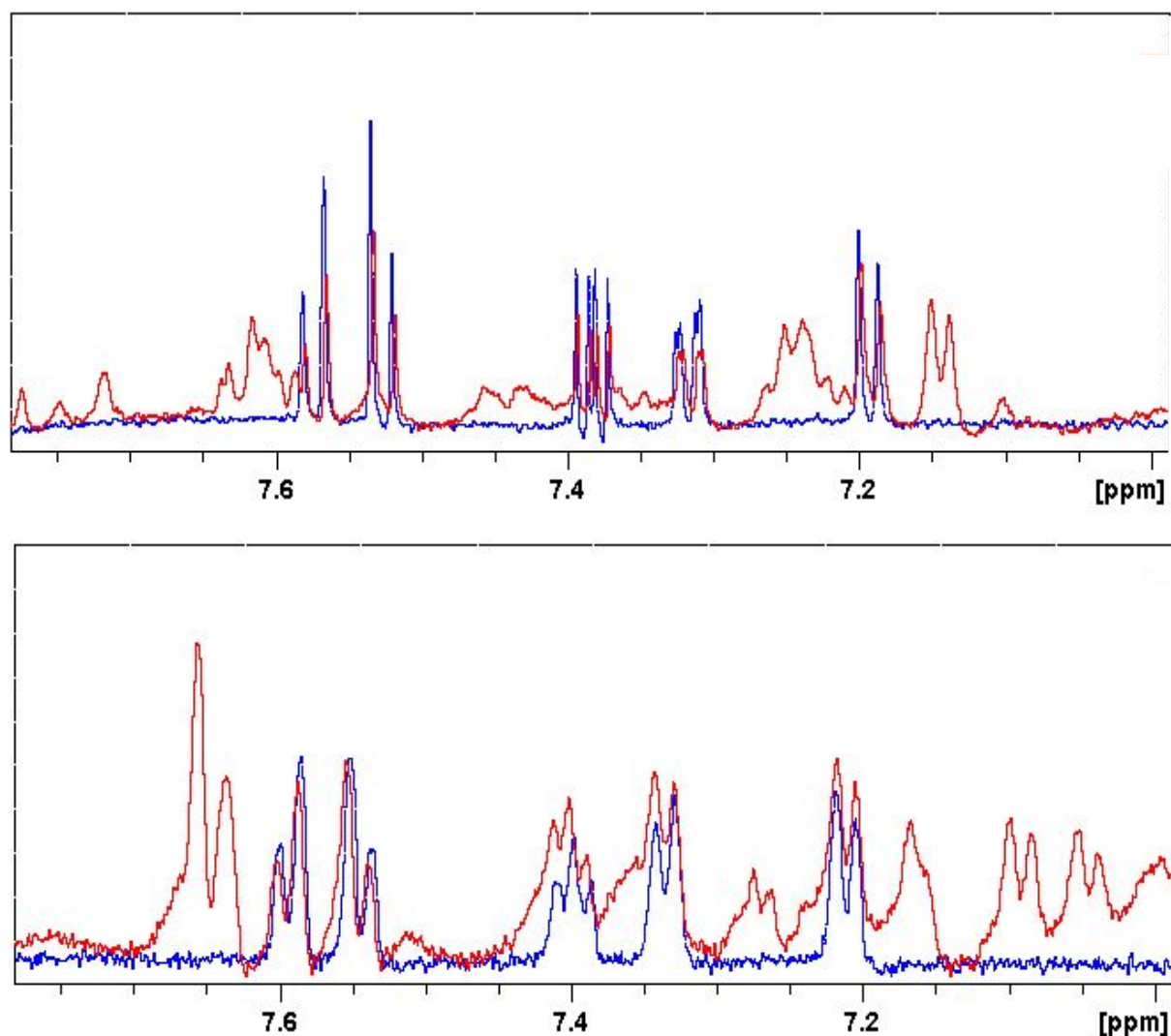


Figure S13. ^1H NMR spectra of compound **5** without (blue) and with (red) protein were collected to determine binding via reduction in peak intensity and line broadening. The top panel refers to **5** with and without human KEAP1 while the bottom panel shows **5** with and without human KLHL2. The compound peak intensity is reduced when human KEAP1 is added. The reduction is about 40%, which is expected for this tight-binding compound and the protein and compound concentrations used. No intensity change or significant broadening was observed for **5** in the presence of KLHL2. The differences in line sharpness and ppm values between the top and bottom panels can be explained by the storage buffers of KLHL2 and KEAP1, which were significantly different and made up a large fraction of the final NMR samples.

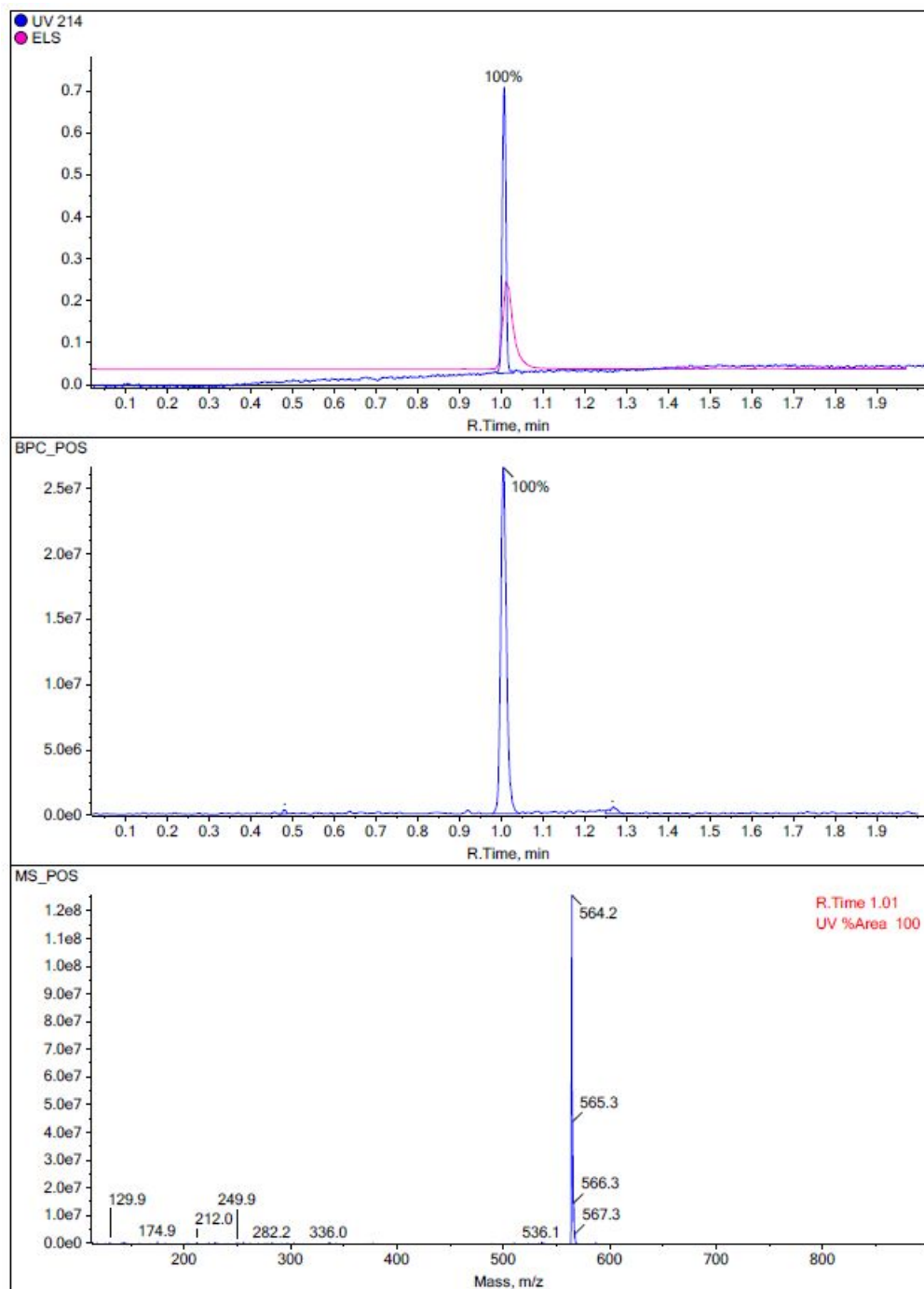
NMR Binding Studies of Compound 5

6His-Avi-TEV-human KEAP1 (321-609) was expressed in *E. coli* and cells were lysed by sonication in 50 mM Tris pH 8.0, 300 mM NaCl, 10% glycerol and 0.1 mM TCEP. The cleared lysate was combined with Ni-NTA superflow resin and allow to batch bind for two hours at 4 °C, before washing with 20 mM imidazole and eluting the protein with lysis buffer containing 300 mM imidazole. The elution fractions were pooled and concentrated before loading into a Superdex 200 column pre-equilibrated with 20 phosphate pH 7.5, 150 mM NaCl, 5% glycerol, 0.1 mM TCEP. The elution fractions were pooled, and the final concentration was 1.2 mg/mL

Flag-6His-Tev-human KLHL2 (294-593) was expressed in *E. coli* and the cells were lysed by sonication in lysis buffer (50 mM HEPES pH 7.5, 500 mM NaCl, 0.1 mM TCEP, 5 mM imidazole and 5% glycerol). The cleared lysate was combined with Ni-NTA superflow resin and allow to batch bind for two hours at 4 °C, before washing with the lysis buffer and eluting the protein with lysis buffer containing 250 mM imidazole. The elution peak was batch bound with anti-flag M2 resin and washed with lysis buffer, before eluting with 50 mM HEPES pH 7.5, 500 mM NaCl, 0.1 mM TCEP and 0.15 mg/mL flag peptide. The elution fractions were pooled and concentrated to 2.3 mg/mL.

¹H NMR spectra of 50 µM (**5**) dissolved in d₆-DMSO without and with 20 µM protein were acquired in 50 mM Tris pH 8.0, 150 mM NaCl, 0.1 mM TCEP at 298 K on a shielded Oxford 600 MHz spectrometer equipped with a Bruker AVANCE HD 600 console and a QCI-F cryoprobe using 5 mm capillary tubes with a sample volume of 500 µL. Data processing was performed using TopSpin 3.6.2 (Bruker).

LCMS Trace of Compound 5



Use of VCD to confirm absolute stereochemistry of compound 5

The absolute stereochemistry of the propanoic acid center in **5** could not be reliably assigned initially using calculated VCD data, due in large part to the availability of one of the isomers at the time this analysis was first performed. This first lot of **5** was purified by chiral HPLC after the last step providing **5** and **22**. Compound **5** was later re-synthesized utilizing **36-S** as an intermediate where the C-5 center was set as S, and **27** as described in experimental is enantiomerically pure.

VCD Analysis of 36-S

A sample of **36-S** was obtained after chiral preparative HPLC of **36** as described in the experimental.

Conformational Search: MOE stochastic search using MMFF94x force field

- Model Chemistry: # opt freq=(noraman,vcd) b3lyp/dgdzvp2 scrf=(solvent=ethanol)
- Conformational Analysis: Fractional populations estimated using Boltzmann statistics
- Lorentzian band width: 8 cm⁻¹
- Frequency scale factor: 0.9825
- Estimation of Confidence Limit: CompareVOA (BioTools, Inc.) analysis.

Experimental:

- Spectrometer: BioTools ChiralIR-2X FT-VCD spectrometer operated at 4 cm⁻¹
- VCD Frequency Range: 2400-800 cm⁻¹
- PEM Calibration: PEM calibrated at 1400 cm⁻¹
- PEM Retardation Settings: PEM1 = 0.250*λ; PEM2 = 0.245*λ
- ChiralIR-2X Scan Method(s): **36-S**: four 4 h block scan (49920 total scans)
- Solvent: CDCl₃
- Concentration: **36-S** ~ 12 mg/120uL
- Baseline Correction (modified half-difference using solvent background VCD spectrum)
$$\text{VCD}_{36-S \text{ 11-B1}} (\text{corr'd}) = \text{VCD}_{36-S \text{ 11-B1}} \text{ minus VCD solvent}$$
- Additional Processing: Savitsky-Golay 9-point smooth.

Results:

Analysis of Experimental and Calculated Data:

The VCD and IR spectra observed for **36-S** are compared with calculated data in Figure S14. The green box in each panel highlights the spectral range over which VCD data were used to assign stereochemistry and estimate the confidence limit for this assignment. Inspection of VCD data in the analysis range indicates that the VCD spectrum of **36-S** is largely coincident with the model spectrum, consistent with this molecule having the same S absolute configuration as the model. The confidence limit in this assignment was estimated to be >95%, consistent with a moderately reliable assignment. This level is typical of that expected for single enantiomer VCD analysis due to greater uncertainty in the experimental VCD intensities used to quantify the correlation between the experimental and calculated VCD spectra. As shown in the bottom panel of Figure S14, the IR spectrum calculated for Model fs1 is in reasonably good agreement with experimental, supporting accurate coverage of this molecule's solution phase conformational space, as required for a reliable VCD assignment.

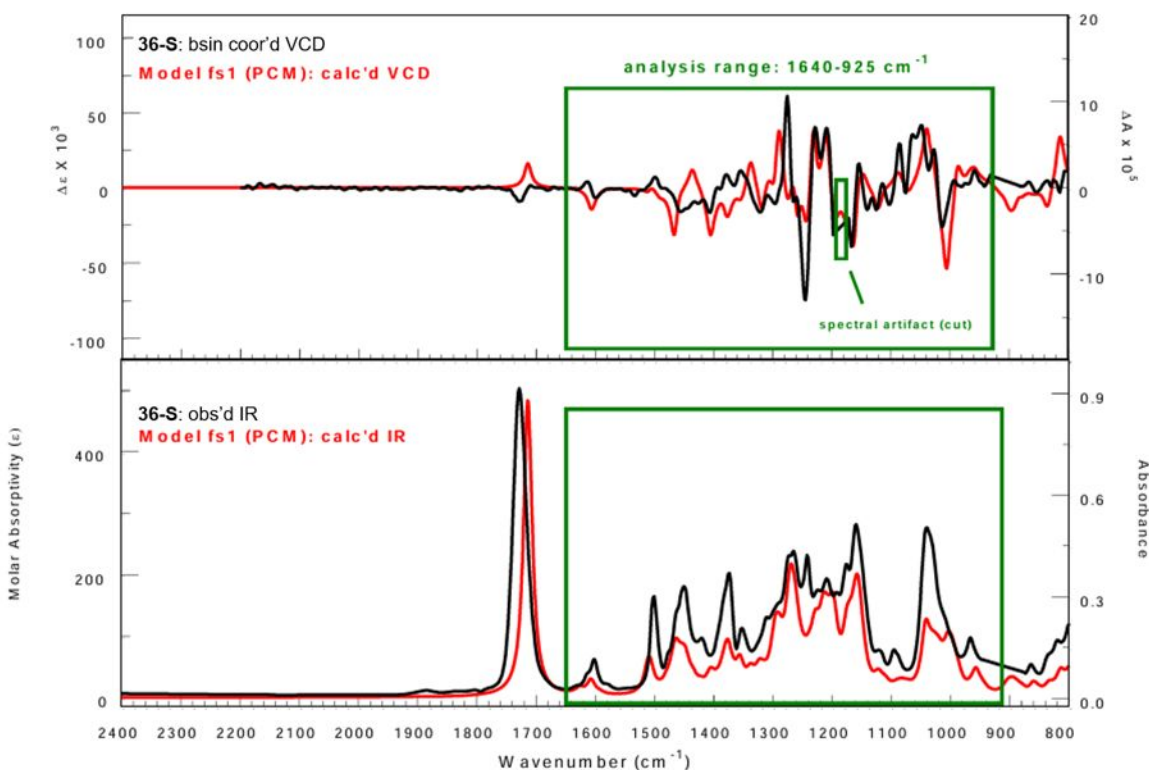


Figure S14: VCD and IR spectra of **36-S** are compared with corresponding spectra calculated for Model fs1.

VCD Analysis of **27**

A sample of **27** was obtained as described in the experimental. Although the synthesis utilized starting material **26** with known absolute stereochemistry, a VCD analysis was run to provide additional VCD data on the series of compounds.

Theoretical Analysis:

- Conformational Search: MOE stochastic csearch using MMFF94x force field
- Model Chemistry: # opt freq=(noraman,vcd) b3lyp/dgdzvp2 scrf=(solvent=ethanol)
- Conformational Analysis: Fractional populations estimated using Boltzmann statistics
- Lorentzian band width: 8 cm⁻¹
- Frequency scale factor: 0.9825
- Estimation of Confidence Limit: CompareVOA (BioTools, Inc.) analysis.

Experimental:

- Spectrometer: BioTools ChiralIR-2X FT-VCD spectrometer operated at 4 cm⁻¹
- VCD Frequency Range: 2400-800 cm⁻¹
- PEM Calibration: PEM calibrated at 1400 cm⁻¹
- PEM Retardation Settings: PEM1 = 0.250* λ ; PEM2 = 0.245* λ
- ChiralIR-2X Scan Method(s): **27**: four 4 h block scan (49920 total scans)
- Solvent: CDCl₃
- Concentration: **27** ~ 8 mg/130 μ L
- Baseline Correction (modified half-difference using solvent background VCD spectrum)
- VCD_{27 11-B1} (corr'd) = VCD_{27 11-B1} minus VCD solvent
- Additional Processing: Savitsky-Golay 9-point smooth.

Results:

Analysis of Experimental and Calculated Data:

The VCD and IR spectra observed for **27** are compared with calculated data in Figure S15. The green box in each panel highlights the spectral range over which VCD data were used to assign stereochemistry and estimate the confidence limit for this assignment. Inspection of VCD data in the analysis range indicates that the VCD spectrum of **27** is largely coincident with the model spectrum, consistent with this molecule having the same R absolute configuration as the model. The confidence limit in this assignment was estimated to be > 95%, consistent with a moderately reliable assignment. This level is good considering the greater uncertainty in observed VCD intensities for single enantiomer analyses and the poor prediction of SO₂ frequencies by the computational method. As shown in the bottom panel of Figure S15, the IR spectrum calculated for Model fs1 is in reasonably good agreement with experimental, with the exceptions of the SO₂ bands, whose stretching frequencies are overestimated due to limitations in the computational method which does not include f-atomic orbitals, leading to poorly constructed molecular orbitals and in turn, mis-predicted frequencies for these modes. Otherwise, good agreement between these IR data demonstrates accurate coverage of the solution phase conformational space occupied by **27**, as required for a reliable VCD assignment. Data reported in Figure S15 confirms the R absolute configuration expected for **27**.

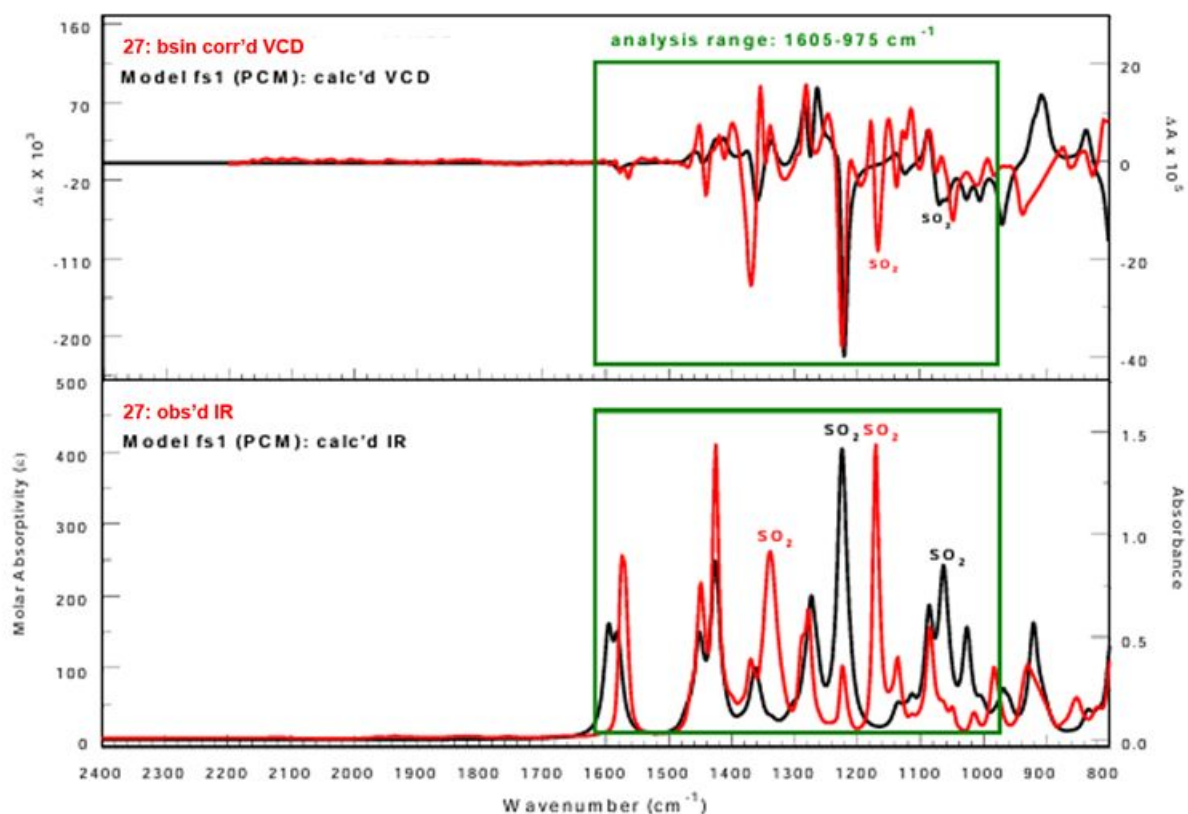


Figure S15: VCD and IR spectra of **27** are compared with corresponding spectra calculated for Model fs1.

VCD Analysis of **5**

Experimental:

- Spectrometer: BioTools ChiralIR-2X FT-VCD spectrometer operated at 4 cm-1
- VCD Frequency Range: 2400-800 cm-1
- PEM Calibration: PEM calibrated at 1400 cm-1
- PEM Retardation Settings: PEM1 = 0.250* λ ; PEM2 = 0.245* λ
- ChiralIR-2X Scan Method(s): **5-lot 1**: seven 4 h block scan (87360 total scans); **5-lot 2**: three 4 h block scan (37440 total scans)
- Solvent: d₆ DMSO
- Concentration(s): **5-lot 1** ~ 12 mg/100 μ L **5-lot 2** ~ 10 mg/100 μ L
- Baseline Correction (Modified Half-Difference): Not required

- Additional Processing: spectral subtraction (IR only).

Results:

The absolute stereochemistry of the propanoic acid center could not be reliably assigned using calculated VCD data, due in large part to the availability of only one sample, **5-lot 1**, at the time this analysis was first performed. As a work around, a re-synthesis was carried out using intermediates whose configurations were reliably assigned by ab initio VCD analysis (**27** and **36-S**). The product from this reaction, **5-lot 2** was assigned with (S,R) absolute configuration. VCD and IR spectra were then acquired for **5-lot 2** and compared with corresponding spectra for **5-lot 1**. The VCD and IR spectra for these compounds were found to be identical, as shown in Figure S16, clearly indicating that both stereo centers in **5-lot 1** have the same absolute configuration as **5-lot 2**. Had the propanoic acid center in **5-lot 1** been inverted relative to **5-lot 2**, the comparison of observed VCD spectra would have exhibited a mixture of bands, with some coincident (due to the shared configuration of the oxathiazepine center) and others inverted (due to the different configurations of the propanoic acid center). Based on these data, **5** was reliably assigned with (S, R) configuration.

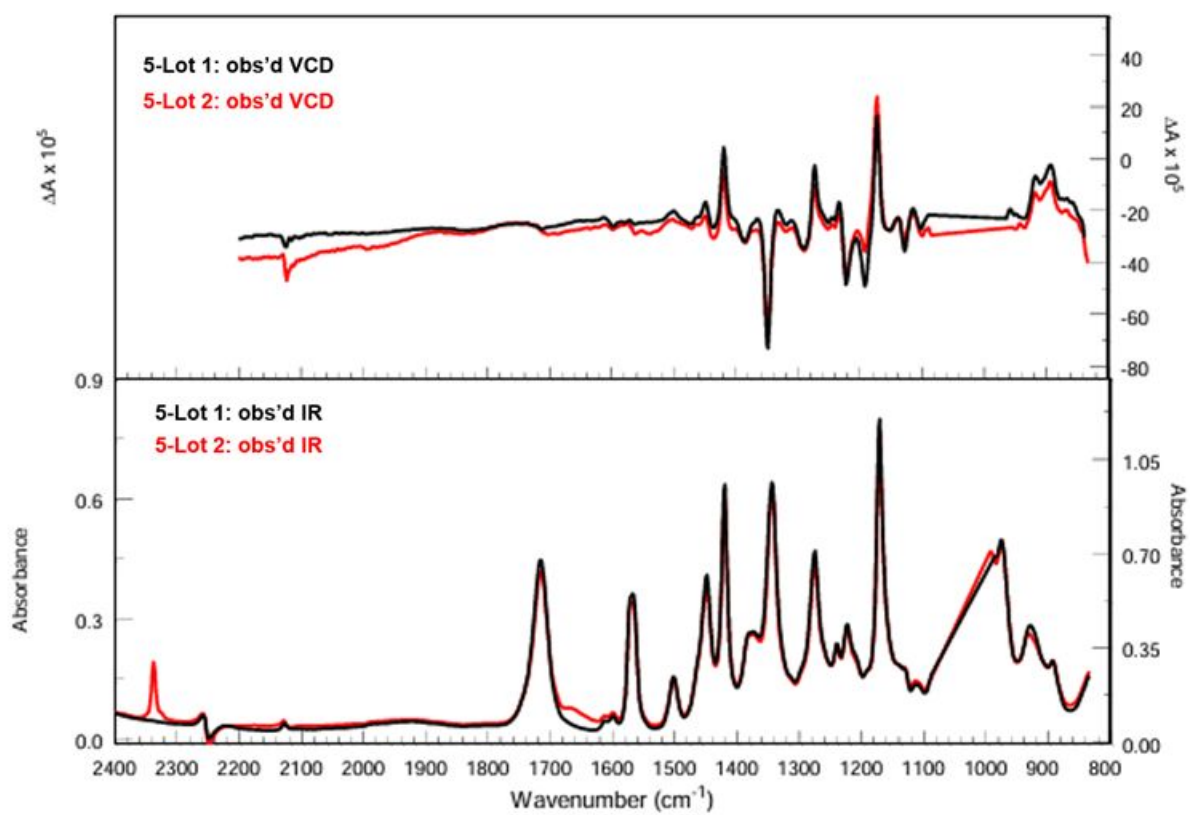


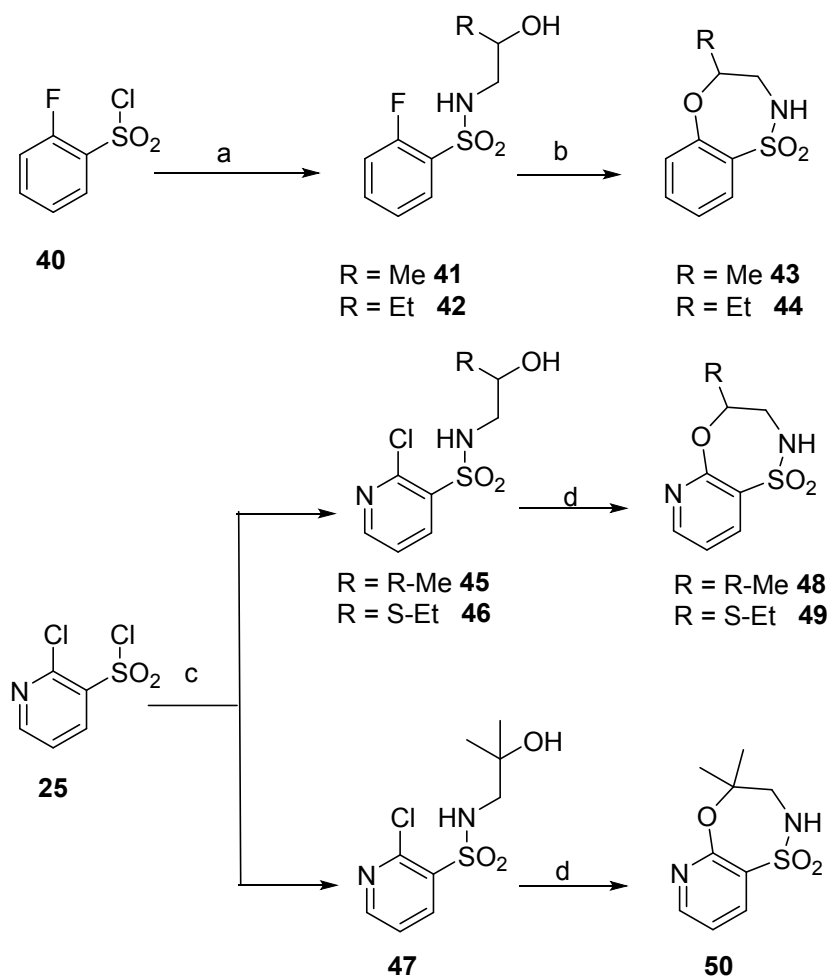
Figure S16: VCD spectra of **5-lot 1** and **5-lot 2** (top); IR spectra of **5-lot 1** and **5-lot 2** (bottom).

Preparation of compounds 7-21, 23-24

All solvents and commercially available reagents were used as received. All reactions were followed by thin-layer chromatography (TLC) analysis (TLC plates GF254, Merck) or LC-MS (liquid chromatography mass spectrometry). Intermediates and final products were purified by either flash column chromatography or semi-prep reverse-phase HPLC. Flash chromatography was performed on columns pre-packed with silica gel from Teledyne ISCO, unless otherwise noted. LC-MS analysis was completed using either a Sciex 7 or Waters 9 LC-MS system. Preparative reverse-phase HPLC was performed using a Gilson Preparative System with variable wavelength UV detection or an Agilent Mass Directed AutoPrep (MDAP) system with both mass and variable wavelength UV detection or Waters Preparative System with UV / PDA detection or a Shimadzu PREP LC 20AP. A variety of reverse phase columns, e.g., Luna 5 μ C18(2) 100A, SunFire C18, XBridge C18, Atlantics T3 were used in the purification with the choice of column support dependent upon the conditions used in the purification. The compounds are eluted using a gradient of CH₃CN and water. Neutral conditions used an CH₃CN and water gradient with no additional modifier, acidic conditions used an acid modifier – 0.1% TFA (added to both the CH₃CN and water) or 0.1 % formic acid – and basic conditions used a basic modifier, 0.1% NH₄OH (added to the water) or 10 mM ammonium bicarbonate. Preparative Chiral SFC was performed using a Thar/Waters Preparative SFC System with single wavelength UV detection system or PDA detector. A variety of chiral SFC columns (e.g., ChiralPak IA, IC, AY, AD, OD, OJ, C2) were used in the purification. The compounds are eluted using supercritical fluid CO₂ and co-solvents such as MeOH, EtOH, IPA or a combination of these solvents in different ratios. Modifiers (0.1% of TFA, NH₄OH, DEA) were used as needed. Routine proton nuclear magnetic resonance (¹H NMR) spectra were recorded with a Bruker NMR spectrometer unless otherwise noted in the indicated solvent. NMR spectra are referenced as follows: ¹H (400 or 500 MHz), internal standard TMS at δ = 0.00. Abbreviations for multiplicities observed in NMR spectra: s; singlet; br s, broad singlet; d, doublet; t, triplet; q, quadruplet; p, pentuplet; spt, septuplet; m, multiplet. VCD analyses were carried out following the guideline from USP Chapters 782 and 1782. Each sample was dissolved in a suitable deuterated solvent (usually CDCl₃, DMSO-d₆ or CD₃CN) at a suitable concentration and placed in a 100 μ m pathlength cell with BaF₂ windows. IR and VCD spectra were recorded on a ChiralIR2XTM VCD spectrometer (BioTools, Inc.)

equipped with dual PEM accessory, with 4 cm⁻¹ resolution, 6-16 hour collection for both sample and solvent, and instrument optimized at 1400 cm⁻¹. Theoretical calculations: a conformational search on each of the modeled structure was carried out using MOE at LowMode using MMFF94x or Amber 12:EHT forcefield with Born solvation, dielectric constant set at 10 or 20 depending on polarity of the structure, and exterior dielectric constant set at 5 for samples measured in CDCl₃, 47 for samples measured in DMSO-d₆ and 36 for samples measured in CD₃CN. The geometry optimization, frequency, and IR and VCD intensity calculations of the 30 lowest-energy conformers resulted from each conformational search were carried out at the DFT level (b3lyp/dgdzvp2 scrf=(solvent=solvent used for VCD measurement)) with Gaussian09 or Gaussian16 (Gaussian Inc., Wallingford, CT). The Gaussian output files were converted to VCD and IR spectra using BLAIR. The calculated frequencies were scaled by 0.98 and the IR and VCD intensities were converted to Lorentzian bands with 6-cm⁻¹ half-width for comparison to experimental spectra. The assignment was evaluated by CompareVOA program (BioTools. Inc., Jupiter, FL) to generate a confidence level based on current database that includes 88 previous correct assignments for different chiral structures. In addition, theoretical calculations were supplemented with VCD/IR results from previous experimental determinations. All final compounds reported are of at least 95% purity according to LC–MS unless stated otherwise.

Scheme S1. Preparation of the intermediate cyclic sulfonamides 43-44, 48-49, 50.^a



^a(a) (R)-2-amino-1-propanol or (R)-1-aminobutan-2-ol, K₂CO₃, THF/H₂O, room temperature, 20 to 66 h, **41**: quant., **42**: 79%; (b) KOtBu, DMSO, heat, 80 °C, 20 h or 100 °C, 20 min., **43**: 58.8, **44**: 74.8%; (c) (R)-2-amino-1-propanol, (R)-1-aminobutan-2-ol, (S)-1-aminobutan-2-ol or 1-amino-2-methylpropan-2-ol, K₂CO₃, THF/H₂O, room temperature, 2 h to 16 h, **45**: 72.7%, **46**: 41.1%, **47**: 78%; (d) KOtBu, DMSO, 75 - 80 °C, 1 h. to 15 h., **48**: 17.9%, **49**: 77%, **50**: 70.1%.

(R)-2-Fluoro-N-(2-hydroxypropyl)benzenesulfonamide (41). (R)-1-Amino-2-propanol (0.386 mL, 5.00 mmol) in tetrahydrofuran (10 mL) and H₂O (2.5 mL) was added K₂CO₃ (0.691 g, 5.00 mmol) and then 2-fluorobenzene-1-sulfonyl chloride **40** (0.662 mL, 5 mmol) slowly. The resulting reaction mixture was stirred at room temperature for 66 h. The reaction mixture was diluted with H₂O (10 mL), extracted with EtOAc (20 + 2 x 10 mL). The combined organic layer was washed with brine (15 mL), dried over MgSO₄, filtered, evaporated down with a rotavap evaporator under vacuum, dried under high vacuum with oil pump to afford desired product (R)-2-fluoro-N-(2-

hydroxypropyl)benzenesulfonamide **41** (1.2159 g, 5.21 mmol, 104 % yield). LC-MS m/z: 234.0 [M+H]⁺, 0.65 min. (ret. time).

(R)-4-Methyl-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepine 1,1-dioxide (43). (R)-2-Fluoro-N-(2-hydroxypropyl)benzenesulfonamide **41** (1,166 mg, 5 mmol) in dimethyl sulfoxide (20 mL) was added potassium tert-butoxide (1,683 mg, 15.00 mmol). The resulting reaction mixture was heated with at 100 °C for 20 min. The reaction mixture was diluted with H₂O (30 mL), acidified with HCl (1 N) to pH ~6, extracted with EtOAc (3 x 50 mL). The combined organic layer was washed with brine (50 mL), dried over MgSO₄, filtered, evaporated with a rotavap evaporator under vacuum, purified with Combiflash chromatograph over silica gel column to afford desired product, (R)-4-methyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide **43** (841.6 mg, 3.95 mmol, 79 % yield). LC-MS m/z: 214.0 [M+H]⁺, 0.73 min. (ret. time).

(R)-2-Fluoro-N-(2-hydroxybutyl)benzenesulfonamide (42). To a solution of (R)-1-aminobutan-2-ol (11.91 g, 134 mmol) in tetrahydrofuran (THF) (50 ml) and water (20 ml) was added K₂CO₃ (17.04 g, 123 mmol) and then 2-fluorobenzene-1-sulfonyl chloride **40** (20 g, 103 mmol) at room temperature and stirred for 20 h. The reaction was quenched with water (100 mL) and extracted with EtOAc (2 x 100 mL). The combined organic layer was washed with brine solution (50 mL) and dried over anhydrous Na₂SO₄, filtered and evaporated the solvent under vacuum to give (R)-2-fluoro-N-(2-hydroxybutyl)benzenesulfonamide **42** (16 g, 60.4 mmol, 58.8 % yield) as a gummy brown liquid. LC-MS m/z: 248.1 [M+H]⁺, 1.67 min. (ret. time); ¹H NMR (400 MHz, chloroform-d) δ: 7.95 – 7.8 (m, 1H), 7.65 – 7.55 (m, 1H), 7.35 – 7.15 (m, 2H), 5.2 (br s, 1 H), 3.7 – 3.6 (m, 1H), 3.2 – 3.1 (m, 1H), 2.9 – 2.8 (m, 1H), 1.9 – 1.8 (m, 1H), 2.55 – 2.4 (m, 2H), 0.95 (t, 3H).

(R)-4-ethyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide (44). To a solution of (R)-2-fluoro-N-(2-hydroxybutyl)benzenesulfonamide **42** (16 g, 64.7 mmol) in dimethyl sulfoxide (DMSO) (150 mL) was added potassium tert-butoxide (21.78 g, 194 mmol) at 10 °C and stirred at 80 °C for 20 h. Reaction mass cooled to room temperature, added cold water (100 mL), diluted with EtOAc (3x100 mL) and washed with brine solution (80 mL) and dried over anhydrous Na₂SO₄, filtered and evaporated under vacuum to obtain a crude product (R)-4-ethyl-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepine 1,1-dioxide **44** (11 g) as a yellow solid. This compound was

blended with product from another reaction (2.4 g) and washed with n-pentane (60 mL), diethyl ether (40 mL) and added diethyl ether (40 mL) stirred for 20 min and then filtered. The solid was dried under vacuum to afford (R)-4-ethyl-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepine 1,1-dioxide **44** (11 g, 48.4 mmol, 74.8 %) as a pale-yellow solid. LC-MS m/z: 228.01 [M+H]⁺, 1.81 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ: 7.8 - 7.7 (m, 2H), 7.6 - 7.5 (m, 1H), 7.3 - 7.2 (m, 2H), 3.75 - 3.6 (m, 1H), 3.45 - 3.2 (m, 2H), 1.65 - 1.55 (m, 2H), 1.05 (t, 3H).

(R)-2-Chloro-N-(2-hydroxypropyl)pyridine-3-sulfonamide (45). To a solution of (R)-1-aminopropan-2-ol (3.72 g, 49.5 mmol) in tetrahydrofuran (30 mL) and water (6.00 mL) was added K₂CO₃ (4.56 g, 33.0 mmol) and then added 2-chloropyridine-3-sulfonyl chloride **25** (7 g, 33.0 mmol) at 20 °C and stirred for 15 h. The reaction mixture was quenched with water (10 mL) and extracted with ethyl acetate (2 x 30 mL). The organic layer was combined and washed with brine (30 mL), dried over Na₂SO₄ and concentrated to give (R)-2-chloro-N-(2-hydroxypropyl)pyridine-3-sulfonamide **45** (6.2 g, 23.99 mmol, 72.7 % yield) as a yellow oil. LC-MS m/z: 251.1 [M+H]⁺, 0.76 min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 8.58 (dd, J = 4.8, 1.9 Hz, 1H), 8.41 (dd, J = 7.8, 1.9 Hz, 1H), 7.46 (dd, J = 7.8, 4.8 Hz, 1H), 3.93 (dq, J = 12.6, 6.3, 3.3 Hz, 1H), 3.10 (dd, J = 13.1, 3.2 Hz, 1H), 2.83 (dd, J = 13.1, 8.0 Hz, 1H), 1.17 (d, J = 6.3 Hz, 3H).

(R)-4-Methyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide (48). To a solution of (R)-2-chloro-N-(2-hydroxypropyl)pyridine-3-sulfonamide **45** (6.2 g, 24.73 mmol) in dimethyl sulfoxide (10 mL). Then potassium tert-butoxide (5.55 g, 49.5 mmol) was added. The reaction was heated at 80 °C and stirred for 15 h. The reaction was purified by Prep-HPLC (0.05% NH₄HCO₃/H₂O:MeOH = 5%~95%) to give (R)-4-methyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **48** as a white solid (950 mg, 4.43 mmol, 17.93 % yield). LC-MS m/z: 215.1 [M+H]⁺, 1.27 min. (ret. time); ¹H NMR (500 MHz, DMSO) δ 8.37 (dd, J = 4.8, 1.9 Hz, 1H), 8.09 (dd, J = 7.6, 1.9 Hz, 1H), 7.31 (dd, J = 7.6, 4.9 Hz, 1H), 4.25 - 4.16 (m, 1H), 3.35 (qd, J = 14.7, 5.4 Hz, 2H), 1.28 (d, J = 6.4 Hz, 3H).

(S)-2-Chloro-N-(2-hydroxybutyl)pyridine-3-sulfonamide (46). To a solution of 2-chloropyridine-3-sulfonyl chloride **25** (16 g, 75 mmol) in a mixture of tetrahydrofuran (30 mL) and water (6.0 mL), (S)-1-aminobutan-2-ol (10.09 g, 113 mmol) and K₂CO₃ (10.43 g, 75 mmol) were added. The

reaction mixture was stirred at room temperature for 16 h. The reaction mixture was partitioned between ethyl acetate and water. The water layer was extracted with ethyl acetate (x2). The combined organic phase was dried over MgSO_4 and concentrated. The crude product was purified by Gilson HPLC under TFA acidic condition to obtain (S)-2-chloro-N-(2-hydroxybutyl)pyridine-3-sulfonamide **46** (8.2 g, 31.0 mmol, 41.1 % yield). LC-MS m/z : 265.0 $[\text{M}+\text{H}]^+$, 0.52 min. (ret. time); ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ ppm 8.62 (dd, $J=4.77$, 2.01 Hz, 1 H), 8.36 (dd, $J=7.91$, 1.88 Hz, 1 H), 8.01 (t, $J=5.90$ Hz, 1 H), 7.64 (dd, $J=7.78$, 4.77 Hz, 1 H), 3.45 - 3.29 (m, 1 H), 2.96 - 2.75 (m, 2 H), 1.52 - 1.34 (m, 1 H), 1.32 - 1.12 (m, 1 H), 0.88 - 0.73 (m, 3 H).

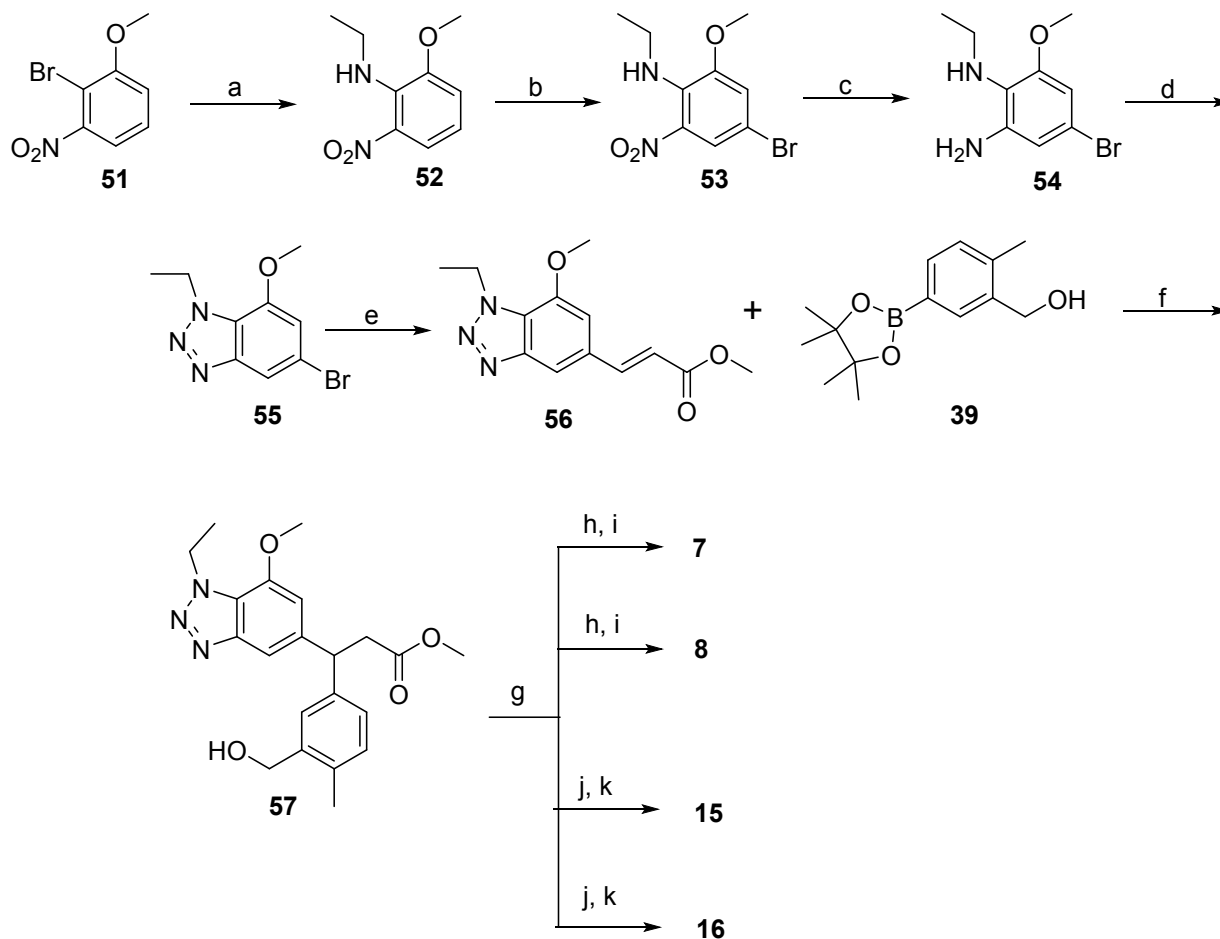
(S)-4-Ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine-1,1-dioxide (49). To a solution of (S)-2-chloro-N-(2-hydroxybutyl)pyridine-3-sulfonamide **46** (6.19 g, 23.38 mmol) in dimethyl sulfoxide (20 mL), was added potassium tert-butoxide (6.56 g, 58.5 mmol). The reaction mixture was stirred at 80 °C for 1 h. The solid was filtered and the mother liquid was purified by Gilson HPLC under acid (TFA) condition to obtain (S)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **49** (4,100 mg, 17.96 mmol, 77 % yield). Analytical chiral SFC confirmed a single enantiomer was present. LC-MS m/z : 229.1 $[\text{M}+\text{H}]^+$, 0.52 min. (ret. time).

2-Chloro-N-(2-hydroxy-2-methylpropyl)pyridine-3-sulfonamide (47). To a solution of 2-chloropyridine-3-sulfonyl chloride **25** (9 g, 42.4 mmol) in tetrahydrofuran (150 mL) was added 1-amino-2-methylpropan-2-ol (3.78 g, 42.4 mmol), potassium carbonate (5.87 g, 42.4 mmol) and water (45 mL). The reaction mixture was stirred for 2 h at 25 °C. The reaction mixture was quenched with water (20 mL) and extracted with EtOAc (3 x 20mL) and combined extracts were dried over anhydrous Na_2SO_4 and concentrated to give 2-chloro-N-(2-hydroxy-2-methylpropyl)pyridine-3-sulfonamide **47** (9 g, 33.1 mmol, 78 % yield) as off-white solid. LC-MS: 97 m/z : 265.0 $[\text{M}+\text{H}]^+$, 1.37 min. (ret. time); ^1H NMR (400 MHz, chloroform- d) δ = 8.57 (dd, J = 1.9, 4.7 Hz, 1H), 8.40 (dd, J = 1.9, 7.8 Hz, 1H), 7.43 (dd, J = 4.8, 7.9 Hz, 1H), 5.54 (br s, 1H), 2.90 (d, J = 6.4 Hz, 2H), 1.26 (s, 6H).

4,4-Dimethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide (50). To a solution of 2-chloro-N-(2-hydroxy-2-methylpropyl)pyridine-3-sulfonamide **47** (9 g, 34.0 mmol) in

dimethyl sulfoxide (90 mL) was added potassium tert-butoxide (11.44 g, 102 mmol) at 10 °C then heated to 75 °C for 1h. The reaction mixture was cooled to room temperature and poured into water (400 mL) and neutralized with 1N HCl and the obtained solid was filtered, washed with water and dried well to give 4,4-dimethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **50** (4.96 g, 21.70 mmol, 63.8 % yield) as light yellow color solid. This material was mixed with 520 mg from another reaction in hexane (50 mL), stirred well and filtered, washed with hexane (10 mL) and dried well to offered 4,4- dimethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **50** (5.45 g, 23.82 mmol, 70.1 % yield) as pale yellow color solid. LC-MS m/z: 229.09 [M+H]⁺, 1.25 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ = 8.44 (dd, *J* = 2.0, 4.8 Hz, 1H), 8.28 (br t, *J* = 6.6 Hz, 1H), 8.11 (dd, *J* = 2.0, 7.7 Hz, 1H), 7.36 (dd, *J* = 4.8, 7.7 Hz, 1H), 3.43 (br d, *J* = 5.7 Hz, 2H), 1.21 (br s, 6H).

Scheme S2. Preparation of compounds 7-8 and 15-16.^a



^a(a) Cu, ethanamine, 1,4-dioxane, 100 °C, 4 h; 64.6 %; (b) NBS, DMF, 0 °C, 5 min, 64.8 %; (c) tin chloride-dihydrate, EtOH, 80 °C, 2 h., 82 %; (d) H₂SO₄, H₂O, 0 °C, 30 min.; NaNO₂, H₂O, 0 to 20 °C, 2 h, 95 %; (e) tri-*o*-tolylphosphine, methyl acrylate, Pd(OAc)₂, K₂CO₃, DMF, 120 °C, 12 h., 83%; (f) Et₃N, chloro (1,5-cyclooctadiene) rhodium(I) dimer, 1, 4-dioxane/H₂O, 90 °C, 16 h, 64.8%; (g) chiral SFC chromatography, Chiralpak IG column; (h) **44**, nBu₃P, ADDP, THF, RT, 24 h; (i) LiOH (aqueous), THF/MeOH, 50 °C, 3 h, **7**: 83%; **8**: 81%; (j) **27**, Bu₃P, ADDP, THF, RT, 24 h., quantitative; (k) LiOH (aqueous), THF, MeOH, 50 °C, 3 h., **15**: 91 %, **16**: 92 %.

***N*-Ethyl-2-methoxy-6-nitroaniline (52).** A solution of 2-bromo-1-methoxy-3-nitrobenzene **51** (14 g, 60.3 mmol) in 1,4-dioxane (30 mL) was added copper (0.192 g, 3.02 mmol) and ethanamine (45.3 mL, 603 mmol) slowly under nitrogen at 20 °C. The mixture reaction was stirred at 100 °C for 4 h. The reaction mixture was added to water (50 mL) and extracted with ethyl acetate (50 mL x 3), concentrated and used in the subsequent reaction without further purification. The desired product, *N*-ethyl-2-methoxy-6-nitroaniline **52** (10 g, 39.0 mmol, 64.6 % yield) was obtained as a

red oil. LC-MS m/z: 197.3 [M+H]⁺, 1.31 min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 7.70 (dd, J = 8.7, 1.3 Hz, 1H), 6.91 (dd, J = 7.8, 0.9 Hz, 1H), 6.67 – 6.51 (m, 1H), 3.85 (s, 3H), 3.62 – 3.50 (m, 2H), 1.24 (t, J = 7.2 Hz, 3H).

4-Bromo-N-ethyl-2-methoxy-6-nitroaniline (53). A solution of N-ethyl-2-methoxy-6-nitroaniline **52** (20 g, 102 mmol) in N,N-dimethylformamide (200 mL) was added NBS (18.14 g, 102 mmol) in N,N-dimethylformamide (20 mL) dropwise at 0 °C. It was stirred at 0 °C for 5 min. It was added 50 ml of water and stirred for 1 h. It was filtered to obtain the desired product, 4-bromo-N-ethyl-2-methoxy-6-nitroaniline **53** (20g, 66.0 mmol, 64.8 % yield) as red solid. LC-MS m/z: 275.2/277.2 [M+H]⁺, 1.47min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 7.87 (d, J = 2.2 Hz, 1H), 6.93 (d, J = 2.2 Hz, 1H), 3.85 (s, 3H), 3.59 (q, J = 7.2 Hz, 2H), 1.24 (t, J = 7.2 Hz, 3H).

4-Bromo-N1-ethyl-6-methoxybenzene-1,2-diamine (54). A solution of 4-bromo-N-ethyl-2-methoxy-6-nitroaniline **53** (34 g, 124 mmol) in ethanol (200 mL) was added tin (II) chloride dihydrate (69.7 g, 309 mmol) slowly under nitrogen at 80 °C. The reaction mixture was stirred at 80 °C for 2 h. The reaction mixture was concentrated, water (500 mL) added and adjusted pH to 14 with 6 N NaOH. It was extracted with ethyl acetate (500 ml x 3), Isolute was added to the combined organic layers, and the mixture was concentrated under a stream of nitrogen at 50 °C and used in subsequent reaction without further purification. The desired product, 4-bromo-N1-ethyl-6-methoxybenzene-1,2-diamine **54** (27g, 101 mmol, 82 % yield) was obtained as black oil. LC-MS m/z: 245.1/247.1 [M+H]⁺, 1.91 min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 6.52 (d, J = 2.0 Hz, 1H), 6.44 (d, J = 2.0 Hz, 1H), 3.78 (s, 3H), 2.88 (q, J = 7.2 Hz, 2H), 1.13 (t, J = 7.1 Hz, 3H).

5-Bromo-1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazole (55). To a solution of 4-bromo-N1-ethyl-6-methoxybenzene-1,2-diamine **54** (27 g, 110 mmol) in water (300 mL), sulfuric acid (43.2 g, 441 mmol) was added and stirred for 30 min. then sodium nitrite (11.40 g, 165 mmol) dissolved in water (30 mL) was added into the reaction mixture at 0 °C, warmed to 20 °C and stirred for 2 h. The reaction mixture was extracted by EtOAc (200 mL) three times, the organic layers were combined and washed by brine (300 mL) and dried with Na₂SO₄ and concentrated in vacuo. The crude product was then purified on a silica cartridge (120 g) with a Combiflash Companion, eluting

at 60 mL/min with a gradient running from 100% hexanes to 20% EtOAc/hexanes over 80 min). The desired product 5-bromo-1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazole **55** (27 g, 105 mmol, 95 % yield) was obtained as yellow solid. LC-MS m/z: 256.0/258.1 [M+H]⁺, 1.92 min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 7.77 (d, J = 1.3 Hz, 1H), 6.87 (d, J = 1.1 Hz, 1H), 4.83 (d, J = 7.2 Hz, 2H), 4.01 (s, 3H), 1.57 (t, J = 7.2 Hz, 3H).

(E)-Methyl 3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)acrylate (56). A mixture of tri-*o*-tolylphosphine (1.902 g, 6.25 mmol), methyl acrylate (10.76 g, 125 mmol), 5-bromo-1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazole **55** (16 g, 62.5 mmol), palladium(II) acetate (0.701 g, 3.12 mmol), potassium carbonate (17.27 g, 125 mmol) and N,N-dimethylformamide (90 mL) was stirred at 120 °C under the N₂ for 12 h. Water (90 mL) was added and the mixture was extracted by ethyl acetate (3 x 1, 50mL). The organic layers were washed with brine (150 mL), dried with Na₂SO₄ and concentrated. The crude product was then purified on a silica cartridge (80 g) with a Combiflash Companion, eluting at 70 mL/min with a gradient running from 100% hexanes to 40% EtOAc/hexanes over 50 min. The desired product (E)-methyl 3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **56** (15 g, 51.7 mmol, 83 % yield) was obtained as yellow solid. LC-MS m/z: 262.1 [M+H]⁺, 1.86 min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 7.79 (d, J = 15.9 Hz, 1H), 7.73 (s, 1H), 6.93 (s, 1H), 6.46 (d, J = 15.9 Hz, 1H), 4.85 (q, J = 7.2 Hz, 2H), 4.05 (s, 3H), 3.83 (s, 3H), 1.59 (t, J = 7.2 Hz, 3H).

Methyl (R)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate 46-R and methyl (S)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (57-S). A solution of (E)-methyl 3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **56** (1,000 mg, 3.83 mmol) in 1,4-dioxane (10 mL) and water (5.00 mL) was added (2-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol **39** (2,849 mg, 11.48 mmol), triethylamine (1.067 mL, 7.65 mmol). It was stirred for 5 min and then added chloro (1,5- cyclooctadiene) rhodium(I) dimer (94 mg, 0.191 mmol) under the protection of nitrogen. The reaction mixture was stirred at 90 °C for 16 h. after the reaction was over. The mixture was extracted by ethyl acetate (3x20 mL), combined the organic layers and washed with brine (20 mL), dried by Na₂SO₄, the solvent was removed *in vacuo*. The crude product was then purified on a silica cartridge (12 g) with a Combiflash

Companion, eluting at 20 mL/min with a gradient running from 100% hexanes to 60% EtOAc/hexanes over 35 min). The desired product, methyl 3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57** (1 g, 2.480 mmol, 64.8 % yield), was isolated as yellow oil. This product was combined with a product from a subsequent reaction (11 g, 28.7 mmol) and purified using chiral SFC with a Chiralpak IG column (20x250mm, 5 μ), 35% EtOH/CHCl₃ with a flowrate of 65 g/min to provide methyl (R)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57-R** 6.14 g), LC-MS m/z: 384 [M+H]⁺, 0.93 min. (ret. time); ¹H NMR (400 MHz, chloroform-d) δ 7.51 (s, 1H), 7.28 (s, 1H), 7.12 (d, J=1.25 Hz, 2H), 6.62 (d, J=0.75 Hz, 1H), 4.82 (q, J=7.28 Hz, 2H), 4.64-4.69 (m, 3H), 3.94 (s, 3H), 3.62 (s, 3H), 3.13 (dq, J=7.91, 15.94 Hz, 2H), 2.32 (s, 3H), 1.71 (br. s., 1H), 1.57 (t, J=7.28 Hz, 3H) and methyl (S)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57-S** (5.44 g), LC-MS m/z: 384 [M+H]⁺, 0.90 min. (ret. time), ¹H NMR (400 MHz, chloroform-d) δ 7.50 (s, 1H), 7.28 (s, 1H), 7.12 (d, J=1.00 Hz, 2H), 6.62 (d, J=0.75 Hz, 1H), 4.81 (q, J=7.19 Hz, 2H), 4.63-4.69 (m, 3H), 3.93 (s, 3H), 3.62 (s, 3H), 3.13 (dq, J=7.91, 15.85 Hz, 2H), 2.31 (s, 3H), 1.75 (br. s., 1H), 1.56 (t, J=7.28 Hz, 3H). The absolute stereochemistry of the enantiomers was confirmed by VCD analysis.

(S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (7). To a solution of methyl (S)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57-S** (150 mg, 0.391 mmol), (R)-4-ethyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide **44** (89 mg, 0.391 mmol), and 1,1'-(azodicarbonyl)dipiperidine (ADDP) (197 mg, 0.782 mmol) in tetrahydrofuran (2.5 mL) at room temperature was added tri-n-butylphosphine (0.193 mL, 0.782 mmol). The reaction mixture was stirring at ambient temperature for 24 h. The reaction was concentrated and purified using a Combiflash companion (12 g silica cartridge, 30 mL/min, 0-100% EtOAc/Hexanes). The desired fractions eluted at ~55% EtOAc/hexane and were concentrated to give the title compound methyl (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (246.1 mg, 0.415 mmol, 106 % yield) as a yellow oil and was used without further purification. LC-MS m/z:

593.5 [M+H]⁺, 1.296 min.(ret. time); ¹H NMR (chloroform-d) δ: 7.86 (dd, J=7.8, 1.8 Hz, 1H), 7.50 (td, J=7.8, 1.8 Hz, 1H), 7.44 (s, 1H), 7.20-7.26 (m, 1H + CDCl₃), 7.10-7.19 (m, 3H), 7.05 (d, J=1.3 Hz, 1H), 6.57 (d, J=0.8 Hz, 1H), 4.80 (q, J=7.1 Hz, 2H), 4.61 (t, J=8.0 Hz, 1H), 4.54 (d, J=13.8 Hz, 1H), 3.86-3.95 (m, 4H), 3.75 (d, J=14.1 Hz, 1H), 3.67 (dd, J=15.1, 10.5 Hz, 1H), 3.60 (s, 3H), 3.00-3.16 (m, 2H), 2.87 (dd, J=15.2, 1.4 Hz, 1H), 2.32 (s, 3H), 1.52-1.57 (m, 3H), 1.17-1.29 (m, 1H + EtOAc), 0.92 (t, J=7.3 Hz, 3H).

Methyl (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (238.7 mg, 0.403 mmol) was dissolved in tetrahydrofuran (3.00 mL) and methanol (3 mL) and LiOH (aqueous) (1 mL, 2.000 mmol) was added. The mixture was heated to 50 °C for 3 h, until the starting material was consumed. The reaction was acidified with HCl and diluted with EtOAc. The mixture was extracted with EtOAc three times, dried over magnesium sulfate, filtered, and concentrated. The crude residue was dissolved in DMSO for purification by Gilson HPLC (10-100% acetonitrile/water, 10 min. gradient time). The desired fractions were concentrated to give (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **7** (199.4 mg, 0.345 mmol, 86 % yield) as a white solid. LC-MS m/z: 579.5 [M+H]⁺, 1.177 min.(ret. time); ¹H NMR (methanol-d₄) δ: 7.81 (dd, J=7.9, 1.6 Hz, 1H), 7.59 (td, J=7.8, 1.5 Hz, 1H), 7.42 (s, 1H), 7.20-7.32 (m, 3H), 7.17 (d, J=7.8 Hz, 1H), 7.12 (d, J=1.5 Hz, 1H), 6.81 (s, 1H), 4.83 (q, J=7.2 Hz, 2H), 4.60 (t, J=8.0 Hz, 1H), 4.55 (d, J=13.8 Hz, 1H), 3.87-3.99 (m, 4H), 3.77 (d, J=14.1 Hz, 1H), 3.63 (dd, J=15.3, 10.3 Hz, 1H), 3.10-3.18 (m, 1H), 3.01-3.10 (m, 1H), 2.79 (dd, J=15.3, 1.0 Hz, 1H), 2.33 (s, 3H), 1.53 (t, J=7.2 Hz, 3H), 1.36-1.49 (m, 1H), 1.04-1.16 (m, 1H), 0.83-0.91 (m, 3H).

(R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (8). To a solution of methyl (R)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57-R** (150 mg, 0.391 mmol), (R)-4-ethyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide (89 mg, 0.391 mmol) **44**, and 1,1'-(azodicarbonyl)dipiperidine (ADDP) (197 mg, 0.782 mmol) in tetrahydrofuran (2.5 mL) at room temperature was added tri-n-butylphosphine (0.193 mL, 0.782 mmol). The reaction mixture was

stirring at ambient temperature for 24 h. The reaction was concentrated and purified using a Combiflash companion (12 g silica cartridge, 30 mL/min, 0-100% EtOAc/hexanes). The desired fractions eluted at ~55% EtOAc/hexane and were concentrated to give methyl (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (263.3 mg, 0.444 mmol, 114 % yield) as a yellow oil. LC-MS m/z: 593.4 [M+H]⁺, 1.296 min.(ret. time); ¹H NMR (chloroform-d) δ: 7.86 (dd, J=7.8, 1.5 Hz, 1H), 7.51 (td, J=7.8, 1.8 Hz, 1H), 7.43 (s, 1H), 7.21-7.25 (m, 1H), 7.19 (dd, J=8.3, 1.0 Hz, 1H), 7.10-7.16 (m, 2H), 7.06 (s, 1H), 6.56 (d, J=1.0 Hz, 1H), 4.80 (q, J=7.1 Hz, 2H), 4.61 (t, J=7.9 Hz, 1H), 4.53 (d, J=14.1 Hz, 1H), 3.89-3.98 (m, 4H), 3.64-3.79 (m, 3H), 3.60 (s, 3H), 3.44-3.48 (m, 1H), 3.00-3.15 (m, 2H), 2.90 (dd, J=15.2, 1.4 Hz, 1H), 2.32 (s, 3H), 1.58-1.65 (m, 1H + H₂O), 1.52-1.56 (m, 3H), 1.25-1.35 (m, 1H + EtOAc), 0.97 (t, J=7.3 Hz, 3H).

Methyl (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (252.5 mg, 0.426 mmol) was dissolved in tetrahydrofuran (3.00 mL) and methanol (3 mL) and LiOH (aq.) (1 mL, 2.000 mmol) was added. The mixture was heated to 50 °C for 3 h, until the starting material was consumed. The reaction was acidified with HCl and diluted with EtOAc. The mixture was extracted with EtOAc three times, dried over magnesium sulfate, filtered, and concentrated. The crude residue was dissolved in DMSO for purification by Gilson HPLC (10-100% acetonitrile/water, 10 min. gradient time). The desired fractions were concentrated to give (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **8** (198.6 mg, 0.343 mmol, 81 % yield) as a white solid. LC-MS m/z: 579.4 [M+H]⁺, 1.176 min. (ret. time); ¹H NMR (methanol-d₄) δ: 7.81 (dd, J=7.8, 1.8 Hz, 1H), 7.59 (td, J=7.8, 1.5 Hz, 1H), 7.37 (s, 1H), 7.25-7.33 (m, 2H), 7.23 (dd, J=8.0, 1.0 Hz, 1H), 7.19 (d, J=7.8 Hz, 1H), 7.15 (d, J=1.8 Hz, 1H), 6.80 (d, J=0.8 Hz, 1H), 4.83 (q, J=7.1 Hz, 2H), 4.61 (t, J=8.0 Hz, 1H), 4.57 (d, J=13.8 Hz, 1H), 3.90-4.00 (m, 4H), 3.78 (d, J=13.8 Hz, 1H), 3.62 (dd, J=15.2, 10.4 Hz, 1H), 3.04-3.15 (m, 2H), 2.81 (dd, J=15.2, 1.1 Hz, 1H), 2.34 (s, 3H), 1.53 (t, J=7.2 Hz, 3H), 1.37-1.50 (m, 1H), 1.14 (ddd, J=14.2, 7.3, 4.0 Hz, 1H), 0.84-0.90 (m, J=7.4, 7.4 Hz, 2H).

(S)-3-(3-(((R)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (15). To a solution of methyl (S)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57-S** (150 mg, 0.391 mmol), (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **27** (89 mg, 0.391 mmol), and 1,1'-(azodicarbonyl)dipiperidine (ADDP) (197 mg, 0.782 mmol) in tetrahydrofuran (2.5 mL) at room temperature was added tri-n-butylphosphine (0.193 mL, 0.782 mmol). The reaction mixture was stirred at ambient temperature for 24 h. The reaction was concentrated and purified using a Combiflash companion (12 g silica cartridge, 30 mL/min, 0-100% EtOAc/hexanes). The desired fractions eluted at ~70% EtOAc/hexane and were concentrated to give methyl (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (235.3 mg, 0.396 mmol, 101 % yield) as a yellow oil. LC-MS m/z: 594.4 [M + H]⁺, 1.167 min.(ret. time); ¹H NMR (chloroform-d) δ: 8.47 (dd, J=4.9, 1.9 Hz, 1H), 8.24 (dd, J=7.5, 2.0 Hz, 1H), 7.42 (s, 1H), 7.22-7.28 (m, 1H + CDCl₃), 7.11-7.18 (m, 2H), 7.10 (s, 1H), 6.59 (d, J=0.8 Hz, 1H), 4.80 (q, J=7.3 Hz, 2H), 4.62 (t, J=7.9 Hz, 1H), 4.45 (d, J=14.1 Hz, 1H), 4.30-4.39 (m, 1H), 4.01 (d, J=14.1 Hz, 1H), 3.93 (s, 3H), 3.53-3.62 (m, 4H), 3.01-3.17 (m, 3H), 2.31 (s, 3H), 1.60-1.71 (m, 1H), 1.52-1.57 (m, 3H + H₂O), 1.44 (ddd, J=14.3, 7.3, 4.5 Hz, 1H), 0.97 (t, J=7.4 Hz, 3H).

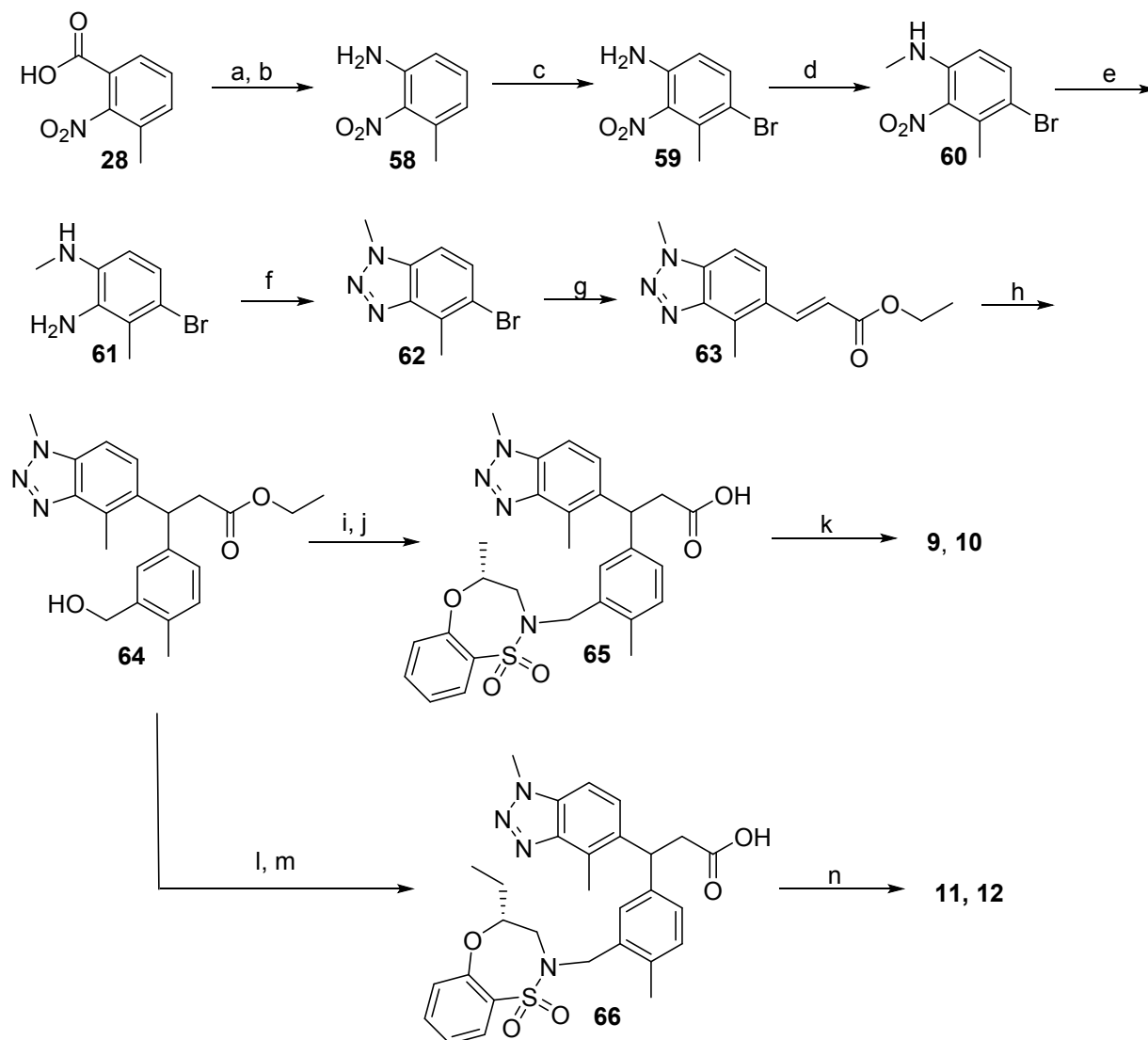
Methyl (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (230.7 mg, 0.389 mmol) was dissolved in tetrahydrofuran (3.00 mL) and methanol (3 mL) and LiOH (aqueous) (1 mL, 2.000 mmol) was added. The mixture was heated to 50 °C for 3 h. The reaction was acidified with HCl and diluted with EtOAc. The mixture was extracted with EtOAc three times, dried over magnesium sulfate, filtered, and concentrated. The crude residue was dissolved in DMSO for purification by Gilson HPLC (10-100% acetonitrile/water, 10 min. gradient time). The desired fractions were concentrated to give (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **15** (204.1 mg, 0.352 mmol, 91% yield) as a white solid. LC-MS m/z: 580.3 [M+H]⁺, 1.047 min. (ret. time); ¹H NMR (400 MHz, methanol-d₄) δ: 8.46 (dd, J=5.0, 1.8 Hz, 1H), 8.29 (dd, J=7.5, 2.0 Hz, 1H), 7.38-7.44 (m, 2H), 7.28 (dd, J=7.8, 1.8 Hz, 1H), 7.18 (d, J=7.8 Hz, 1H), 7.14 (d, J=1.5 Hz, 1H), 6.81 (s, 1H), 4.83 (q, J=7.3 Hz,

2H), 4.61 (t, J=8.0 Hz, 1H), 4.57 (d, J=14.1 Hz, 1H), 4.17-4.27 (m, 1H), 3.91-3.99 (m, 4H), 3.68 (dd, J=15.3, 10.5 Hz, 1H), 3.02-3.19 (m, 2H), 2.92 (dd, J=15.3, 1.3 Hz, 1H), 2.33 (s, 3H), 1.44-1.57 (m, 4H), 1.19-1.30 (m, 1H), 0.87 (t, J=7.4 Hz, 3H).

(R)-3-(3-(((R)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (16). To a solution of methyl (R)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **57-R** (150 mg, 0.391 mmol), (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **27** (89 mg, 0.391 mmol), and 1,1'-(azodicarbonyl)dipiperidine (ADDP) (197 mg, 0.782 mmol) in tetrahydrofuran (2.5 mL) at room temperature was added tri-n-butylphosphine (0.193 mL, 0.782 mmol). The reaction mixture was stirred at ambient temperature for 24 h. The mixture was concentrated and the resulting residue was purified using a Combiflash companion (12 g silica cartridge, 30 mL/min, 0-100% EtOAc/hexanes). The desired fractions eluted at ~55% EtOAc/hexanes and were concentrated to give methyl (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (232 mg, 0.391 mmol, 100 % yield) as a colorless oil. LCMS m/z: 594.4 [M+H]⁺, = 1.166 min. (ret. time); ¹H NMR (chloroform-d) δ: 8.48 (dd, J=4.8, 2.0 Hz, 1H), 8.24 (dd, J=7.5, 2.0 Hz, 1H), 7.42 (s, 1H), 7.23-7.28 (m, 1H + CDCl₃), 7.09-7.18 (m, 3H), 6.59 (d, J=0.8 Hz, 1H), 4.80 (q, J=7.3 Hz, 2H), 4.62 (t, J=7.9 Hz, 1H), 4.44 (d, J=14.1 Hz, 1H), 4.32-4.40 (m, 1H), 4.01 (d, J=14.1 Hz, 1H), 3.93 (s, 3H), 3.56-3.65 (m, 4H), 3.01-3.16 (m, 3H), 2.31 (s, 3H), 1.63-1.76 (m, 1H), 1.52-1.57 (m, 3H), 1.41-1.52 (m, 1H), 0.98 (t, J=7.4 Hz, 3H). Methyl (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (229.2 mg, 0.386 mmol) was dissolved in tetrahydrofuran (3.00 mL) and methanol (3 mL) and LiOH (aqueous) (1 mL, 2.000 mmol) were added. The reaction mixture was heated to 50 °C for 3 h. The reaction was acidified with HCl and diluted with EtOAc. The mixture was extracted with EtOAc three times, dried over magnesium sulfate, filtered, and concentrated. The crude residue was dissolved in DMSO for purification by GILSON HPLC (10-100% acetonitrile/water, 10 min. gradient time). The desired fractions were concentrated to give (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-

methoxy-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **16** (204.8 mg, 0.353 mmol, 92% yield) as a white solid that was immediately carried forward for crystallization using MeCN/water to give crystalline **16** (148.7 mg, 0.257 mmol, 66.4% yield). LC-MS m/z: 580.5 [M+H]⁺, 1.048 min (ret. time); ¹H NMR (400 MHz, methanol-d₄) δ: 8.46 (dd, J=4.9, 1.9 Hz, 1H), 8.29 (dd, J=7.8, 2.0 Hz, 1H), 7.36-7.42 (m, 2H), 7.26-7.31 (m, 1H), 7.15-7.22 (m, 2H), 6.80 (d, J=0.8 Hz, 1H), 4.83 (q, J=7.3 Hz, 2H + methanol-d₄), 4.54-4.66 (m, 2H), 4.20-4.30 (m, 1H), 3.92-3.99 (m, 4H), 3.67 (dd, J=15.2, 10.4 Hz, 1H), 3.11 (d, J=8.0 Hz, 2H), 2.95 (dd, J=15.3, 1.5 Hz, 1H), 2.33 (s, 3H), 1.46-1.59 (m, 4H), 1.30 (s, 1H), 0.88 (t, J=7.4 Hz, 3H); melting point: 158-162 °C.

Scheme S3. Preparation of compounds 9-12.^a



^a(a) (COCl)₂, CH₂Cl₂; NH₄OH, CH₂Cl₂, RT, 30 min., 60.6%; (b) NaOH (aq.), Br₂, 0 °C to 70 °C, 1 h.; NaOH (aq.), 80 °C, 1 h., 74.6%; (c) NBS, AcOH, 110 °C, 1 h., 78%; (d) NaH, DMF, 25 °C, 30 min, MeI, 12 h., 59.4%; (e) SnCl₂, EtOH, 75 °C, 2 h., 39.5 %; (f) 10% H₂SO₄, 0 °C, NaNO₂, 50 min., 21.57%; (g) tri-*o*-tolylphosphine, methyl acrylate, Pd(OAc)₂, DIPEA, DMF, 100 °C, 12 h., 76%; (h) **39**, chloro(1,5-cyclooctadiene)rhodium(I) dimer, triethylamine, 1,4-dioxane, H₂O, 23 °C, 3 h., 91%; (i) **43**, PS-PPh₃, DIAD, THF, RT, 40 min, 89%; (j) 3N NaOH, MeOH, 60 °C, 3 h., 85%; (k) chiral SFC chromatography, AD column, **9**: 29.1%, **10**: 30.1%; (l) **44**, Bu₃P, (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone), THF, 3 h, 98%; (m) LiOH, H₂O, MeOH, 2 h.; (n) combined lots of **66**, chiral SFC chromatography, Chiralpak AD.

3-Methyl-2-nitroaniline (58). To a solution of 3-methyl-2-nitrobenzoic acid **28** (100 g, 552 mmol) in dichloromethane (1000 mL) was added oxalyl chloride (72.5 mL, 828 mmol) at 25 °C. The solvent was removed under reduced pressure. The residue was dissolved in CH₂Cl₂ (100 mL). The

solvent was added to ammonium hydroxide (1,000 mL, 7704 mmol) at room temperature and stirred for 30 min. The water was extracted with ethyl acetate (3 x 500 ml). The combined organic layer was dried and concentrated to get the desired target, 3-methyl-2-nitrobenzamide (67 g, 335 mmol, 60.6%). LC-MS m/z: 181.1 [M+H]⁺, 1.67 min. (ret. time). To a mixture of NaOH (2.220 g, 55.5 mmol) in water (12 mL) was added Br₂ (0.322 mL, 6.26 mmol) at 0 °C. The 3-methyl-2-nitrobenzamide (1 g, 5.55 mmol) was added in one portion and the mixture is warmed slowly in a water bath, with mechanical stirring. The material soon darkens in color, and at 50-55 °C (internal temperature) oily droplets begin to separate. The temperature is raised gradually to 70 °C and maintained at this point for 1 h. A solution of 0.7 g. of sodium hydroxide in 4 mL of water is added slowly, and the temperature is increased to 80°C for an additional hour. The reaction was cooled to room temperature and extracted with ethyl acetate (3x50 mL). The combined organic layer was dried and concentrated to get desire target, 3-methyl-2-nitroaniline **58** (0.7 g, 4.14 mmol, 74.6%). LC-MS m/z 153.1 [M+H]⁺, 1.65 min. (ret. time).

4-Bromo-3-methyl-2-nitroaniline (59). A mixture of NBS (51.5 g, 289 mmol), 3-methyl-2-nitroaniline **58** (44 g, 289 mmol) and acetic acid (450 mL) was stirred at 110 °C for 1 h. The mixture was cooled to room temperature and poured into water (100 mL). The solid was collected to obtain the desired target as 4-bromo-3-methyl-2-nitroaniline **59** (55g, 226 mmol, 78 % yield). LC-MS m/z: 230.9, 232.9 [M+H]⁺, 1.78 min. (ret. time).

4-Bromo-N,3-dimethyl-2-nitroaniline (60). To a solution of 4-bromo-3-methyl-2-nitroaniline **59** (20 g, 87 mmol) in N,N-dimethylformamide (200 mL) was added NaH (3.81 g, 95 mmol) at 25 °C. The reaction was stirred at 25 °C for 30 min. Then to the mixture was added iodomethane (12.90 g, 91 mmol) and the reaction was stirred for 12 h. The reaction was poured into water and the solid was collected as 4-bromo-N,3-dimethyl-2-nitroaniline **60** (18 g, 51.4 mmol, 59.4 % yield). LC-MS m/z: 245.0, 247.0 [M+H]⁺, 1.90 min. (ret. time).

4-Bromo-N¹,3-dimethylbenzene-1,2-diamine (61). A mixture of tin (II) chloride (93 g, 490 mmol), 4-bromo-N,3-dimethyl-2-nitroaniline **60** (30 g, 122 mmol) and ethanol (600 mL) was stirred at 75 °C for 2 h. The solvent was adjusted to pH = 14 by using 40% NaOH (aq.). The water layer was extracted with ethyl acetate. The organic layer was dried and concentrated to give crude

4-bromo-N¹,3-dimethylbenzene-1,2-diamine **61** (26 g, 48.4 mmol, 39.5%) that was used in the next step without further purification.

5-Bromo-1,4-dimethyl-1Hbenzo[d][1,2,3]triazole (62). The 4-bromo-N-1,3-dimethylbenzene-1,2-diamine **61** (30 g, 139 mmol) was dissolved in 17 mL of 10% H₂SO₄ and cooled to 0 °C. Sodium nitrite (13.47 g, 195 mmol) was added in small portions over a 20 min. period. After stirring 30 min. further, 200 mL water was added. The resulting precipitate was collected by filtration, washed with water and dried. The mother liquors were left to stand overnight, and a second batch of precipitate formed, which was collected as before. The combined solids were columned with EtOAc to remove inorganic salts, giving 5-bromo-1,4-dimethyl-1H-benzo[d][1,2,3]triazole **62** (10 g, 30.1 mmol, 21.57 % yield). LC-MS m/z: 226.0, 228.0 [M+H]⁺, 1.71 min. (ret. time).

(E)-Ethyl 3-(1-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)acrylate (63). A mixture of tri-*o*-tolylphosphine (2.69 g, 8.85 mmol), methyl acrylate (7.62 g, 88 mmol), 5-bromo-1,4-dimethyl-1H-benzo[d][1,2,3]triazole **62** (10 g, 44.2 mmol), Pd(OAc)₂ (0.993 g, 4.42 mmol), DIPEA (23.18 mL, 133 mmol) and N,N-dimethylformamide (20 mL) was stirred at 100 °C for 12 h. The mixture was poured into water and extracted with ethyl acetate (30 mL). The organic layer was dried and concentrated to get crude. The crude was purified through silica gel chromatography column (petroleum ether: ethyl acetate = 4:1) to obtain a grey solid. The grey solid was recrystallized from a mixture solvent (30 mL, petroleum ether: ethyl acetate=4:1) to obtain (E)-ethyl 3-(1-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)acrylate **63** (8.2 g, 33.7 mmol, 76%). LC-MS m/z 246.1 [M+H]⁺, 1.68 min. (ret. time); ¹H NMR (DMSO-d₆) δ: 8.04 – 7.68 (m, 2H), 7.71 (d, 1H, J = 8.8 Hz), 6.65 (d, 1H, J = 15.6 Hz), 4.3 (s, 3H), 4.22 (q, 2H, J = 6.8 Hz), 3.33 (s, 3H), 1.28 (t, 3H, J = 7.2 Hz).

Ethyl 3-(1,4-dimethyl-1Hbenzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (64). (E)-ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **63** (1.95 g, 7.95 mmol), (3-(hydroxymethyl)-4-methylphenyl) boronic acid **35** (1.979 g, 11.93 mmol), chloro(1,5- cyclooctadiene)rhodium(I) dimer (0.196 g, 0.398 mmol), and triethylamine (1.108 mL, 7.95 mmol) were added to 1,4-dioxane (46 mL), and water (28 mL) in a Schlenk flask and a stream of Ar was bubbled through the mix for ~2 minutes. The reaction was stirred under argon for 3 h. at 23 °C. The mixture was diluted with water (10 mL) and

EtOAc (25 mL). Phases were shaken together and the aqueous was extracted again with EtOAc (25 mL). The combined EtOAc was washed with saturated aqueous NaCl, dried (Na₂SO₄) and concentrated. The residue was preabsorbed on Isolute and purified on a silica cartridge (80 g) with a Combiflash Companion, eluting at 60 mL/min with a gradient running from DCM to EtOAc over 28 min. The desired fractions were pooled and concentrated to afford ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **64** (2.65 g, 7.21 mmol, 91 % yield) as a crunchy off white foam. LC-MS m/z: 368.1 [M+H]⁺, 0.88 min. (ret. time); ¹H NMR (400 MHz, chloroform-d) δ 7.36-7.48 (m, 1H), 7.16-7.38 (m, 3H), 6.96-7.18 (m, 2H), 4.94-5.12 (m, 1H), 4.59-4.74 (m, 2H), 4.22-4.36 (m, 3H), 3.92-4.15 (m, 2H), 2.98-3.26 (m, 2H), 2.88 (br. s., 3H), 2.24-2.39 (m, 3H), 1.45-1.77 (m, 9H), 1.04-1.25 (m, 3H).

(S)-3-(1,4-dimethyl-1Hbenzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid (9) and (R)-3-(1,4-dimethyl-1Hbenzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid (10). Ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **64** (0.735 g, 2 mmol) in tetrahydrofuran (30 mL) was added (R)-4-methyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide **43** (0.640 g, 3.00 mmol), PS-PPh₃ (2.500 g, 4.00 mmol) and DIAD (0.778 mL, 4.00 mmol). The resulting reaction mixture was stirred at room temperature for 40 min. The reaction mixture was filtered and the filtrate was evaporated under vacuum, purified with Combiflash chromatograph over silica gel column to afford desired crude intermediate ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoate **65** . (1.0031 g, 1.783 mmol, 89 % yield, 87% purity) that was used without further purification. LC-MS m/z: 563.2 [M+H]⁺, 1.12 min. (ret. time).

Ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoate **65** (1.00 g, 1.777 mmol) in methanol (40 mL) was added NaOH (3 N) (2.96 mL, 8.89 mmol). The resulting reaction mixture was stirred at 60 °C for 3 h. The reaction mixture was acidified to pH ~3 with HCl (1 N), evaporated down under vacuum, extracted with EtOAc (3 x 30 mL), dried over MgSO₄,

filtered, evaporated down under vacuum, purified with Combiflash chromatograph over silica gel column to afford desired product 3-(1,4-dimethyl-1Hbenzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid (812.3 mg, 1.519 mmol, 85 % yield) as a mixture of diastereomers. This diastereomer mixture was purified first using Gilson reverse phase chromatography and then by chiral SFC (AD column, 30% modifier, 1:1 IPA:MeOH) to afford the desired enantiomeric pure products (S)-3-(1,4-dimethyl-1Hbenzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid **9** (276.3 mg, 0.517 mmol, 29.1 % yield), LC-MS m/z: 534.8 [M+H]⁺, 0.93 min. (ret. time); ¹H NMR (400 MHz, chloroform-d) δ ppm 1.19 (d, J=6.53 Hz, 3 H) 2.30 (s, 3 H) 2.81 (s, 3 H) 2.88 (d, J=15.06 Hz, 1 H) 3.01 - 3.21 (m, 2 H) 3.69 (dd, J=15.31, 10.54 Hz, 1 H) 3.78 (d, J=14.05 Hz, 1 H) 4.17 - 4.28 (m, 4 H) 4.52 (d, J=14.31 Hz, 1 H) 4.92 (t, J=7.91 Hz, 1 H) 7.04 - 7.12 (m, 3 H) 7.18 (d, J=8.03 Hz, 1 H) 7.21 - 7.26 (m, 1 H) 7.32 (q, J=8.70 Hz, 2 H) 7.51 (t, J=7.65 Hz, 1 H) 7.87 (d, J=7.78 Hz, 1 H) and (R)-3-(1,4-dimethyl-1Hbenzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid **10** (286.2 mg, 0.535 mmol, 30.1 % yield), LC-MS m/z: 534.9 [M+H]⁺, 0.94 min. (ret. time); ¹H NMR (400 MHz, chloroform-d) δ ppm 1.14 (d, J=6.27 Hz, 3 H) 2.31 (s, 3 H) 2.78 - 2.85 (m, 4 H) 2.99 - 3.19 (m, 2 H) 3.63 (dd, J=15.18, 10.42 Hz, 1 H) 3.76 (d, J=14.05 Hz, 1 H) 4.16 - 4.26 (m, 4 H) 4.55 (d, J=14.05 Hz, 1 H) 4.93 (t, J=7.78 Hz, 1 H) 7.01 - 7.14 (m, 3 H) 7.17 (d, J=8.03 Hz, 1 H) 7.22 - 7.27 (m, 1 H) 7.30 (s, 2 H) 7.47 - 7.55 (m, 1 H) 7.88 (dd, J=7.78, 1.25 Hz, 1 H). The absolute stereochemistry of the two diastereomers was confirmed by VCD analysis.

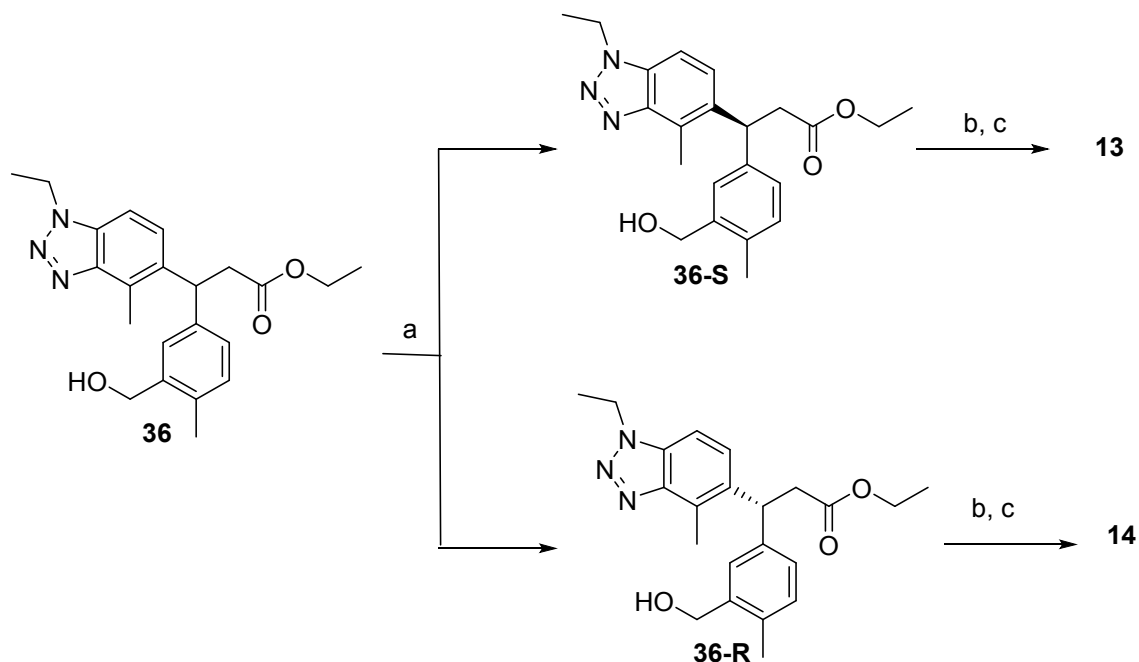
(S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid (11) and (R)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid (12). Ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **64** (510 mg, 1.388 mmol), (R)-4-ethyl-3,4-dihydro-2Hbenzo [b][1,4,5]oxathiazepine 1,1-dioxide **44** (331 mg, 1.457 mmol), and (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone) (700 mg, 2.78 mmol) were dissolved in tetrahydrofuran (15 mL) and tributylphosphine (0.693 mL, 2.78 mmol)

was added. Stirred in a sealed vial for 3 h. The reaction was concentrated and re-dissolved in DCM and preabsorbed on Isolute. The crude product was purified on a silica cartridge (40 g) with a Combiflash Companion, eluting at 40 mL/min with a gradient running from hexanes to 80% EtOAc/hexanes over 26 min. The desired fractions were pooled and concentrated to afford ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoate **66** (782 mg, 1.356 mmol, 98 % yield). LC-MS m/z: 577.5 [M+H]⁺, 1.20 min. (ret. time); ¹H NMR (400 MHz, chloroform-d) δ = 7.90 (br d, J = 8.0 Hz, 1H), 7.53 (t, J = 7.8 Hz, 1H), 7.41 - 7.18 (m, 4H), 7.12 (br s, 1H), 7.02 (d, J = 11.0 Hz, 1H), 5.33 (s, 1H), 4.97 (br t, J = 7.9 Hz, 1H), 4.61 - 4.52 (m, 1H), 4.27 (s, 2H), 4.04 (q, J = 7.2 Hz, 2H), 3.98 - 3.88 (m, 1H), 3.80 - 3.61 (m, 2H), 3.51 - 3.47 (m, 1H), 3.17 - 3.00 (m, 2H), 2.93 - 2.82 (m, 3H), 2.34 (d, J = 4.8 Hz, 2H), 2.07 (s, 1H), 1.73 (br s, 1H), 1.69 - 1.63 (m, 1H), 1.54 (br dd, J = 7.2, 15.2 Hz, 1H), 1.32 - 1.10 (m, 4H), 1.02 - 0.91 (m, 2H).

Ethyl 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoate **66** (2.58 g, 4.47 mmol) was dissolved in tetrahydrofuran (40 mL) and a solution of lithium hydroxide (2.143 g, 89 mmol) dissolved in water (40.0 mL) was added. Methanol (40 mL) was added to the resulting 2 phase mixture sufficient to form a single cloudy solution and the mixture was stirred 2 h. TLC and LC-MS of an aliquot partitioned between 1M HCl and EtOAc showed the desired product and no starting material in the EtOAc. The reaction was diluted with EtOAc (200 mL) and 1M HCl (100 mL) and the phases were shaken. (aqueous pH checked with pH paper and was < 2). The aqueous phase was extracted again with EtOAc and the combined EtOAc was washed with water (50 mL) and saturated aqueous NaCl (50 mL), dried (MgSO₄) and concentrated to afford 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid. This material along with previous batches of 3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid were purified by chiral SFC chromatography (Chiralpak AD, 20x250 mm, 5 μ , co-solvent: 30% MeOH) to afford (S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid **11**. LC-MS m/z: 549.3 [M+H]⁺, 1.02 min. (ret. time); ¹H NMR (500

MHz, DMSO-d₆) δ = 12.15 (br s, 1H), 7.78 (d, J = 7.6 Hz, 1H), 7.67 (t, J = 7.7 Hz, 1H), 7.55 (d, J = 8.4 Hz, 1H), 7.49 (d, J = 8.5 Hz, 1H), 7.36 (t, J = 7.7 Hz, 1H), 7.31 (d, J = 7.9 Hz, 1H), 7.24 (s, 1H), 7.21 (d, J = 7.9 Hz, 1H), 7.12 (d, J = 7.9 Hz, 1H), 4.79 (t, J = 7.9 Hz, 1H), 4.42 (d, J = 14.2 Hz, 1H), 4.24 (s, 3H), 4.17 - 4.08 (m, 1H), 3.81 (d, J = 13.9 Hz, 1H), 3.59 (dd, J = 10.2, 15.3 Hz, 1H), 3.04 (br d, J = 7.9 Hz, 2H), 2.80 - 2.74 (m, 4H), 2.25 (s, 3H), 1.49 (quind, J = 7.5, 14.6 Hz, 1H), 1.27 - 1.16 (m, 1H), 0.97 (t, J = 7.3 Hz, 3H) .and (R)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid **12**. LC-MS m/z: 549.3 [M+H]⁺, 1.02 min. (ret. time); ¹H NMR (500 MHz, DMSO-d₆) δ = 7.77 (d, J = 7.6 Hz, 1H), 7.66 (t, J = 7.7 Hz, 1H), 7.54 (d, J = 8.5 Hz, 1H), 7.40 (d, J = 8.8 Hz, 1H), 7.36 (t, J = 7.6 Hz, 1H), 7.30 - 7.20 (m, 3H), 7.13 (d, J = 8.2 Hz, 1H), 4.81 (t, J = 7.7 Hz, 1H), 4.42 (d, J = 13.6 Hz, 1H), 4.32 - 4.21 (m, 3H), 4.16 - 3.99 (m, 1H), 3.79 (d, J = 13.9 Hz, 1H), 3.56 (dd, J = 10.4, 15.1 Hz, 1H), 3.23 - 2.95 (m, 3H), 2.86 - 2.75 (m, 3H), 2.69 (d, J = 15.1 Hz, 1H), 2.30 - 2.23 (m, 3H), 1.39 (quind, J = 7.5, 14.6 Hz, 1H), 1.13 - 1.04 (m, 1H), 0.81 (t, J = 7.3 Hz, 3H). The absolute stereochemistry of the two diastereomers was confirmed by VCD analysis.

Scheme S4. Preparation of compounds 13-14.^a



^a(a) chiral HPLC, Chiralpak IC; (b) **44**, Bu₃P, ADDP, THF, RT, 20 h., 89 – 79 %; (c) 2 N LiOH (aqueous), MeOH, RT, 18 h., **13**: 48.1 %, **14**: 60.2 %.

(S)-Ethyl-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (36-S) and (R)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (36-R). Ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36** was separated by chiral preparative HPLC using a Chiralpak IC column (20 x 250 mm, 5μ) using a mobile phase: 20% EtOH (A); 80% hexane (B) to provide (S)-ethyl-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-S**, LC-MS m/z: 382.1 [M+H]⁺, 0.98 min. (ret. time) and (R)-ethyl-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-R**, LC-MS m/z: 382.1 [M+H]⁺, 0.98 min. (ret. time). .

(S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid (13). To a solution of (S)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-

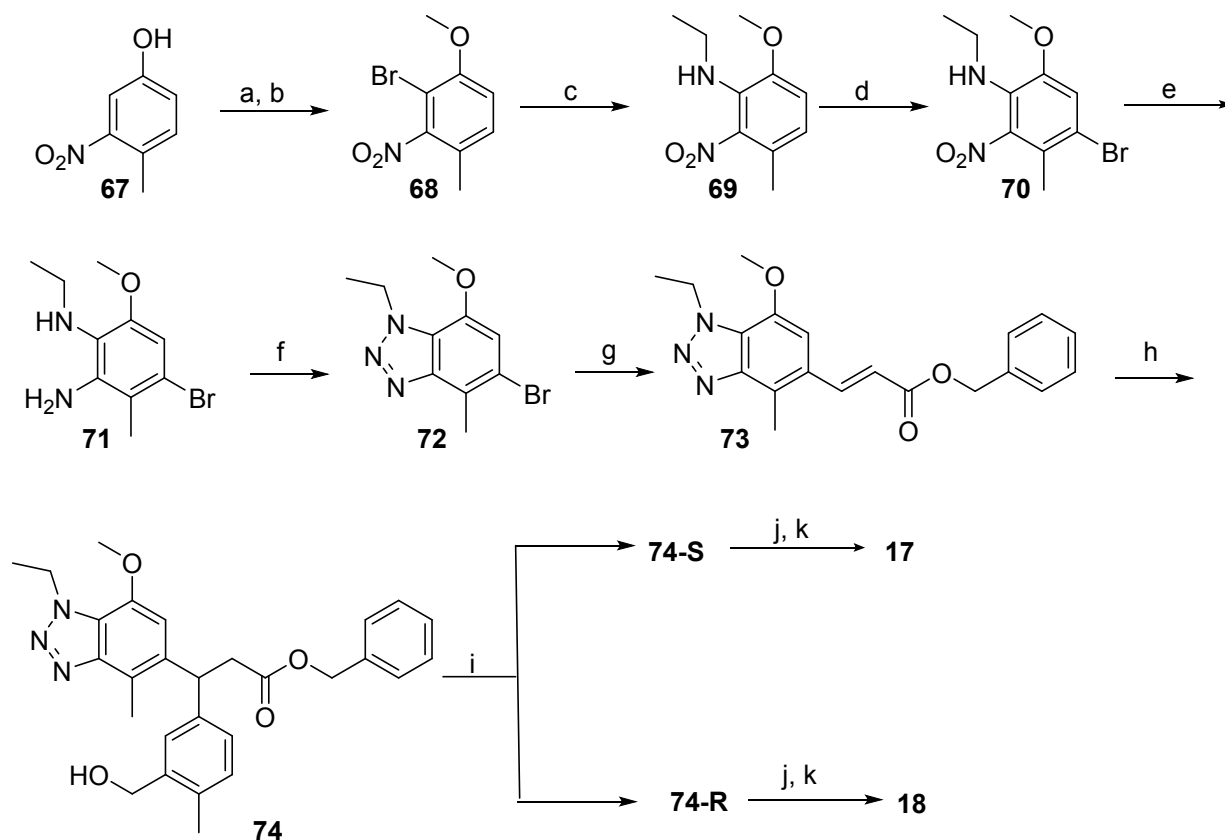
4-methylphenyl)propanoate **36-S** (80 mg, 0.210 mmol), (R)-4-ethyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide **44** (47.7 mg, 0.210 mmol), and 1,1'-(azodicarbonyl)dipiperidine (106 mg, 0.419 mmol) in tetrahydrofuran (3 mL) at room temperature was added tri-n-butylphosphine (0.103 mL, 0.419 mmol). The reaction mixture was stirred at ambient temperature for 20 h. The reaction was quenched with saturated aqueous NH₄Cl and extracted with ethyl acetate twice. The combined organic layer was washed with brine, dried and concentrated. It was purified by silica gel chromatography (Combiflash) (product eluted at 50 % ethyl acetate in hexane). Desired fractions were concentrated to give (S)-ethyl 3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (110 mg, 0.186 mmol, 89 % yield).

(S)-Ethyl-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2Hbenzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate was dissolved in methanol (2.00 mL), 2 M aqueous LiOH (0.629 mL, 1.258 mmol) was added and stirred at room temperature for 18 h. Solvent was concentrated; 0.8 mL of 1 N HCl and 3 mL of DMSO were added and most of the solvents were dried under vacuum (V10). The residual was filtered and then purified by Gilson HPLC (Atlantis T3, 5 μ , mobile phase: A: water; B: MeCN, gradient: 20% B to 60% B in 10 min.) to give (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **13** (56.8 mg, 0.101 mmol, 48.1 % yield) was obtained as yellow solid. LC-MS m/z: 563.1 [M+H]⁺, 1.12 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ = 7.77 (br d, J = 7.8 Hz, 1H), 7.65 (br t, J = 7.7 Hz, 1H), 7.52 (br d, J = 8.5 Hz, 1H), 7.45 - 7.26 (m, 2H), 7.15 - 7.03 (m, 2H), 4.82 (br t, J = 7.3 Hz, 1H), 4.66 (q, J = 7.1 Hz, 1H), 4.40 (br d, J = 13.6 Hz, 1H), 4.09 - 4.01 (m, 1H), 3.76 (br d, J = 14.1 Hz, 1H), 3.55 (br dd, J = 10.8, 14.8 Hz, 1H), 3.39 (br s, 1H), 2.78 - 2.67 (m, 3H), 2.63 - 2.54 (m, 8H), 2.24 (s, 2H), 1.51 - 1.41 (m, 3H), 1.26 - 1.15 (m, 1H), 0.91 (br t, J = 7.3 Hz, 2H). The absolute stereochemistry was confirmed by VCD analysis.

(R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid (14). To a solution of (R)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-

4-methylphenyl)propanoate **36-R** (80 mg, 0.210 mmol), (R)-4-ethyl-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepine 1,1-dioxide **44** (47.7 mg, 0.210 mmol), and 1,1'-(azodicarbonyl)dipiperidine (106 mg, 0.419 mmol) in tetrahydrofuran (3 mL) at room temperature was added tri-*n*-butylphosphine (0.103 mL, 0.419 mmol). The reaction mixture was stirred at ambient temperature for 20 h. It was then quenched with saturated aqueous NH₄Cl and extracted with ethyl acetate twice. The combined organic layer was washed with brine, dried and concentrated. It was purified by silica gel chromatography (Combiflash) (product eluted at 50 % ethyl acetate in hexane). Desired fractions were concentrated to give (R)-ethyl 3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (98 mg, 0.166 mmol, 79 % yield). LC-MS *m/z*: 591.4 [M+H]⁺, 1.33 min (ret. time). The (R)-ethyl 3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate was dissolved in methanol (2.00 mL), 2 M LiOH (0.629 mL, 1.258 mmol) was added and stirred at room temperature for 18 h. Solvent was concentrated; 0.8 mL of 1 N HCl and 3 mL of DMSO were added. Most solvents were dried under vacuum (V10). The residue was filtered and then purified by prep-HPLC to provide (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-benzo[b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (**14**) (71 mg, 0.126 mmol, 60.2 % yield) as yellow solid. LC-MS *m/z*: 563.1 [M+H]⁺, 1.10 min (ret. time); ¹H NMR (400 MHz, DMSO-*d*₆) δ = 12.67 - 11.88 (m, 1H), 7.91 - 7.71 (m, 2H), 7.70 - 7.62 (m, 1H), 7.58 (br d, *J* = 8.5 Hz, 1H), 7.47 - 7.32 (m, 3H), 7.32 - 7.19 (m, 3H), 7.14 (br s, 1H), 4.66 (br d, *J* = 7.0 Hz, 4H), 2.78 (s, 4H), 2.73 - 2.54 (m, 2H), 2.26 (s, 3H), 2.08 (s, 1H), 1.61 - 1.33 (m, 5H), 0.91 - 0.77 (m, 3H). The absolute stereochemistry was confirmed by VCD analysis.

Scheme S5. Preparation of compounds 17-18.^a



^a(a) K_2CO_3 , MeI, acetone, 65 °C, 4 h., 98 %; (b) H_2SO_4 , 20 °C, 25 min.; 1-bromopyrrolidine-2,5-dione, 20 °C, 1 h., 52.1 %; (c) Cu, ethanamine, DMSO, 20 °C, then 100 °C, 16 h., 83 %; (d) NBS, DMF, 0 °C, 3 h., 82 %; (e) H_2NNH_2 , EtOH, DCE, 0 °C, 2 h., 66.1 %; (f) H_2SO_4 , H_2O , 30 min.; NaNO_2 , 0 °C to 20 °C, 2 h., 54.4 %; (g) benzyl acrylate, N-ethyl-N-isopropylpropan-2-amine, tri-*o*-tolylphosphane, $\text{Pd}(\text{OAc})_2$, DMF, 150 °C (microwave), 2 h., 93 %; (h) **39**, $[\text{RhCl}(\text{cod})]_2$, Et_3N , 1,4-dioxane, H_2O , 120 °C (microwave), 1 h., 77 %; (i) Chiral SFC, Chiralpak IA column; (j) **27**, 1,1'-(azodicarbonyl)dipiperidine, Bu_3P , THF, RT, 18-20 h., quantitative, 68.4 %; (k) 2 M LiOH (aqueous), MeOH, RT, 5-8 h., **17**: 95 %, **18**: 48.9 %.

2-Bromo-1-methoxy-4-methyl-3-nitrobenzene (68). A solution of 4-methyl-3-nitrophenol **67** (30 g, 196 mmol) in acetone (200 mL) was added K_2CO_3 (54.2 g, 392 mmol) and MeI (24.50 mL, 392 mmol) slowly under nitrogen and stirrer at 65 °C for 4 h. The reaction mixture was concentrated to give 4-methoxy-1-methyl-2-nitrobenzene (32 g, 191 mmol, 98 % yield) as a brown oil that is used in the next step without further purification. ^1H NMR (500 MHz, CDCl_3) δ 7.50 (d, J = 2.7 Hz, 1H), 7.23 (d, J = 8.5 Hz, 1H), 7.06 (dd, J = 8.5, 2.7 Hz, 1H), 3.85 (s, 3H), 2.53 (s, 3H). 4-Methoxy-1-methyl-2-nitrobenzene (32 g, 191 mmol) was added slowly into sulfuric acid (163 mL,

3063 mmol) at 20 °C and stirred at 20 °C for 25 min. Then cooled down to 20 °C, 1-bromopyrrolidine-2,5-dione (44.3 g, 249 mmol) was added portion wise to the resulting solution and the reaction mixture was warmed to 20 °C over 1 h. Ice water (100 mL) was added and the resulting suspension extracted with ethyl acetate (3 x 200 mL). Combined the organic phase was separated and dried with Na₂SO₄, filtered and concentrated *in vacuo*. The crude product was then purified on a silica cartridge (120 g) with a Combiflash Companion, eluting at 60 mL/min with a gradient running from 100% hexanes to 5% EtOAc/hexanes over 70 min) to give 2-bromo-1-methoxy-4-methyl-3-nitrobenzene **68** (25 g, 100 mmol, 52.1 % yield) as light yellow solid. ¹H NMR (500 MHz, CDCl₃) δ 7.20 (d, J = 8.6 Hz, 1H), 6.91 (d, J = 8.5 Hz, 1H), 3.92 (s, 3H), 2.26 (s, 3H).

N-Ethyl-6-methoxy-3-methyl-2-nitroaniline (69). A solution of 2-bromo-1-methoxy-4-methyl-3-nitrobenzene **68** (55 g, 224 mmol) in dimethyl sulfoxide (100 mL) was added copper (0.710 g, 11.18 mmol) and ethanamine (168 mL, 2235 mmol) slowly under nitrogen at 20 °C. The reaction mixture was stirred at 100 °C for 16 h. The reaction mixture was added to water (200 mL), extracted with ethyl acetate (300mL x 3), and concentrated to give N-ethyl-6-methoxy-3-methyl-2-nitroaniline **69** (41 g, 185 mmol, 83 % yield) as a red oil that was used in the next step without further purification. LC-MS m/z: 211.0 [M+H]⁺, 2.00 min. (ret. time); ¹H NMR (400 MHz, CDCl₃) δ 6.74 (d, J = 8.2 Hz, 1H), 6.58 (d, J = 8.2 Hz, 1H), 3.85 (s, 3H), 3.17 – 3.02 (m, 2H), 2.23 (s, 3H), 1.18 (t, J = 7.1 Hz, 3H).

4-Bromo-N-ethyl-6-methoxy-3-methyl-2-nitroaniline (70). A solution of N-ethyl-6-methoxy-3-methyl-2-nitroaniline **69** (41 g, 195 mmol) in N,N-dimethylformamide (300 mL) was added NBS (34.7 g, 195 mmol) in N,N-dimethylformamide (30 mL) dropwise at 0 °C. The reaction mixture was stirred at 0 °C for 3 h. Water (500 mL) was added and stirred for 1 h. The reaction mixture was filtered and the solid was dissolved in 3 x 300 mL of ethyl acetate, washed with brine (300 mL), dried with Na₂SO₄, concentrated to give 4-bromo-N-ethyl-6-methoxy-3-methyl-2-nitroaniline **70** (52g, 159 mmol, 82 % yield) as a red solid. LC-MS m/z: (79 Br/ 81 Br) 289.1/291.1 [M+H]⁺, 1.51 min (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 6.98 (s, 1H), 3.86 (s, 3H), 3.06 (q, J = 7.1 Hz, 2H), 2.25 (s, 3H), 1.17 (s, 3H).

4-Bromo-N1-ethyl-6-methoxy-3-methylbenzene-1,2-diamine (71). A mixture of nickel (1.056 g, 17.99 mmol) and 4-bromo-N-ethyl-6-methoxy-3-methyl-2-nitroaniline **70** (52 g, 180 mmol) in ethanol (500 mL) and 1,2-dichloroethane (500 mL) was added hydrazine (20.34 g, 540 mmol) dropwise slowly. It was stirred at 0 °C for 2 h. The precipitates were removed by passing through a plug of Celite, washed with 500 ml of ethyl acetate and concentrated to give 4-bromo-N1-ethyl-6-methoxy-3-methylbenzene-1,2-diamine **71** (42 g, 119 mmol, 66.1 % yield) as black oil. LC-MS m/z: (79 Br/81 Br) 259.1/261.1 [M+H]⁺, 2.06 min.(ret. time); ¹H NMR (500 MHz, CDCl₃) δ 6.57 (s, 1H), 3.77 (s, 3H), 2.87 (q, J = 7.1 Hz, 2H), 2.22 (s, 3H), 1.14 (t, J = 7.1 Hz, 3H).

5-Bromo-1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazole (72). To a solution of 4-bromo-N1-ethyl-6-methoxy-3-methylbenzene-1,2-diamine **71** (42 g, 162 mmol) in water (300 mL), sulfuric acid (127 g, 1297 mmol) was added and stirred for 30 min. Then, sodium nitrite (16.77 g, 243 mmol) dissolved in water (30 mL) was added into the reaction mixture at 0 °C, warmed to 20 °C and stirred for 2 h. The reaction mixture was extracted by EtOAc (600 mL) three times, the organic layers was combined and washed by brine (500 mL) and dried with Na₂SO₄ and concentrated in vacuo. The crude product was then purified on a silica cartridge (120 g) with a Combiflash Companion, eluting at 60 mL/min with a gradient running from 100% hexanes to 15% EtOAc/hexanes over 80 min) to give 5-bromo-1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazole **72** (25 g, 88 mmol, 54.4 % yield) as yellow solid. LC-MS m/z: (79 Br/ 81 Br) 270.3/272.3 [M+H]⁺, 1.37 min. (ret. time); ¹H NMR (500 MHz, CDCl₃) δ 6.89 (s, 1H), 4.82 (q, J = 7.2 Hz, 2H), 3.97 (s, 3H), 2.71 (s, 3H), 1.56 (t, J = 7.2 Hz, 3H). .

Benzyl (E)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate (73). A solution of 5-bromo-1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazole **72** (1.5 g, 5.55 mmol) in N,N-dimethylformamide (8 mL) at ambient temperature was added benzyl acrylate (3.40 mL, 22.21 mmol) and N-ethyl-N-isopropylpropan-2-amine (3.88 mL, 22.21 mmol), tri-*o*-tolylphosphane (0.507 g, 1.666 mmol), followed by palladium(II) acetate (0.187 g, 0.833 mmol). The reaction mixture was heated in microwave under high absorption at 150 °C for 2 h. It was passed through Celite and washed with EtOAc. The filtrate was washed with water and brine. The organic layer was collected and concentrated to give the crude product. The crude product was purified on a silica cartridge (40 g) with a Combiflash Companion, eluting at 40 mL/min with a

gradient running from 0 % EtOAc/hexanes to 60 % EtOAc/hexanes over 35 min (product came out at 25 % ethyl acetate in hexane). The desired fractions were concentrated under reduced pressure to give benzyl (E)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **73** (1812 mg, 5.16 mmol, 93 % yield) as solid. LC/MS m/z: 352.2 (M+H)⁺, 1.25 min.(ret. time); ¹H NMR (400MHz, DMSO-d₆) δ = 8.05 (d, J=15.8 Hz, 1H), 7.54 - 7.27 (m, 6H), 6.83 (d, J=15.8 Hz, 1H), 5.26 (s, 2H), 4.78 (q, J=7.0 Hz, 2H), 4.03 (s, 3H), 2.70 (s, 3H), 1.57 - 1.40 (m, 3H).

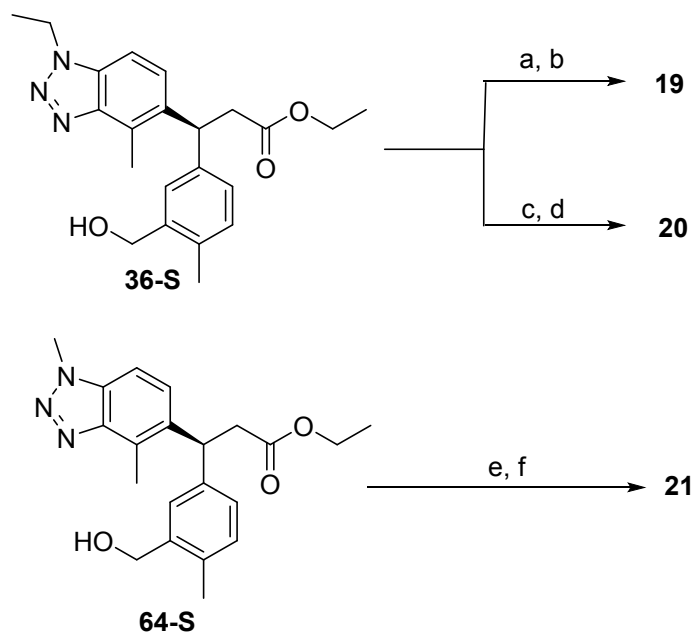
(S)-Benzyl-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (74-S) and (R)-benzyl 3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (74-R). To a suspension of benzyl (E)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **73** (1,800 mg, 5.12 mmol), (2-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol **39** (1,906 mg, 7.68 mmol), and [RhCl(cod)]₂ (253 mg, 0.512 mmol) in 1,4-dioxane (8 mL) and water (2.000 mL) at ambient temperature was added triethylamine (2.142 mL, 15.37 mmol). The reaction was degassed for 10 min. and heated in a Biotage microwave at high absorption for 1 h. at 120 °C. The reaction mixture was passed through Celite and washed with EtOAc. The organic layer was collected and concentrated to give the crude product. The reaction mixture was purified by silica gel chromatography (Combiflash) (product came out at 50 % ethyl acetate in hexane). Desired fractions were concentrated to give benzyl 3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **74** (1,866 mg, 3.94 mmol, 77 % yield). This material was resolved by Chiral SFC (Column: Chiralpak IA 20x250mm, 5μ; co-solvent: 25% EtOH; Flowrate: 50 g/min; Back pressure: 100 Bar) to give single enantiomerically pure benzyl (S)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **74-S** (804.6 mg, 1.699 mmol, 33.2 % yield), chiral SFC ret. time: 2.31 min; LC-MS m/z: 474.4 [M+H]⁺, 1.15 min. (ret. time) and single enantiomerically pure benzyl (R)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **74-R** (778.9 mg, 1.645 mmol, 32.1 % yield), chiral SFC ret. time: 2.96 min; LC-MS m/z: 474.5 [M+H]⁺, 1.15 min (ret. time) . The absolute stereochemistry of the enantiomers was confirmed by comparison by

SFC chiral chromatography of the R-enantiomer to a chiral known reference whose structure was determined by VCD analysis.

(S)-3-(3-(((R)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid (17). To a solution of benzyl (S)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **74-S** (80 mg, 0.169 mmol), (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **27** (38.6 mg, 0.169 mmol), and 1,1'-(azodicarbonyl)dipiperidine (85 mg, 0.338 mmol) in tetrahydrofuran (3 mL) at room temperature was added tri-n-butylphosphine (0.083 mL, 0.338 mmol). The reaction mixture was stirring at ambient temperature for 18 h. It was concentrated and purified by silica gel chromatography (Combiflash) (product came out at 50 % ethyl acetate in hexane). Pure desired fractions were concentrated to give benzyl (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (119 mg, 0.174 mmol, 103 % yield). LC-MS *m/z*: 684.6 [M+H]⁺, 1.35 min. (ret. time). The benzyl ester was dissolved in methanol (3.00 mL), 2M LiOH (0.507 mL, 1.014 mmol) was added and the reaction mixture and stirred at ambient temperature for 5 h. Solvent was concentrated and 1 mL 1N HCl and 3 mL DMSO were added. It was concentrated, filtered and purified with preparative HPLC under neutral condition to give (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **17** (95 mg, 0.160 mmol, 95 % yield). LC-MS *m/z*: 594.4 [M+H]⁺, 1.06 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ = 8.54 (d, J=4.8 Hz, 1H), 8.22 (d, J=7.5 Hz, 1H), 7.45 (dd, J=5.0, 7.5 Hz, 1H), 7.23 - 7.03 (m, 3H), 6.94 (s, 1H), 4.83 - 4.68 (m, 3H), 4.39 (d, J=14.1 Hz, 1H), 4.25 (br. s., 1H), 3.99 - 3.90 (m, 4H), 3.64 (dd, J=10.2, 15.4 Hz, 1H), 2.85 (d, J=15.1 Hz, 1H), 2.55 (s, 4H), 2.23 (s, 3H), 2.08 (s, 1H), 1.51 - 1.36 (m, 4H), 1.29 - 1.17 (m, 1H), 0.86 (t, J=7.3 Hz, 3H). The absolute stereochemistry was confirmed by VCD analysis.

(R)-3-(3-(((R)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (18). To a solution of benzyl (R)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **74-R** (80 mg, 0.169 mmol), (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **27** (38.6 mg, 0.169 mmol), and 1,1'-(azodicarbonyl)dipiperidine (85 mg, 0.338 mmol) in tetrahydrofuran (3 mL) at room temperature was added tri-n-butylphosphine (0.083 mL, 0.338 mmol). The reaction mixture was stirred at ambient temperature for 20 h. It was quenched with water and extracted with ethyl acetate twice. The combined organic layer was washed with brine and concentrated. It was purified by silica gel chromatography (Combiflash) (product came out at 30 % ethyl acetate in hexane). Pure desired fractions were concentrated to give benzyl (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (79 mg, 0.116 mmol, 68.4 % yield). The benzyl ester was dissolved in methanol (3.00 mL), 2M LiOH (0.507 mL, 1.014 mmol) was added and the reaction mixture and stirred at ambient temperature for 8 h. Solvent was concentrated and 1 mL 1N HCl and 3 mL DMSO were added. It was concentrated, filtered and purified with neutral preparative HPLC to give (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-7-methoxy-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid **18** (49 mg, 0.083 mmol, 48.9 % yield). LC-MS m/z 594.5 (M+H)⁺, 1.06 min (ret. time); ¹H NMR (400MHz, DMSO-d₆) δ = 12.17 (br. s., 1H), 8.55 (d, J=4.8 Hz, 1H), 8.23 (d, J=7.5 Hz, 1H), 7.52 - 7.39 (m, 1H), 7.32 - 7.21 (m, 2H), 7.15 (d, J=8.0 Hz, 1H), 6.80 (s, 1H), 4.76 (qd, J=7.4, 14.5 Hz, 3H), 4.43 (d, J=14.1 Hz, 1H), 4.29 (br. s., 1H), 4.02 - 3.88 (m, 4H), 3.65 (dd, J=10.3, 15.1 Hz, 1H), 3.18 - 2.97 (m, 2H), 2.80 (d, J=15.1 Hz, 1H), 2.65 (s, 3H), 2.25 (s, 3H), 1.51 - 1.36 (m, 4H), 1.17 (d, J=5.8 Hz, 1H), 0.88 - 0.78 (m, 3H). The absolute stereochemistry was confirmed by VCD analysis.

Scheme S6. Preparation of compounds 19-21.^a



^a(a) **48**, Bu₃P, (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone, THF, RT, 3 h., quantitative; (b) 2M LiOH (aqueous), MeOH, RT, 18 h., 73.4 %; (c) **50**, Bu₃N, 1,1'-(azodicarbonyl)dipiperidine, THF, RT, 20 h., 91 %; (d) 2M LiOH (aqueous), MeOH, RT, 18 h., 78 %; (e) **27**, Bu₃P, (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone), THF, RT, 20 h.; (f) LiOH, MeOH, H₂O, 25 °C, 20 h., 48.9 %, two steps.

(S)-3-(1-Ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid (19). To a solution of ethyl (S)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-S** (99.6 mg, 0.261 mmol) and (R)-4-methyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine-1,1-dioxide **48** (67.1 mg, 0.313 mmol) in tetrahydrofuran (2 mL), (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone (132 mg, 0.522 mmol) and tributylphosphine (0.130 mL, 0.522 mmol) were added. The reaction mixture was stirred at room temperature for 3 h. The reaction mixture was then evaporated and dissolved in EtOAc. The organic phase was washed with water (x 2) and the aqueous phase was extracted with EtOAc (x 2) and the combined organic layers were dried over sodium sulfate, filtered, and concentrated. The crude product was purified via flash chromatography to give ethyl (S)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoate (160.8 mg, 0.278 mmol, 107 %

yield). Some tributyl phosphine oxide still remained but the product was carried forward to the next step with no additional purification. LC-MS *m/z*: 578.3 [M+H]⁺, 1.13 min. (ret. time).

To a solution of ethyl (S)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoate (151 mg, 0.261 mmol) in methanol (2 mL) was added 2M LiOH (1.307 mL, 2.61 mmol). The reaction was allowed to stir at room temperature for 18 h. 1M HCl was added until a pH of ~2 was reached. It was then extracted with EtOAc (x 3) and the combined organic layers were dried over sodium sulfate, filtered and concentrated to dryness. The crude product was dissolved in DMSO and purified under acidic conditions (formic acid) via preparative HPLC to give (S)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(4-methyl-3-(((R)-4-methyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)phenyl)propanoic acid **19** (105.4 mg, 0.192 mmol, 73.4 % yield). LC-MS *m/z*: 550.2 [M+H]⁺, 0.95 min. (ret. time); ¹H NMR (400 MHz, methanol-d₄) δ ppm = 8.48 (d, *J*=4.77 Hz, 1 H), 8.31 (d, *J*=7.53 Hz, 1 H), 7.50 - 7.62 (m, 2 H), 7.43 (dd, *J*=7.40, 5.14 Hz, 1 H), 7.11 - 7.25 (m, 3 H), 5.51 (s, 1 H), 4.96 (t, *J*=8.03 Hz, 1 H), 4.73 (q, *J*=7.36 Hz, 2 H), 4.44 - 4.60 (m, 2 H), 3.96 (d, *J*=14.05 Hz, 1 H), 3.70 (dd, *J*=15.43, 10.42 Hz, 1 H), 3.02 - 3.19 (m, 2 H), 2.91 (d, *J*=15.31 Hz, 1 H), 2.77 (s, 3 H), 2.32 (s, 3 H), 1.60 (t, *J*=7.28 Hz, 3 H), 1.17 (d, *J*=6.27 Hz, 3 H). The absolute stereochemistry was confirmed by VCD analysis.

(S)-3-(3-((4,4-dimethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (**20**). To a solution of (S)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-S** (80 mg, 0.210 mmol), 4,4-dimethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **50** (47.9 mg, 0.210 mmol), and 1,1'-(azodicarbonyl)dipiperidine (106 mg, 0.419 mmol) in tetrahydrofuran (3 mL) at room temperature was added tri-*n*-butylphosphine (0.103 mL, 0.419 mmol). The reaction mixture was stirred at ambient temperature for 20 h. It was quenched with saturated NH₄Cl and extracted with ethyl acetate twice. The combined organic layer was washed with brine, dried and concentrated. It was purified by silica gel chromatography (Combiflash) (product came out at 70 % ethyl acetate in hexane). Desired fractions were concentrated to give (S)-ethyl 3-(3-((4,4-dimethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-

methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate (113 mg, 0.191 mmol, 91 % yield). LC-MS m/z: 592.4, 1.16 min. (ret. time). The (S)-ethyl 3-(3-((4,4-dimethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate was dissolved in methanol (2.00 mL), 2 M LiOH (0.629 mL, 1.258 mmol) was added and stirred at room temperature for 18 h. Solvent was concentrated and then 0.8 mL of 1 N HCl and 3 mL of DMSO were added. Most solvents were evaporated under vacuum (V10). The residue was filtered and then submitted to purification for neutral Prep-HPLC to give (S)-3-(3-((4,4-dimethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **20** (92 mg, 0.163 mmol, 78 % yield) as yellow solid. LC-MS m/z: 564.2 [M+H]⁺, 0.96 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ = 8.48 (dd, J = 2.0, 4.9 Hz, 1H), 8.15 (dd, J = 2.0, 7.8 Hz, 1H), 7.55 (d, J = 8.3 Hz, 1H), 7.42 (s, 1H), 7.41 - 7.39 (m, 1H), 7.16 - 7.00 (m, 3H), 4.87 (t, J = 7.3 Hz, 1H), 4.66 (q, J = 7.0 Hz, 2H), 4.48 - 4.28 (m, 2H), 2.74 - 2.60 (m, 5H), 2.55 (s, 4H, some overlap with DMSO peaks), 2.14 (s, 3H), 1.46 (t, J = 7.1 Hz, 3H), 1.05 (s, 6H). The absolute stereochemistry was confirmed by VCD analysis.

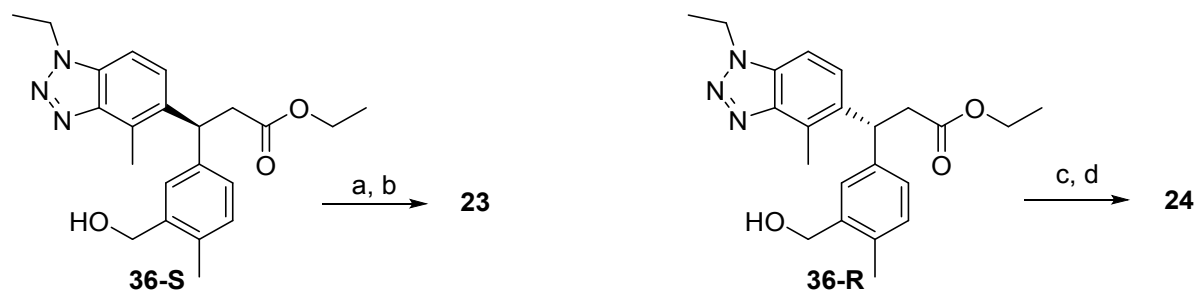
(S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid (21).

Enantiomerically pure **64-S** was obtained by the chiral purification of **64** into the two isomers, **64-S** and **64-R** whose absolute stereochemistry was confirmed by VCD. To a solution of ethyl (S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **64-S** (270.0 mg, 0.735 mmol) in tetrahydrofuran (10 mL), (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **27** (201 mg, 0.882 mmol), (E)-diazene-1,2-diylbis(piperidin-1-yl)methanone) (371 mg, 1.470 mmol) and tributylphosphane (0.367 mL, 1.470 mmol) were added. The reaction mixture was stirred at room temperature for 20 h. The reaction mixture was then combined with material from a previous experiment and the solvent was evaporated and the combined compound was purified via flash chromatography and dried under high vacuum to give ethyl (S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoate (0.5319 g, 0.921 mmol, 125 % yield, there appears by LC-MS and ¹H NMR to be some impurity from tributyl phosphine oxide; the product was carried forward to the next step without any additional

purification). LC-MS m/z : 578.5 $[M+H]^+$, 1.14 min. (ret. time); 1H NMR (400 MHz, methanol- d_4) δ = 8.51 - 8.41 (m, 1H), 8.33 - 8.25 (m, 1H), 7.57 - 7.50 (m, 2H), 7.45 - 7.35 (m, 1H), 7.25 - 7.15 (m, 2H), 7.08 (s, 1H), 5.51 (s, 1H), 4.99 - 4.87 (m, 1H), 4.55 (d, J = 14.1 Hz, 1H), 4.30 (s, 3H), 4.21 (br s, 1H), 4.12 (q, J = 6.9 Hz, 1H), 4.05 - 3.93 (m, 3H), 3.71 - 3.56 (m, 1H), 3.38 - 3.35 (m, 2H), 3.19 - 3.07 (m, 2H), 2.92 (br d, J = 15.3 Hz, 1H), 2.76 (s, 3H), 2.33 (s, 3H), 2.03 (s, 2H), 1.85 - 1.71 (m, 1H), 1.67 (br s, 1H), 1.62 - 1.42 (m, 3H), 1.32 - 1.22 (m, 3H), 1.17 - 1.06 (m, 4H), 1.01 - 0.86 (m, 5H).

To a solution of ethyl (S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoate (0.5300 g, 0.917 mmol) in methanol (2.0 mL) and water (1.0 mL) was added LiOH (0.220 g, 9.17 mmol). The reaction was allowed to stir for 20 h at 25 °C. The reaction mixture was concentrated, and water was added along with 1M HCl until the pH was $\sim$ 1. The aqueous layer was then extracted with EtOAc (3 x 10mL) and dried over sodium sulfate, filtered and concentrated. The crude product was then purified by prep. HPLC under acidic conditions to give (S)-3-(1,4-dimethyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)propanoic acid **21** (246.7 mg, 0.449 mmol, 48.9 % yield). A portion of the product was then recrystallized in a mixture of MeOH and water. LC-MS m/z : 550.4 $[M+H]^+$, 0.94 min (ret. time); 1H NMR (400 MHz, DMSO- d_6) δ = 8.56 (br d, J = 5.0 Hz, 1H), 8.24 (d, J = 7.8 Hz, 1H), 7.62 - 7.54 (m, 1H), 7.50 - 7.45 (m, 2H), 7.23 (t, J = 7.0 Hz, 2H), 7.13 (d, J = 7.2 Hz, 1H), 4.80 (br t, J = 7.9 Hz, 1H), 4.50 - 4.41 (m, 1H), 4.33 (br s, 1H), 4.27 - 4.23 (m, 3H), 3.93 (d, J = 13.8 Hz, 1H), 3.66 (br dd, J = 10.3, 15.1 Hz, 1H), 3.04 (br d, J = 8.3 Hz, 2H), 2.83 (br d, J = 15.3 Hz, 1H), 2.75 (s, 3H), 2.25 (s, 3H), 1.62 - 1.44 (m, 1H), 1.44 - 1.24 (m, 2H), 0.96 (t, J = 7.3 Hz, 3H).

Scheme S7. Preparation of compounds 23-24.^a



^a(a) **49**, Bu₃P, (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone), THF, 0 °C to RT, 1 h., 86 %; (b) LiOH, MeOH, H₂O, RT, 6 h., 53.6 %; (c) **49**, Bu₃P, (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone), THF, 0 °C to RT, 1 h., 99 %; (d) LiOH, MeOH, H₂O, RT, 6 h., 70.4 %.

(S)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2Hpyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid (23). To a solution of (S)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-S** (54 mg, 0.142 mmol) and (S)-4-ethyl-3,4-dihydro-2Hpyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **49** (38.8 mg, 0.170 mmol) in tetrahydrofuran (2 mL) and (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone) (71.4 mg, 0.283 mmol) was added. Then the reaction mixture was cooled at 0 °C, tributylphosphine (57.3 mg, 0.283 mmol) was added. The reaction mixture was stirred at room temperature for 1 h. All the solvent was evaporated under Biotage-V10 and purified by Combiflash (Hexane/Ethyl acetate) to provide 72 mg (0.122 mmol, 86 %) of ethyl (S)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2Hpyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoate. LC-MS m/z: 592.5 [M+H]⁺, 1.11 min. (ret. time). To a solution of ethyl (S)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2Hpyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoate in methanol (2.000 mL), LiOH (33.9 mg, 1.416 mmol) was added, followed by 5 drops of water. Then the reaction mixture was stirred at room temperature for 6 h. 1N of HCl was added until pH = 1. All the solvent was evaporated under reduced pressure (V-10) and was purified by reverse phase HPLC (Atlantics T3, 19 x 100 mm, 5μ; mobile phase: water: MeCN; flow rate: 18 mL/min; gradient: 35 % B to 65 % B in 10 min.) to give (S)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2Hpyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid **23** (42.8 mg, 0.076 mmol, 53.6 %). LC-MS m/z: 564.3 [M+H]⁺, 0.94 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ = 8.56 (dd, J = 1.6, 4.6 Hz, 1H), 8.23 (d, J = 7.1 Hz, 1H), 7.59 (d, J = 8.5 Hz, 1H), 7.47 (dd, J = 4.9, 7.4 Hz, 1H), 7.39 (d, J = 8.8 Hz, 1H), 7.26 - 7.19 (m, 2H), 7.15 (br d, J = 8.0 Hz, 1H), 4.82 (br t, J = 7.4 Hz, 1H), 4.71 - 4.63 (m, 2H), 4.45 (br d, J = 13.8 Hz, 1H), 4.33 - 4.26 (m, 1H), 3.92 (br d, J = 14.1 Hz, 1H), 3.63 (br dd, J = 10.7, 14.7 Hz, 1H), 3.03 (br dd, J = 7.8, 17.3 Hz, 2H), 2.82 - 2.72 (m, 4H), 2.26 (s, 3H), 1.54 - 1.37 (m, 4H), 1.17 - 1.06 (m, 1H), 0.86 - 0.77 (m, 3H). The absolute stereochemistry was confirmed by VCD analysis.

(R)-3-(3-(((S)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid (24). To a solution of (R)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-R** (52 mg, 0.136 mmol) in tetrahydrofuran, (S)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepine 1,1-dioxide **49** (37.3 mg, 0.164 mmol) and (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone) (68.8 mg, 0.273 mmol) were added. Then the reaction mixture was cooled at 0 °C, tributylphosphine (55.2 mg, 0.273 mmol) was added. The reaction mixture was stirred at room temperature for 1 h. All the solvent was evaporated under reduced pressure (Biotage-V10) and purified by Combiflash (hexane/ethyl acetate) to obtain 86 mg (0.135 mmol, 99 %) of ethyl (R)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate. LC-MS m/z: 592.5 [M+H]⁺, 1.11 min. (ret. time).

To a solution of ethyl (R)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate in methanol (2.000 mL), LiOH (32.6 mg, 1.363 mmol) was added followed by 5 drops of water. Then the reaction mixture was stirred at room temperature for 6 h. 1N HCl was added until pH = 1. All the solvent was evaporated under reduced pressure (Biotage V-10) and was purified by reverse phase HPLC (Column: Atlantis T3, 19 x 100 mm, 5μ; mobile phase: water: MeCN; flow rate: 18 mL/min; gradient: 35 % B to 65 % B in 10 min.) to give (R)-3-(3-(((S)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-b][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1Hbenzo[d][1,2,3]triazol-5-yl)propanoic acid **24** (54.1 mg, 0.096 mmol, 70.4 %). LC-MS m/z: 564.4, 0.95 min. (ret. time); ¹H NMR (400 MHz, DMSO-d₆) δ = 8.60 - 8.55 (m, 1H), 8.26 - 8.22 (m, 1H), 7.60 (d, J = 8.5 Hz, 1H), 7.50 - 7.45 (m, 1H), 7.27 - 7.21 (m, 1H), 7.15 - 7.12 (m, 1H), 4.81 (br t, J = 8.3 Hz, 1H), 4.68 (q, J = 7.1 Hz, 1H), 4.46 (s, 1H), 4.43 (s, 1H), 4.34 (br dd, J = 1.9, 3.6 Hz, 1H), 4.11 - 4.05 (m, 1H), 3.96 (s, 1H), 3.93 (s, 1H), 3.66 (br dd, J = 10.5, 14.6 Hz, 1H), 3.18 (br d, J = 4.0 Hz, 4H), 3.06 - 3.01 (m, 1H), 2.86 (s, 1H), 2.82 (s, 1H), 2.75 (s, 2H), 2.25 (s, 2H), 1.56 - 1.45 (m, 3H), 1.34 (br s, 1H), 0.95 (t, J = 7.3 Hz, 2H). The absolute stereochemistry was confirmed by VCD analysis.